Refresher Training through Gamified Activities on Maternal and Newborn Care for Frontline Healthcare Workers

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by

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Dedicated to my beloved parents.

Thesis Approval

This thesis entitled **Refresher Training through Gamified Activities for Front-line Healthcare Workers on Maternal and Newborn care** by **Arka Majhi** is approved for the degree of **Doctor of Philosophy**.

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CERTIFICATE OF COURSE WORK

This is to certify that **Arka Majhi** (Roll No. 184350001) was admitted to the candidacy of Ph.D. degree on 12 July 2019, after successfully completing all the courses required for the Ph.D. programme. The details of the course work done are given below.

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2	TD 621	Food Processing and Nutrition Delivery	6
3	DE 668	Instructional Design	4
4	TD 694	Seminar	4
5	TD 610	Contemporary Critical Issues in Technology and Development	AU
6	HS 791	Communication Skills -I	PP
7	TD 792	Communication Skills -II	PP
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Abstract

With the increasing demand for universal healthcare coverage, targeted towards mothers and children under different healthcare programmes in India, there has been a rapid increase in the number of Frontline Workers (FLWs) or Community Healthcare Workers (CHWs), i.e., the Anganwadi Workers (AWWs) and the Accredited Social Health Activists (ASHAs). The job training and the refresher training of these workers leave a lot to be desired. The capacity deficit of field functionaries has been identified as a major factor affecting the effectiveness of state health programmes. Given the massive numbers of field functionaries and the continuously advancing base of knowledge and techniques, conventional training pedagogy is ineffective in building and updating the capacity of the CHWs and their supervisors. With the improvement of technology and the widespread availability of smartphones in cities as well as rural and tribal villages of India, access to digital training systems can potentially benefit the healthcare workers and the healthcare system to a large extent, resulting in an increase in the delivery of maternal and child health services and consequent reduction in undernutrition, morbidity, and child mortality countrywide. The conventional training methods are proving to be inadequate for the task. The gamification of training material for refresher training and the use of digital tools for its dissemination offer one potential solution to this problem, which is the subject matter of the doctoral research. The child and mother care protocol, which includes the immunization schedule, medicine supplements for children, and scheduled visits, check-ups, and medication for antenatal and postnatal mothers, is gamified through various gaming options. Digital versions of these games have been developed, which are playable on Android smartphones. Card-based games turn out to be well-suited for this purpose, as they are easily customizable to accommodate the training content. Besides, they are also easily playable in the physical mode as well, given the familiarity of the field functionaries with traditional card games. The study compares the effectiveness of game-based learning pedagogy and conventional classroom instruction among a group of CHWs, and discusses ways to make it a scalable

and effective refresher training tool. Given its design and scope, we observed that the gamified training is best suited as a scalable training tool for the refresher training of frontline workers to reinforce and update existing skills, rather than as a substitute for the initial, in-person job training that CHWs require for hands-on competence.

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List of Abbreviations

ASHA	Accredited Social Health Activist
AWW	Anganwadi Worker
AWH	Anganwadi Helper
AWC	Anganwadi Centre
CHW	Community Health Worker
NFHS	National Family Health Survey
CNNS	Comprehensive National Nutrition Survey
ICDS	Integrated Child Development Services
IFA	Iron and Folic Acid
MWCD	Ministry of Women and Child Development
LMIC	Low and Middle income Countries
LISA	Local Indicators of Spatial Association
PHC	Primary Healthcare Centre
mHealth	Mobile Health
FDG	Focused Group Discussion
MCPC	Mother and Child Protection Card
IAP	Indian Academics of Paediatrician
ORS	Oral Rehydration Solution
ANC	Antenatal Care
GLO	General Learning Objectives
SLO	Specific Learning Objectives

MCQ	Multiple Choice Question
CINI	Child in Need Institute

Chapter 1

Introduction

Maternal and child health and nutrition are significant indicators of the growth of a nation and the well-being of its population. Healthcare systems worldwide are struggling with a shortage of Community Health Workers (CHWs) or trained health workers, especially in low and middle-income countries (LMICs), reaching an estimated deficit of 18 million by 2030 (range 16-19) (WHO, 2020). Despite a myriad of interventions and policy actions, these are still chronic problems, including in India, where malnutrition, morbidity, and mortality among children are strong public health concerns. Of the many government schemes addressing these issues, the Integrated Child Development Services (ICDS) scheme under the Ministry of Women and Child Development (MWCD) focuses on reducing malnutrition and overall development of children (below 6 years), as well as improving maternal health of pregnant women and lactating mothers. However, while there has been a noticeable improvement in the rate of child mortality, morbidity, high and persistent child malnutrition levels with their tardy reduction, as seen in successive health surveys, continue to be a matter of concern in India (International Institute for Population Sciences (IIPS) and Ministry of Health and Family Welfare, 2021).

One of the prominent features of the state health programmes is the role played by CHWs or frontline personnel, such as Anganwadi Workers (AWWs) and Accredited Social Health

Activists (ASHAs), who are the interface between the health and nutrition system and the community, where they usually reside. ASHAs are state-government-employed CHWs tasked with delivering last-mile healthcare. They are not salaried employees but considered community volunteers incentivized for their work on a milestone basis. ASHAs' activities involve mobilizing and supporting women for antenatal care (ANC) check-ups, ensuring timely postnatal care (PNC) visits, facilitating routine immunization for children and pregnant women, promoting institutional deliveries, tracking high-risk cases, and guiding households on nutrition, hygiene, and disease prevention.

The state health programme's implementation depends on their ability to converge in community healthcare planning and actions, and deliver essential healthcare services efficiently, which in turn, depends heavily on their training, expertise, and competence (Gujral et al., 1991; Agarwal et al., 2019). Training and supervision have been identified as one of the most ignored intervention-related aspects in systematic reviews of CHW success factors (Kok et al., 2015). However, given the magnitude of the issue and the annual growth in the number of field functionaries, traditional training and capacity-building methods have faced significant limitations. It remains a challenge for LMICs to train or educate CHWs (Sultan et al., 2025), ensuring the most vulnerable population's good health and well-being.

Capacity Building of CHWs

Refresher training for field functionaries, particularly in terms of capacity building, has traditionally been de-emphasized in terms of funding and practice. Traditionally, field workers are trained in a physical classroom environment, following the cascading model of teaching, where the state Master Trainer trains the district trainers, who in turn train the field functionaries. Training program budgets are low, and refresher training, which is essential for retaining and updating knowledge, gets disproportionately impacted by budget cuts. The trainers and learners also complain about inadequate infrastructure for conducting training and assessing the knowledge gains in crowded classrooms (Yadav et al., 2021). These are further compounded by the logistical difficulties of coordinating large-scale, face-to-face training sessions in widely dispersed and geographically distant locations. Conventional training methods have also proven ineffective in motivating field workers and facilitating long-term knowledge retention.

There is a pressing need to study alternative and emerging pedagogies that can bridge such gaps and render attempts at capacity building more affordable, participatory, and efficient. The

rapid proliferation of digital technologies and smartphones in India, especially following the COVID-19 pandemic, has opened up new possibilities for leveraging digital tools to enhance the training of field staff. Interestingly, the post-pandemic period has witnessed a considerable surge in digital uptake, and most field functionaries now have basic digital competence and exposure to smartphones. This development has paved the way for building digital training solutions tailored to their specific requirements.

1.1 Research Questions

This research addresses several key questions, focusing on the effectiveness of game-based training, knowledge retention, user engagement, and data collection efficiency. These questions aim to explore the potential of content gamification and digital tools in improving healthcare training contributing to the growing body of evidence supporting innovative pedagogical approaches in resource-constrained settings. These research questions are as follows:

1. **Effectiveness of Game-Based Training**

   RQ1 To what extent are game-based training methods more effective than traditional classroom-based refresher training for community healthcare workers (CHWs) in terms of knowledge acquisition and retention?

   RQ2 How do digital and physical gameplay modalities compare in improving CHWs' knowledge and performance?

2. **Knowledge Retention**

   RQ3 Are there significant differences in long-term knowledge retention between physical and digital gameplay modes for CHWs?

   RQ4 How does knowledge retention from game-based training compare to that achieved through traditional training methods after an extended period?

3. **Engagement and User Experience**

   RQ5 How do CHWs perceive and engage with digital and physical game-based training tools in terms of ease of use, motivation, and satisfaction?

1.2 Why haven't the educational paradigms in training materialized?

Over the years, researchers, government programmes, and non-governmental organizations have attempted to revolutionize training approaches for CHWs in India. A wide range of mobile phone-based interventions has been introduced at different points in time, each with varying degrees of effectiveness in promoting learning and knowledge retention (Bassi et al., 2018; Bashingwa and et.al., 2021; Joshi et al., 2023; Thomas et al., 2024). Some of these approaches, however, proved to be less adaptable to field conditions. We briefly review past instances to gain a better understanding of the problem involved.

In the 1930s, proponents of radio broadcasting claimed that it could replace teachers by providing expert instruction on a large scale. Researchers have even recently experimented with community radio like Mobile Vaani, an interactive voice response (IVR)–based mHealth intervention and multimedia content delivered through feature phones and smartphones to train CHWs (Yadav and et.al., 2017; Yadav et al., 2019; Yadav and et.al., 2019). By the 1950s, television broadcasting was considered an advanced educational medium, combining moving images with sound. Lectures were recorded and broadcast simultaneously to multiple classrooms. Research revealed that learners who viewed the broadcast from other rooms achieved the same level of knowledge gain as those who attended face-to-face in-person sessions.

Critics might argue that, though the medium was visual, the incentive was not aligned and motivating enough for people to learn. Researchers implemented an incentive-based training approach for CHWs using basic mobile phones (Shah et al., 2017). The intervention involved preloading the phones with video modules on maternal and child healthcare. Each video was followed by a set of related questions, with their correct answers. CHWs received follow-up phone calls during which they were quizzed on the material. Correct responses were rewarded with cellular talk-time credits, serving as an incentive to reinforce learning and participation. However, evidence suggests that once the incentive is withdrawn, the interest and engagement of CHWs tend to decline. The collection of offline-accessible videos functioned in a manner similar to earlier offline encyclopedic resources distributed by libraries and schools for students—useful as reference material but limited in sustaining long-term motivation without

external reinforcement.

Critics have also argued that the lack of interactivity might be an important aspect in such interventions. More recently, in a study conducted in Dharavi, Mumbai (n = 37), instructional illustrations were found to be significantly more effective than the video-based training materials distributed by the Ministry of Women and Child Development (MWCD) for training CHWs (Tulaskar, 2020).

Researchers have also experimented with quizzes and trivia as a teaching method (Majhi et al., 2021). Answering MCQs did not teach CHWs how to create knowledge, but rather how to choose answers. It also encourages guessing, and sometimes they receive rewards for answering something they are unaware of, simply by chance. The multiple answers comprise the wrong ones, too, exposing them to the incorrect content while they read all the options. The MCQs encourage memorizing the questions to beat the system. Quizzes were often part of MOOCs.

Massive Open Online Courses (MOOCs) were also considered transformative for education. Platforms such as Coursera, EdX, IITBombayX, Swawam, and NPTEL have provided access to knowledge and delivered education at scale. However, their effectiveness remains unclear, with studies critiquing high dropout rates (Gupta and Jain, 2014), limited learner engagement (Nieder et al., 2022), and non-consistent learning outcomes across different populations. Spoken Tutorial has experimented with extending this model beyond technical domains on topics such as exclusive breastfeeding, latch and positioning, complementary feeding, and maternal nutrition, designed for self-learning by CHWs and other field health workers (Rupal Dalal, 2019). While learning through MOOCs and other digital technologies generally has delivered factual knowledge effectively, they have yielded mixed results in developing hands-on skills (Mulcahy et al., 2022; Jensen et al., 2023). Scaling up the reach of training through technological solutions might enable wide dissemination and access to expert knowledge. Technology can provide strong lower-level scaffolds (structured, high-quality input at scale), but without deliberate design of interactive, contextual, and mentorship layers around it, the full scaffold needed for complex CHW hands-on skills and judgment remains incomplete.

With recent advancements in artificial intelligence, researchers have experimented with deploying chatbots, voice interfaces, and AI agents to support CHWs (Ramjee et al., 2025; Sivasubramanian and Raval, 2025). These tools serve as external scaffolding, assisting CHWs with daily tasks such as retrieving information for advocacy, estimating child anthropometrics

through image processing, classifying images to recognize symptoms, and providing access to relevant knowledge resources. AI can help offload cognitively demanding processes and provide timely guidance. Excessive dependence on AI might limit the development of CHWs' expertise (UNESCO, 2025). The challenge lies in designing gradual phasing-out strategies, where AI scaffolding initially supports beginners but reduces its assistance as competence grows, allowing CHWs to internalize knowledge, strengthen cognitive processes, and ultimately achieve mastery.

1.3 Gamification as an Innovation in Pedagogy

Gamification, the application of game design in non-game contexts, has been recognized as one of the most effective strategies for optimizing participation, motivation, and learning outcomes (Deterding et al., 2011b; Hamari et al., 2014). Field workers, who typically undergo tedious and monotonous training protocols, can benefit from gamification as a means of making learning more enjoyable and engaging. Gamification offers opportunities for competition, cooperation, and reward, which can lead to increased continuity and retention of knowledge (Hamari et al., 2014; Hamari and Koivisto, 2015).

Multiple studies and training documents report that CHWs, including ASHAs in India, often perceive traditional classroom-based training as boring, lecture-heavy, or non-engaging due to its rigid, instructor-focused format lacking interactivity (B and Kumar, 2017; Guleri et al., 2014). It prompted many researchers to explore strategies for making training and learning more enjoyable. While learning through games may not guarantee complete comprehension of a subject, it often serves as a significant motivator. Although the literature on creating enjoyable learning experiences is extensive, there is limited direct evidence connecting fun and enjoyment in game-based learning to improved learning outcomes. However, game-based learning is believed to enhance engagement, which in turn can foster better learning (Li et al., 2023). Various strategies have been employed to achieve this, including the use of diverse media, entertaining instructional methods, and hands-on constructivist approaches. The boundary between enjoyment and education is often blurred. Interestingly, in many cases, learning becomes a byproduct of enjoyable activities, where engagement in fun mechanics, such as challenges, rewards, and

narratives, drives knowledge acquisition without explicit instruction (Soyoof et al., 2024).

We therefore looked at gamification and gameful activities as an effective option for exploration. The three studies in this thesis outline these efforts, which involved developing a digital game-based solution for field worker refresher training. Before arriving at the solution, we underwent an iterative design process, where we experimented with various playful activities, as described in the three studies. The details are appended in the appendix. [1]

The objective is to design an Android-centric game as a refresher module that complements the in-person initial core training, alleviating its limitations. This approach leverages the widespread adoption and popularity of smartphones to deliver training content in an efficient and scalable manner.

The game design borrows from traditional card games, which are simple to learn, straightforward to play, and inherently engaging. The physical card game serves as a precursor and introduction to the digital one, offering a seamless transition for those who are upcoming adopters of smartphones and other technologies. The card game format also facilitates the easy incorporation of key elements of gamification, such as rules, procedures, and structured gameplay, making it an ideal choice for this situation.

1.4 ASHA and Anganwadi Workers: Team Building

ASHA workers and Anganwadi workers (AWWs) collaborate to organize Village Health and Nutrition Days (VHNDs), where services such as immunization, antenatal care (ANC), postnatal care (PNC), and nutrition support are provided. They also conduct home visits to ensure that pregnant women and children receive proper care and follow-up. Both workers play a key role in mobilizing the community to participate in health programs and address vaccine hesitancy. Their combined efforts significantly contribute to improving immunization coverage, reducing maternal and child mortality, and raising awareness about maternal and child health, especially in underserved communities. They are essential to the success of national health programs, such as the National Health Mission (NHM) and Poshan Abhiyan (Nutrition Mission). We therefore

[1] Choice of game mechanics, drafting rules of play, and the digital game app are annexed in appendix A.5 and A.6

trained both of these workers, as the game content was relevant to both of them in implementing the MCH programme.

We found that the dynamics of cooperation and competition come into play when AWWs join teams with ASHA workers, either collaborating or competing with each other. This interaction provides valuable insights into inter-cadre teamwork and coordination. It not only increases participation but also helps both cadres better understand each other's roles and responsibilities in community healthcare delivery.

We briefly describe below how the field experiment evolved, building upon the experience of each experiment incrementally. In hindsight, this sequence could not have been anticipated, and in some sense, the flexibility exercised made the research more relevant to the co-creation and improvement of the game's design and the nature of the experiment.

1.5 Field Trials and Methodological Considerations

The first field trial [Study 1] (n=100 sampled and 86 of them retained) was conducted in urban areas of Kolkata and rural areas of West Bengal. We conducted play sessions, tests, and interviews in Bengali, which is the local language. We worked in partnership with an NGO named Child in Need Institute (CINI), which also serves as a State Training Centre (STC) for training the ASHAs in the surrounding districts.

The second field trial [Study 2] was conducted at three places. Prior notification of the field testing sessions was given to the ASHA and AWW coordinators, as well as the Health Officers of Burhanpur Urban (Burhanpur, Madhya Pradesh), Joka village (Kolkata, West Bengal), and Mandala village (Slum near BARC, Mankhurd, Mumbai). Out of 400 CHWs selected through convenience sampling, n=368 (184 ASHAs and 184 AWWs) were retained till the end of the study. We conducted play sessions, tests, and interviews in the local languages, i.e., Hindi, Bengali, and Marathi.

The third field trial [Study 3] was conducted at different places in Madhya Pradesh, comprising urban, rural, and tribal areas. Ninety participants (45 ASHAs and 45 AWWs), each from three groups, were retained until the end of the study, totaling 270 participants across the three

groups.

Some ASHAs, particularly those with limited formal education, faced challenges reading Hindi numbers on cards and filling out survey questionnaires. To address this, numbers were changed to English while retaining Hindi words. However, ASHAs educated in Madrasas, familiar only with English and Urdu numbers, required English translations to complete forms, but could still play cards using other information. Consequently, all numbers in the cards and the app were updated to English. During questionnaire reviews, discrepancies were identified, including instances where ASHAs skipped Rotavirus-related questions. This was due to the non-operational status of Rotavirus vaccines in Madhya Pradesh, which resulted in their exclusion from training materials. Additionally, confusion arose around the term "booster" for the PCV vaccine, as it was not included in their training schedules, although most ASHAs eventually understood and answered correctly.

Large-scale field trials in multicultural linguistic and geographic settings have guided the development and piloting of the gamified training module. The field trials have provided information about the operational challenges and potentialities entailed in implementing gamification in a field training program. An instance would be how the interrelation between teamwork and collaboration was brought out through observations of ASHAs' and AWWs' interaction when playing the game in cooperation. Additionally, the design process involved iterative prototyping and feedback from field workers to ensure that the game was both culturally appropriate and easy to use.

What emerged during these sessions was the players' intuitive understanding of the card game and their ability to make quick adjustments to new rules or information. The case of Madarsa-educated field workers or those unfamiliar with a new protocol, such as Roravirus, demonstrated that they were able to quickly adjust to the game. This was not anticipated earlier, but it is operationally important in that the game can help them internalize new changes and protocols quickly through a peer-to-peer exchange of notes.

We conducted the experiments in phases of study, each study addressing different yet connected aspects of training and knowledge enhancement for Community Healthcare Workers (CHWs) in India. The first phase was on comparing physical and augmented reality (AR)-based playful activities for refresher training, which was conducted in rural and semi-urban areas of West Bengal. This phase laid the foundation for the effectiveness of game-based training com-

pared to traditional methods. The second phase expanded the scope by integrating digital and physical card games, with a larger cohort of CHWs from various states across India, to test for the retention of knowledge and interest in the long term. Each phase built on the findings of the previous one to create a cohesive narrative that addressed the issues of training, retention, and implementation. The three paper format thesis structure allows for the exploration, analysis, and presentation of each phase separately. This allows for the integration of various methodologies, from qualitative co-design work to quantitative quasi-experimental studies, and provides an overall picture of the effectiveness of game-based training tools. Through the three separate studies, the dissertation explores the multifaceted nature of the problem, contributing to the body of knowledge on gamified training methodologies for CHWs in low-resource settings.

The three studies present a detailed discourse on the implementation of gamified learning, assess its effectiveness, and address the training demands of CHWs. These studies respond to the gaps in the training and performance of CHWs, who are the primary contact points between the health system and rural communities. The studies emphasize the importance of interactive and user-oriented learning methods, stressing the facilitation of engagement and motivation to achieve better learning outcomes. They also emphasize the importance of reinforcement learning and feedback mechanisms in promoting the long-term retention of health information among CHWs. The studies differ in their focus on specific game modalities. While the first study focuses on the comparative effectiveness of physical and computer games, the second and third studies focus on the long-term retention gains of smartphone-based games. This variety of approaches demonstrates the adaptability of gamified learning and its applicability to diverse training needs and settings. The findings of these articles provide valuable insights into the potential of game-based training to enhance knowledge acquisition, retention, and interest among CHWs, as well as the utility of these instruments in practical applications of data collection and healthcare delivery.

The first study, "Physical and Augmented Reality-based Playful Activities for Refresher Training of ASHA Workers in India", shows the drawbacks of the traditional classroom training model and suggests the concept of playful, interactive training tools. Through a comparison of physical playing cards with an AR-based smartphone game, the study demonstrates that AR-based training leads to significant improvements in knowledge retention and interest among CHWs. The immersive and interactive nature of AR games makes them particularly suitable for

visual learners, while physical card games are more suitable for auditory learners who acquire knowledge through verbal communication while playing the games. This study highlights the need to tailor training strategies to the diverse learning styles of CHWs, ensuring that training programs are not only engaging but also effective.

However, field observations identified key barriers to scaling AR training for CHWs in resource-limited environments. First, AR required high-end smartphones and reliable data connectivity, which were often unavailable in rural and tribal areas, leading to frequent disruptions, slow performance, and sometimes complete failure to load content. Second, CHWs encountered usability problems with AR on small screens, which distracted them from the core learning material. Third, sustaining AR long-term proved difficult due to high costs for device repairs, software updates, and technical support, which we struggled to provide at scale.

In comparison, physical playing cards offered greater practicality for these settings. They needed no power or internet, were simple to print and share, and encouraged natural group discussion and collaborative play, making them well-suited to CHWs' capacity levels and local constraints.

Following these findings, we further focused on developing the card game for both digital and physical play and tested it with CHWs. The development of the physical card game was not an easy process in itself. Segmentation of the topics of maternal and child care into four 'suites' was the first. The unequal number of cards in each suit was a strong psychological block that took time to overcome. The issue of numbering the cards, particularly when we had multiple actions, e.g., 3 doses of DPT, was also debated with the workers, and we finally decided to give sub-numbers to each sub-activity to emphasize the bundled nature of the activity. These are described in some detail in the appendix [2]. The digital version subsequently presented technical challenges, albeit not a significant one, with different game modes posing their own algorithmic challenges. While we have focused on a traditional team game, 'Satti', the game can also be played in solitaire and rummy style.

The second study, "Refresher Training through Digital and Physical, Card-Based Game for Accredited Social Health Activists (ASHAs) and Anganwadi Workers (AWWs) in India", further explores the relative effectiveness of physical and digital card games. The study finds that while digital games yield higher short-term knowledge gains, long-term knowledge retention

[2]The development of the game content is detailed in A.5 and A.6

is equally high, with no significant difference between digital and physical play modes. This suggests that both modalities are viable options for refresher training, with digital games having the added advantage of scalability and accessibility. The study also highlights the necessity for ongoing reinforcement in an attempt to preserve knowledge retention, as the control group, without training, performed significantly worse than the game-based groups. These results highlight the importance of ongoing training and support for CHWs in delivering high-quality healthcare services.

Following these findings from the second study, we attempted to deploy and test both modalities, physical and digital card games, in different locations to evaluate the product and determine if the results are generalizable. The third study, "Replay, Revise, and Refresh: Smartphone-Based Refresher Training for Community Healthcare Workers in India", examines the long-term effect of smartphone-based refresher training. The study compares digital and physical card games with conventional classroom teaching and finds that both game-based methods outperform conventional training in knowledge retention. While the digital game shows higher short-term knowledge gain, long-term knowledge retention is similar for both digital and physical gameplay modes. The qualitative answers provided by CHWs indicate a strong preference for digital games due to their convenience and interactivity, yet highlight the need for physical card decks for purposes of learning and reference when undertaking home visits, where individual mothers can't be expected to be familiar or experienced with the digital mode. This study emphasizes the importance of incorporating game-based training into existing programmes, while addressing the practical needs and interests of CHWs.

In light of the findings from these articles, several key themes emerge. Firstly, game-based training methods, with digital and AR capabilities, are better than classroom training in facilitating learning and knowledge retention among CHWs. Their interactive and immersive nature makes them highly engaging, responsive to different learning styles, and makes training fun and effective. Second, while digital games offer access and scalability, physical card games are also an appropriate choice in resource-constrained contexts, or/where mobile phone access is restricted. The choice between them on which mode to deploy must be tailored to the specific requirements and contexts of the targeted population. These studies focus on knowledge gain and knowledge retention over time through game-based training. While these tools may have the ability to maximize short-term knowledge acquisition, long-term effectiveness is contingent

upon regular use and reinforcement.

Overall, the evidence from the following articles provides strong support for game-based training methods' ability to improve the knowledge, and engagement of CHWs in India. These game-based trainings are an effective alternative to traditional training methods, counteracting the challenge of limited literacy, resource constraints, and poor retention. They are, however, dependent upon careful planning, ongoing assistance, and alignment within existing healthcare structures.

Study-1 Physical and Augmented Reality based Playful Activities for Refresher Training of ASHA Workers in India			
Aim	**Methods**	**Results**	**Conclusions**
To compare the effectiveness of physical and augmented reality (AR) card games for improving knowledge gain and retention on child immunization among ASHA workers.	Focus group discussions (FGDs) and participatory design exercises with CHWs. Field trials were conducted to evaluate the effectiveness of both physical and AR-based games.	AR-based games demonstrated higher engagement, knowledge gain, and retention compared to physical card games, mainly due to their interactive and intuitive nature.	AR-based games were found to be more effective in learning enhancement and knowledge retention, suggesting potential as a complementary training tool for ASHA workers.
Study-2 Refresher Training through Digital and Physical, Card-Based Game for Accredited Social Health Activists (ASHAs) and Anganwadi Workers (AWWs) in India			
Aim	**Methods**	**Results**	**Conclusions**
To investigate the effectiveness of digital and physical card-based games in enhancing knowledge of child immunization among community healthcare workers (CHWs) in India.	A quasi-experimental study with 368 participants (n=92 for each group: digital, physical, traditional classroom, and control). Quantitative gameplay data and qualitative feedback were collected through interviews.	Both digital and physical card games showed significant knowledge improvements compared to traditional classroom and control groups. Digital gameplay demonstrated slightly higher knowledge retention.	Game-based refresher training can be a scalable, cost-effective, and engaging tool for improving CHWs' knowledge and practices in resource-constrained settings.
Study-3 Replay, Revise, and Refresh: Smartphone-Based Refresher Training for Community Healthcare Workers in India			
Aim	**Methods**	**Results**	**Conclusions**
To assess the differences in knowledge gain and retention among digital, physical, and traditional classroom training methods for CHWs.	270 CHWs (n=90 per group) were divided into three groups: digital game, physical card game, and traditional classroom training. Pre- and post-intervention tests and retention tests after six months were conducted.	All groups demonstrated significant knowledge improvement, with digital gameplay yielding higher post-intervention scores and superior long-term knowledge retention.	Smartphone-based training outperformed both physical and classroom methods, indicating its potential for improving long-term knowledge retention through frequent, interactive learning sessions.

Table 1.1: Summary of Aims, Methods, Results, and Conclusions from Three Studies

Chapter 2

Review of Literature

2.1 Maternal and Child Health and the Role of Community Health Workers

Maternal, newborn, and child health remains an important area of public health in India, particularly in communities where access to healthcare services, health information, and timely care are limited. Community Health Workers (CHWs) play an important role in connecting communities with the formal healthcare system. Their responsibilities often include communicating health information, encouraging the use of maternal and child health services, supporting immunization, identifying potential health risks, and guiding families towards appropriate healthcare facilities. Consequently, the knowledge and preparedness of CHWs can influence how effectively health information and services reach mothers and children in need.

This thesis focuses particularly on three sequential interconnected areas of maternal and child healthcare: Antenatal Care (ANC), Postnatal Care (PNC), and childhood immunization. These areas require not only the availability of healthcare services but also awareness among families and the ability of frontline workers to communicate appropriate information and en-

courage timely utilization of services.

2.1.1 Antenatal and Postnatal Care

Antenatal Care (ANC) and Postnatal Care (PNC) constitute important components of maternal and newborn healthcare. Early registration of pregnancy, preferably during the first trimester, enables timely identification of health risks and provides opportunities for mothers to access appropriate healthcare services. ANC involves a range of activities, including monitoring blood pressure and weight, hemoglobin testing, tetanus vaccination, nutritional counseling, and birth preparedness. PNC extends this continuum of care beyond childbirth and addresses the health of both the mother and newborn. Important components include early initiation of breastfeeding, identification of post-delivery complications, family planning counseling, and linkage to immunization and other child health services.

For CHWs, knowledge of these services is important because they frequently act as an intermediary between healthcare systems and families. Their role is not limited to providing information. They also need to recognize situations that require referral, explain the importance of recommended services, and reinforce the information provided during clinical consultations.

2.1.2 Childhood Immunization

Immunization is another important component of child healthcare. Vaccination prevents a substantial burden of childhood morbidity and mortality, with vaccines estimated to avert approximately 2-3 million deaths each year (Andre et al., 2008). Despite the availability of effective vaccines, substantial inequalities in immunization coverage remain. Globally, approximately 14.5 million children under 12 months of age were reported to have missed essential vaccinations, including approximately 1.6 million children in India (WHO, 2024).

Evidence from India demonstrates considerable geographical and socioeconomic variation in childhood immunization. Using NFHS-4 data, Panda et al. (Panda et al., 2020) found that immunization coverage was below 50% in 163 districts. Immunization inequalities are also

associated with household socioeconomic status. According to NFHS-4 estimates, 47% of children from households in the lowest wealth quantile were not fully immunized, while 71% of families lacked health insurance coverage (International Institute for Population Sciences (IIPS) and Ministry of Health and Family Welfare, 2016). Such disparities can contribute to substantial household healthcare expenditure, with previous research estimating that healthcare payments push millions of people below the poverty line each year (van Doorslaer et al., 2006; Garg and Karan, 2009).

These findings suggest that immunization gaps cannot be understood solely as a problem of vaccine availability. Although vaccination services are provided free of charge through government hospitals and Primary Healthcare Centres (PHCs), and initiatives such as Mission Indradhanush have attempted to improve coverage (MoHFW, 2016), complete immunization remains difficult to achieve in many communities (International Institute for Population Sciences (IIPS) and Ministry of Health and Family Welfare, 2021). Knowledge, awareness, socioeconomic conditions, healthcare access, and families' ability to navigate health services may all influence vaccination behavior.

Panda et al. (Panda et al., 2020), using Local Indicators of Spatial Association (LISA) and spatial regression models, reported that mothers receiving antenatal and postnatal care were more likely to have fully immunized children. The study also identified associations between child immunization and indicators of women's empowerment, including maternal literacy and women's authority within the household. These findings are particularly relevant to CHW-mediated healthcare because CHWs can facilitate communication and reinforce the information mothers need to make informed decisions about maternal and child health services.

The same body of evidence also points to the importance of awareness. Limited knowledge among family members regarding vaccination can contribute to inadequate healthcare utilization and may be associated with poor child health outcomes. Therefore, improving access to health information and strengthening frontline workers' capacity to communicate it are important components in addressing immunization inequalities. mHealth interventions offer a potential mechanism to support these efforts (Peek, 2017).

2.2 Knowledge and Training Gaps among Community Health Workers

Although CHWs serve as an important link between communities and healthcare systems, their effectiveness depends on the knowledge and skills they possess and their opportunities to refresh and reinforce those skills. Maternal and child healthcare involves a wide range of information, procedures, warning signs, and service-related knowledge. Retaining this information and applying it appropriately in field situations can be challenging, particularly when workers operate in resource-constrained environments.

An exploratory focus group discussion conducted by the authors with mothers (n=9) in a rural setting in the South 24-Parganas district of West Bengal revealed gaps in awareness regarding healthcare delivery, medical care, check-up services, and family planning. These observations are consistent with findings from other parts of India. Mani et al. (Mani et al., 2020) found limited awareness among mothers regarding tetanus vaccination, immunization, and iron and folic acid (IFA) supplementation. More than half of the pregnant women in their study were unaware of several important components of ANC, including blood pressure and weight measurement, and abdominal examination. Knowledge of pregnancy-related high-risk conditions and danger signs was also limited, while awareness of institutional delivery procedures and maternity benefit schemes was also low.

These findings illustrate an important distinction between the availability of healthcare services and the ability of communities to make effective use of them. Information may be formally available through health facilities, government programs, and printed materials, but this does not necessarily mean that it is understood, remembered, or acted upon. CHWs therefore require not only access to information but also opportunities to develop practical knowledge and confidence in communicating that information.

Traditional training approaches can provide structured instruction, but they may provide limited opportunities for repeated practice and contextual decision-making. A classroom session may explain what a CHW should do when encountering a particular situation, but knowing the correct procedure in principle is not necessarily equivalent to being able to recognize and apply it during fieldwork. This distinction between acquiring information and developing us-

able knowledge provides an important motivation for exploring more active forms of refresher training.

2.3 Conventional and Technology-Supported Training for CHWs

Training frontline health workers presents several practical challenges. CHWs often work within low-resource community settings and may have limited opportunities to attend extended classroom-based training sessions. Training content may also cover a broad range of topics, making repeated reinforcement difficult. Consequently, there is a need for training approaches that can provide accessible, contextualized, and repeatable learning opportunities without requiring extensive disruption to routine work.

The increasing availability of mobile technologies provides one potential opportunity. According to a Telecom Regulatory Authority of India (TRAI) press release, India had approximately 1,205.64 million telephone subscribers as of June 30, 2024, of whom 538.51 million were rural subscribers. Rural wireless subscribers accounted for 535.53 million of these connections (Telecom Regulatory Authority of India, 2021). Although telephone ownership and internet access should not be treated as equivalent, these figures demonstrate the extensive reach of mobile communication technologies.

Internet access has expanded as well, although substantial disparities remain. Data from the 75th Round of the National Sample Survey, conducted between July 2017 and June 2018, indicated that 35.8% of Indian households had internet access, with access substantially higher in urban households (51.3%) than rural households (24.8%) (MoSPI, 2015). These differences are important when considering smartphone-based training in rural and resource-constrained contexts. Interventions cannot simply assume continuous high-speed internet access or high digital literacy.

At the same time, increasing access to mobile devices creates opportunities for supporting healthcare workers through mobile learning. Previous research has identified mHealth as a potentially useful mechanism for improving healthcare information delivery and engagement, particularly in settings with limited conventional infrastructure (Ganapathy and Ravindra, 2008; Bassi et al., 2018; Madanian et al., 2019). The challenge is therefore not simply whether mobile

technology can deliver information, but how it can be used to create learning experiences that are accessible, engaging, and educationally effective.

2.4 mHealth and Mobile Learning

The World Health Organization defines mobile health (mHealth) as the practice of medicine and public health supported by mobile devices (WHO, 2011). The widespread availability of mobile devices offers an opportunity to supplement traditional approaches to maternal and child healthcare education, such as printed brochures, job cards, and face-to-face instruction.

One advantage of mHealth is the potential to distribute information in a relatively accessible and scalable manner. Mobile interventions can also be designed around the linguistic, socioeconomic, and cultural characteristics of particular populations. However, simply transferring existing educational material to a mobile platform does not necessarily improve learning. The design of the learning experience remains important, particularly when learners have different levels of literacy, prior knowledge, and digital familiarity.

Several mHealth interventions have attempted to address these issues. Kumar et al. (Kumar and Anderson, 2015) identified approaches, including radio, text messages, and other forms of mobile communication, for engaging communities with healthcare-related information. Ramachandran et al. (Ramachandran et al., 2010) explored the use of short videos delivered through mobile phones to improve health workers' motivation and knowledge. Kumar et al. (Kumar and Anderson, 2015) similarly reported that short films concerning maternal and child care could empower health workers and mothers to manage pregnancy-related issues more proactively.

Mobile-based interventions have also incorporated motivational mechanisms. Shah et al. (Shah et al., 2017), for example, examined an incentive-based mobile video training approach for Anganwadi Workers (AWWs). Workers received videos on maternal and child health via feature phones and were subsequently asked questions about the video content via a toll-free telephone service. Correct answers were rewarded with mobile talk time. Such interventions illustrate how mobile technologies can combine content delivery, assessment, and motivation.

Other approaches have focused on accessibility for users with limited literacy. Perez et al. (Pérez et al., 2020) evaluated "Tika Vaani," an Interactive Voice Response (IVR)-based mHealth intervention designed to improve immunization knowledge in India. The study demonstrated the potential of customized audio-based interventions to improve healthcare knowledge among less-literate populations.

Collectively, these studies demonstrate that mobile technologies can support healthcare education through multiple modalities, including text, video, audio, assessment, and incentives. However, much of this work focuses primarily on delivering information. This raises a further question: can mobile technologies be used not merely as information-delivery channels, but as environments in which learners actively practice, make decisions, receive feedback, and learn from peers?

Game-based learning provides one possible response to this question.

2.5 e-Learning and Game-Based Learning in Healthcare Education

e-Learning refers broadly to teaching and learning approaches that use electronic technologies to support access to training, communication, interaction, and knowledge development (Sangrà et al., 2012). Blended learning combines electronic learning with face-to-face instruction, allowing digital resources to supplement rather than replace conventional teaching.

Within healthcare education, games represent one approach to increasing learner participation. Unlike conventional instructional materials, educational games require learners to interact with a structured environment, make choices, respond to challenges, and experience consequences based on their actions. This can create opportunities for learning through practice rather than a one-way reception of information.

A Cochrane review by Akl et al. (Akl et al., 2010) examined physical games used to educate healthcare professionals and concluded that they can be an effective educational strategy. More broadly, games designed with purposes beyond entertainment have been applied in areas including healthcare, education, training, defense, aviation, city planning, politics, and ecology.

Different terms have been used to describe these approaches, including Game-Based Learning, Games for Good, Alternative Purpose Games (Sawyer and Smith, 2008), and Edugaming (Angarita et al., 2005).

The growing use of games in healthcare education is particularly relevant to CHW training because they can reproduce simplified versions of situations that workers encounter in practice. Rather than presenting a rule in isolation, a game can place the learner in a scenario where they must decide what to do, observe the consequences, receive feedback, and try again.

2.5.1 Serious Games

The concept of the Serious Game was articulated by Clark Abt in his 1970 book *Serious Games*. Abt described serious games as games designed with an explicit educational purpose rather than primarily for amusement (Abt, 1970). The concept has subsequently been applied to a wide range of learning and training contexts.

Serious games combine game structures with educational objectives. Their potential value lies partly in their ability to provide learners with opportunities to interact with problems rather than simply read about them. Research has suggested that serious games can support cognitive, social, and health-related skills beyond the immediate gaming environment (Ritterfeld et al., 2009; Giunti et al., 2015). The increasing number of reviews examining serious games in healthcare education further demonstrates the growing interest in this approach (Graafland et al., 2014; de Ribaupierre et al., 2014; Laamarti et al., 2014; De Wit-Zuurendonk and Oei, 2011).

Importantly, serious games are not necessarily limited to digital environments. Physical board games, card games, role-playing activities, and other forms of structured gameplay can also be designed around serious educational objectives. This distinction is relevant to the present research, which examines both digital and physical game-based learning approaches.

2.5.2 Game-Based Learning and Gamification

Game-based learning is used in this thesis as a broad term for learning that occurs through gameplay. A serious game can therefore be considered one form of game-based learning when the game itself constitutes the primary learning intervention.

Gamification is conceptually different. Deterding et al. defined gamification as the "use of game design elements in non-game contexts" (Deterding et al., 2011b,a; Deterding, 2011, 2012). Rather than transforming an activity into a complete game, gamification incorporates selected elements such as points, badges, progression systems, challenges, or leaderboards into an existing activity.

These distinctions are important because game-based learning and gamification involve different design assumptions. A serious game creates a structured environment in which learning occurs through gameplay, whereas gamification modifies an existing non-game activity using selected game elements. In the context of CHW training, both approaches may contribute to engagement and motivation, but the present thesis primarily investigates game-based learning as a training modality.

Game design can incorporate both intrinsic and extrinsic motivational mechanisms. Self-Determination Theory emphasizes the importance of autonomy, competence, and relatedness in supporting intrinsic motivation (Ryan and Deci, 2000). Game elements such as points, badges, progress indicators, and leaderboards can provide forms of extrinsic feedback, while opportunities to develop mastery, exercise choice, and interact with others can support intrinsic motivation. Deterding et al. (Deterding et al., 2011b) similarly emphasized the motivational potential of gameful experiences.

However, motivation alone does not establish educational effectiveness. A game can be engaging without producing meaningful learning. Consequently, the design of a serious game must align game mechanics with explicit learning objectives and appropriate learning theories.

2.6 Learning Theories Relevant to Game-Based CHW Training

The effectiveness of game-based learning depends not only on the presence of game elements but also on how the learning experience is structured. Several educational theories provide complementary explanations for why gameplay may support learning. In this thesis, Adult Learning Theory, Experiential Learning Theory, Cognitive Theory of Multimedia Learning, constructivist perspectives, and Social Learning Theory are used to inform the design and interpretation of game-based CHW training. Dual-process theory provides an additional cognitive perspective on how repeated practice may contribute to increasingly efficient decision-making.

2.6.1 Adult Learning Theory

Knowles' theory of adult learning emphasizes several characteristics that distinguish adult learners from children (Knowles, 1980). Adults tend to become increasingly self-directed, draw on previous experiences, focus on problems relevant to their lives, and become more oriented toward practical outcomes.

These characteristics have direct implications for CHW training. CHWs bring prior experiences from their community work, and training may be more meaningful when it builds upon these experiences rather than presenting information in isolation. Similarly, training activities that reproduce practical problems can make learning more relevant to their responsibilities.

Games can support these principles by allowing learners to make decisions independently, draw upon prior knowledge, and solve contextually meaningful problems. Progression through small tasks can also provide a clear sense of goal orientation. Rather than receiving information passively, learners can encounter a problem and use their existing knowledge to determine an appropriate response.

2.6.2 Experiential Learning

Kolb's Experiential Learning Theory proposes a cyclical process involving concrete experience, reflective observation, abstract conceptualization, and active experimentation (Kolb, 1984). Learning therefore involves more than acquiring information. Learners develop understanding by experiencing a situation, reflecting on it, developing conceptual understanding, and applying that understanding in subsequent situations.

Gameplay can provide a structured environment for this cycle. A learner encounters a scenario, makes a decision, observes its outcome, receives feedback, reflects on the result, and applies the resulting understanding to another situation. Repeated gameplay can therefore create opportunities for practice and experimentation without exposing real patients or families to the consequences of incorrect decisions.

Ritterfeld et al. (Ritterfeld and Weber, 2006) proposed three ways of combining learning and entertainment: reinforcement, motivation, and blended models. In the reinforcement model, learning is rewarded through entertainment. In the motivation model, entertainment encourages interest and attention toward learning. In the blended model, learning and entertainment are integrated, making the learning process itself engaging. The blended approach is particularly relevant to serious games because it attempts to make learning intrinsically connected to the gameplay experience rather than treating learning as an activity that occurs before or after play (Breuer and Bente, 2010; Koster, 2010).

2.6.3 Cognitive Theory of Multimedia Learning

Mayer's Cognitive Theory of Multimedia Learning proposes that people learn by processing information through visual and auditory channels, while recognizing that each channel has limited processing capacity (Mayer, 1997). This perspective is relevant to digital learning environments because multimedia can convey information through complementary media such as text, images, animations, and audio.

However, multimedia does not automatically improve learning. When unnecessary infor-

mation or poorly designed interfaces overload learners, additional media may increase rather than reduce cognitive demands. Consequently, multimedia elements in a serious game should be selected according to their instructional function rather than simply to make the game more visually attractive.

2.6.4 Constructivism, Scaffolding, and Guided Learning

Constructivist perspectives view learners as active participants in the construction of knowledge. Rather than simply receiving information, learners interpret new experiences in relation to their existing understanding. Games can support this process by presenting problems that require learners to make decisions and construct solutions.

However, unrestricted discovery does not necessarily lead to effective learning. Learners with limited prior knowledge may lack the conceptual structures necessary to interpret a complex environment. Scaffolding is therefore important. Early stages of training may require more explicit guidance, demonstrations, prompts, or feedback, while later stages can gradually reduce this support as learners become more capable.

This principle can be implemented in game design through progressive difficulty. Initial scenarios can provide more information and guidance, while later scenarios can require learners to make decisions with less assistance. Such progression provides a balance between supporting the learner and allowing independent problem-solving.

2.6.5 Social Learning

Learning is not exclusively an individual cognitive process. Bandura's Social Learning Theory emphasizes that people learn through observation, imitation, modeling, and interaction with others (Bandura, 1977). This social dimension is particularly relevant to CHWs because community health work itself involves communication, interaction, and collaboration.

Game-based training can incorporate social learning through peer discussion, collaborative problem-solving, competition, observation of other players, and explanation of decisions.

Physical games may provide particularly direct opportunities for face-to-face interaction, while digital games can support structured collaboration or competition depending on their design.

Social interaction may therefore contribute to learning beyond its motivational value. When learners explain why they selected an answer, observe alternative strategies, or discuss a difficult scenario with peers, they are required to articulate and reconsider their understanding.

2.7 Theoretical Framework

Although several learning theories inform this thesis, dual-process theory provides an overarching cognitive perspective for understanding the role of repeated practice in CHW training. Kahneman's framework distinguishes between two broad modes of thinking (Kahneman, 2011). System 1 (S1) is characterized as fast, intuitive, and relatively automatic, whereas System 2 (S2) is slower, more deliberate, and effortful.

S1 can rapidly recognize familiar patterns and produce responses based on previously developed knowledge. S2, in contrast, is more strongly involved when a problem is unfamiliar, complex, or requires conscious analysis. These processes should not be understood as two completely independent systems; rather, they provide a useful framework for describing different characteristics of human cognition.

The distinction is relevant to CHW training because many field situations require workers to respond efficiently while recognizing when a situation requires additional deliberation or referral. A novice CHW may need to consciously recall a procedure step by step, whereas a more experienced worker may recognize familiar patterns and respond with less deliberate effort.

Repeated practice can contribute to the development of more efficient and increasingly automatic performance. Through repeated exposure to relevant situations, learners may strengthen their ability to recognize patterns and select appropriate responses. From this perspective, training should not focus exclusively on transmitting declarative information. It should also provide opportunities to apply knowledge repeatedly in meaningful situations.

This does not imply that expert CHWs should rely exclusively on intuitive responses.

Novel or high-risk situations may require deliberate reasoning and careful verification. Rather, the theoretical implication is that effective training should help learners develop both a foundation of knowledge and the ability to apply that knowledge efficiently in familiar situations.

For CHW training, this provides a rationale for experiential and practice-based learning. A game can repeatedly expose learners to representative scenarios in which they must recognize a situation, decide what to do, receive feedback, and attempt similar problems again. Over time, such practice may support the development of more efficient decision-making while retaining opportunities for deliberate reflection.

2.8 Design Implications for CHW Training Games

The literature and theoretical perspectives discussed above suggest several principles for designing game-based training interventions for CHWs.

2.8.1 Managing Cognitive Load

Working memory has limited capacity, making unnecessary complexity an important concern in instructional design. Game interfaces should therefore minimize extraneous cognitive load through clear layouts, intuitive interactions, concise instructions, and appropriate visual organization.

At the same time, training should not remove all difficulty. Intrinsic task complexity should be appropriate to learners' prior knowledge, while activities should encourage learners to connect new information to existing knowledge. Game difficulty can therefore be increased progressively rather than presenting all concepts simultaneously.

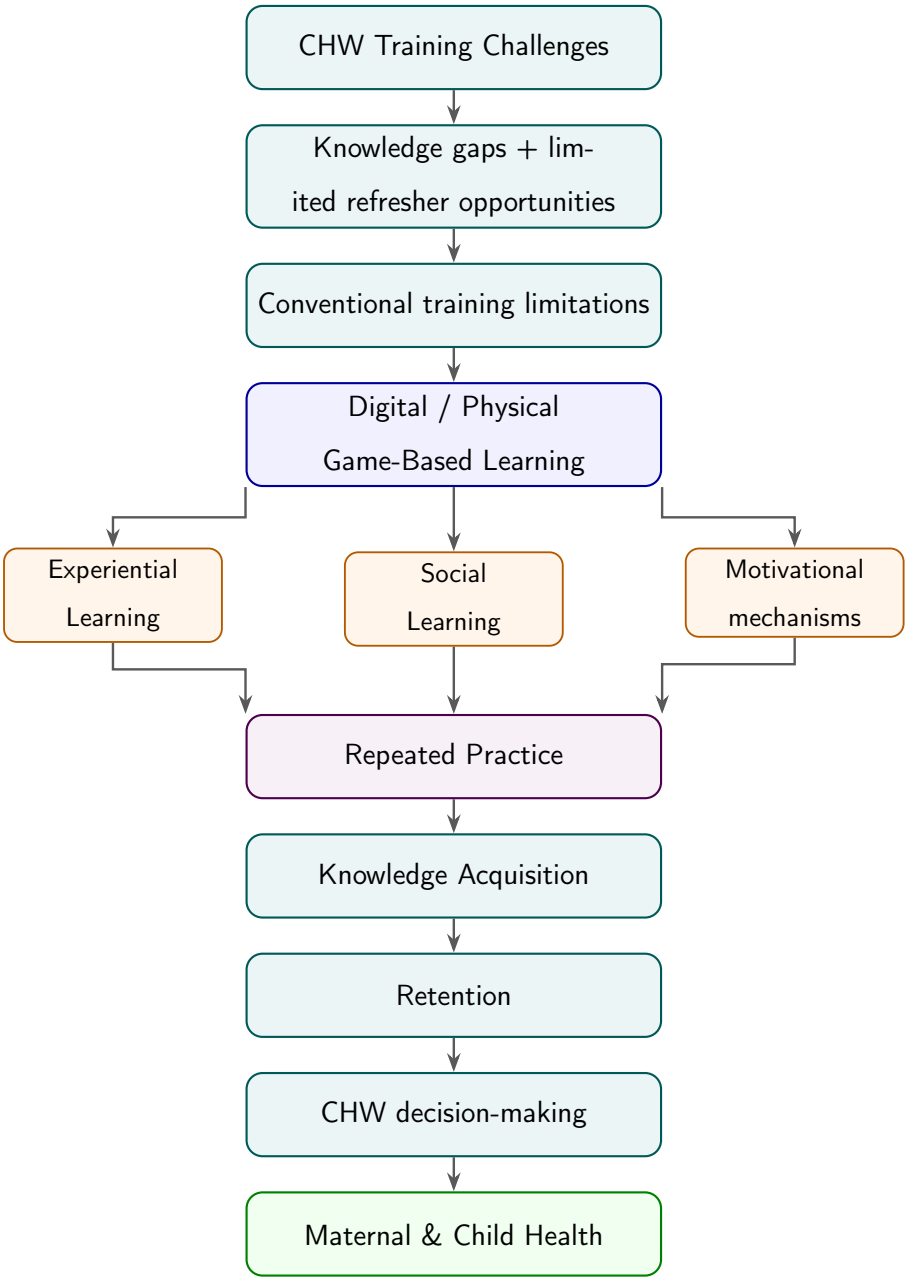


Figure 2.1: Conceptual Framework figure

2.8.2 Providing Repeated Practice and Feedback

Learning requires opportunities to attempt tasks, make mistakes, receive feedback, and try again. A serious game can provide this cycle without the practical risks associated with making mistakes in real healthcare situations.

Immediate or timely feedback can help learners identify incorrect reasoning and understand why a particular response is appropriate. Repeated practice can then reinforce this un-

derstanding. Importantly, repetition should involve meaningful variation rather than simply requiring learners to memorize identical questions. Presenting similar principles across different scenarios may encourage learners to recognize the underlying concept rather than memorize a particular answer.

2.8.3 Using Scaffolding and Progressive Difficulty

Game-based training should provide sufficient support for learners to understand the task without making the process so constrained that meaningful decision-making disappears. Early levels can provide instructions, examples, hints, or more explicit feedback. As learners become more familiar with the content, these supports can be gradually reduced.

This approach is analogous to learning to navigate a new city. Initially, a learner may rely heavily on directions or a navigation system. With experience, fewer instructions are required because the learner develops an internal understanding of the environment. Similarly, training should provide guidance initially while creating opportunities for learners to become progressively more independent.

2.8.4 Connecting Learning to practical CHW Situations

Adult learning and experiential learning theories suggest that training becomes more meaningful when it is connected to practical problems. Consequently, game scenarios should reflect situations that CHWs may realistically encounter.

Rather than presenting isolated factual questions, the game can place learners in situations involving maternal health visits, immunization, danger signs, family counseling, or referral decisions. Such scenarios can require learners to interpret information, select appropriate actions, and consider the consequences of their decisions.

2.8.5 Supporting Social and Peer Learning

The social nature of CHW work creates an additional opportunity for collaborative learning. Game-based training can encourage players to discuss answers, explain their reasoning, compare strategies, and learn from peers' mistakes.

Competitive elements such as scores or leaderboards may increase engagement, but they should be used carefully. Competition should not undermine collaboration or discourage learners who perform less well. Cooperative challenges, group discussion, and shared problem-solving can provide alternative mechanisms for social engagement.

2.8.6 Balancing Digital and Physical Interaction

The availability of mobile devices makes digital game-based learning attractive because digital interventions can potentially be distributed at scale and updated more easily. However, digital delivery also depends on device availability, digital literacy, connectivity, and user acceptance.

Physical games can avoid some of these technological barriers and may provide direct opportunities for face-to-face interaction. At the same time, they may require physical distribution and facilitator support. These differences suggest that digital and physical games should not necessarily be viewed as competing technologies. They represent different delivery formats with distinct affordances and constraints.

This distinction is particularly relevant to the present thesis, which examines both digital and physical game-based interventions. Comparing these formats can provide insight into whether the learning benefits associated with gameplay depend primarily on the underlying game-based pedagogy or are influenced substantially by the delivery medium.

Table 2.1: Theoretical Foundations and Design Implications

Literature	**Finding**	**Design Implication**
Cognitive Load Theory	Working memory is limited	Reduce unnecessary information
Experiential Learning	Learning develops through experience/reflection	Provide repeated scenarios
Adult Learning	Adults value relevant problems	Use authentic CHW situations
Social Learning	Learning occurs through observation	Enable peer interaction
Dual-process theory	Practice can support automaticity	Use repeated decision-making
Constructivism	Learners construct knowledge	Provide problem-solving situations
Gamification	Game elements can support motivation	Use feedback/progression appropriately

2.9 Learning as a Social and Contextual Activity

The increasing availability of educational technology sometimes leads to the assumption that the primary barrier to learning is limited access to information. Access is important, particularly in resource-constrained environments, but information availability alone does not guarantee learning.

Learning takes place within social and contextual environments. CHWs bring prior experiences, interact with peers, and work within specific communities and health systems. Consequently, training should account for these social dimensions rather than treating learners as isolated recipients of information.

Bandura's Social Learning Theory (Bandura, 1977) provides a useful basis for understanding this process. Through observation, interaction, modelling, and feedback, learners can develop knowledge that may not emerge from individual information consumption alone. Game-based learning can incorporate these mechanisms by allowing learners to observe how others

approach problems, discuss decisions, and receive feedback from peers.

This social dimension may be particularly valuable in physical game-based interventions, where players can interact directly around a shared game environment. Digital games can also support social learning when designed to facilitate collaboration, competition, discussion, or shared decision-making.

Therefore, the value of game-based CHW training should not be understood solely in terms of individual engagement with a game. The broader learning environment, including peer interaction and opportunities for reflection, may influence how effectively gameplay translates into knowledge and practical understanding.

2.10 Research Gap

The literature reviewed in this chapter establishes several important points. First, maternal and child healthcare outcomes depend partly on effective communication, service utilization, and timely decision-making at the community level. CHWs play an important role in this process, yet evidence indicates persistent knowledge and awareness gaps among frontline workers and the communities they serve.

Second, conventional training approaches may not provide sufficient opportunities for repeated, contextualized, and practice-based learning. Mobile technologies offer opportunities to extend training beyond conventional classroom environments, and prior mHealth research has demonstrated the potential of mobile videos, text, IVR, and other approaches to deliver healthcare information.

Third, game-based learning offers a distinct pedagogical approach. Unlike information-delivery interventions, games can create environments in which learners actively make decisions, experience consequences, receive feedback, practice repeatedly, and interact with peers. Serious and educational games have shown promise in healthcare education, while learning theories offer several mechanisms by which gameplay may support learning.

However, several gaps remain.

First, much of the existing mHealth literature focuses on the delivery of health information

rather than on interactive practice and decision-making. There is comparatively less evidence on the use of game-based approaches specifically for CHW refresher training in low-resource settings.

Second, evidence concerning the relative effectiveness of different game-based formats remains limited. Digital games offer scalable, interactive learning, whereas physical games can facilitate face-to-face interaction and may be more feasible in contexts with limited technological infrastructure. It is therefore important to examine whether these different formats produce comparable or different learning outcomes.

Third, educational effectiveness should not be evaluated solely through immediate post-intervention knowledge. Knowledge retention is particularly important for refresher training because the ultimate purpose is to support knowledge that remains available for later application. Evidence comparing immediate knowledge acquisition and subsequent retention across different game-based formats is therefore needed.

Fourth, the social mechanisms of game-based CHW training require more focus. Competition, collaboration, peer discussion, and shared problem-solving may influence how learners engage with training, yet these mechanisms are often treated as secondary features rather than as components of the learning process.

Finally, there is a need to connect game design more explicitly with theories of adult learning, experiential learning, social learning, cognitive processing, and practice-based skill development. A theoretically grounded approach can help move game-based healthcare training beyond the assumption that making training "fun" will automatically improve learning.

These gaps motivate the studies presented in this thesis. The research investigates game-based refresher training for CHWs across digital and physical formats and compares these approaches with more conventional classroom-based training. Across the studies, the research examines knowledge acquisition, retention, and engagement, as well as the role of game-based interaction in supporting learning.

The proposition emerging from the literature is therefore not simply that games are more enjoyable than conventional training. Rather, game-based interventions may provide a different learning environment in which CHWs can repeatedly encounter relevant problems, apply knowledge, receive feedback, interact with peers, and gradually develop more efficient re-

sponses. The studies in this thesis empirically examine whether these theoretical advantages translate into measurable improvements in CHW learning outcomes.

2.11 Summary

The literature reviewed in this chapter provides the foundation for investigating game-based refresher training among CHWs. Persistent gaps in maternal and child health knowledge demonstrate the continuing need for effective frontline health worker training. At the same time, the expansion of mobile technologies creates opportunities to deliver learning beyond conventional classroom environments.

Previous mHealth interventions demonstrate that mobile devices can support the delivery of healthcare information through video, text, audio, IVR, and incentive mechanisms. However, these approaches do not necessarily provide the active practice and contextual decision-making required for knowledge retention. Game-based learning offers a complementary approach by embedding educational content within interactive, playful activities.

Theoretical perspectives further explain why such an approach may be effective. Adult learning emphasizes relevance and practical problem-solving. Experiential learning emphasizes learning through experience, reflection, and experimentation. Cognitive theories highlight the importance of effective information management. Constructivist perspectives emphasize active knowledge construction and scaffolding. Social learning highlights the role of observation and peer interaction. Dual-process theory provides a complementary explanation for how repeated practice may support increasingly efficient decision-making.

These perspectives collectively suggest that an effective CHW training game should not merely contain health information or game elements. It should provide meaningful scenarios, appropriate challenges, repeated practice, feedback, scaffolding, and opportunities for social interaction. It should also be designed with consideration for the technological and contextual realities of the intended users.

The research presented in this thesis builds on these insights by empirically examining whether game-based refresher training can improve CHW learning outcomes and by inves-

tigating how digital and physical game formats compare with conventional classroom-based training.

Chapter 3

Methods

3.1 Research Problem to a Methodological Strategy

The preceding chapter established the broader problem that motivates this research. Community Health Workers (CHWs), particularly Accredited Social Health Activists (ASHAs), occupy a critical position in the delivery of maternal and child health services in India, yet their training and refresher-training systems continue to rely heavily on conventional, information-intensive approaches. The literature reviewed in the previous chapter indicates that CHWs are expected to retain and apply a considerable amount of procedural and factual knowledge while working in environments characterized by limited resources, varying educational backgrounds, substantial workloads, and geographically dispersed communities. This creates a methodological challenge for both research and practice. It is not sufficient to determine whether CHWs can access training material and examine whether an alternative learning format can help them acquire, retain, and engage with the knowledge required in their everyday work.

The methodological approach adopted in this thesis emerged from this problem rather than from a commitment to a particular technology. The central question was not whether

digital technology, augmented reality (AR), or games are inherently better than conventional training. Instead, the research sought to investigate whether transforming refresher learning into an interactive and playful activity could produce measurable improvements in learning and engagement, and whether the form of engagement influenced these outcomes.

This distinction is important because the research progressed iteratively. The initial intervention explored a physical card-based activity and an augmented-reality-supported version. Subsequent studies examined whether the same underlying learning approach could be implemented through physical and digital card-based formats and compared these approaches with conventional classroom refresher training. A later study further examined smartphone-based gameplay as a more scalable form of refresher training. Thus, the methodological trajectory of the thesis was not a single experiment repeated three times. Rather, it represents a sequence of related studies in which the findings and practical experiences from one stage informed the design and evaluation of subsequent interventions.

The methodology consequently combines principles from Human-Computer Interaction (HCI), game-based learning, health education, and experimental evaluation. The studies use quantitative measures of knowledge acquisition and retention as their primary indicators of learning effectiveness, complemented by usability, game experience, satisfaction, and, where appropriate, gameplay observations. This combination was necessary because the effectiveness of a learning intervention cannot be understood solely through test scores. An intervention may improve knowledge but be difficult to use, poorly understood, insufficiently engaging, or unsuitable for the practical context of CHWs' work. On the other hand, an intervention may be enjoyable without producing meaningful learning. The methodological framework, therefore, treats learning outcomes and user experience as related but different dimensions.

The research also deliberately compares different modes of delivery. Physical cards, digital cards, AR-supported gameplay, and conventional classroom training differ in technological complexity, interaction possibilities, social dynamics, and accessibility. Comparing these formats provides a way to examine not only whether game-based learning works, but also which aspects of the intervention may contribute to its effectiveness. In particular, the later studies distinguish immediate knowledge acquisition from longer-term retention. A refresher training intervention is of limited value if its effects disappear shortly after the training session, undermining its purpose.

The resulting methodological strategy can therefore be understood as a progression:

Identify a training problem → Design a playful learning intervention → Test feasibility and experience → Compare different formats of learning → Measure immediate learning → Measure knowledge retention → Refine the intervention and examine scalability.

This chapter describes this methodological strategy and provides the common foundation for the three empirical studies presented in the subsequent publication chapters.

3.2 Research Aim and Methodological Objectives

The overall aim of the research was to investigate how game-based and technology-enabled learning interventions can support refresher training and knowledge development among CHWs in India.

To address this aim, the research pursued four interconnected methodological objectives:

1. To design and develop a game-based refresher-training intervention appropriate for CHWs and their working context
2. To evaluate whether playful and interactive learning formats improve immediate knowledge acquisition compared with conventional training
3. To examine whether knowledge acquired through game-based interventions is retained over time, and
4. To investigate how different delivery formats, including physical cards, digital games, and augmented reality, influence learning experience, usability, and engagement.

These objectives correspond to two broad outcome domains. The first is *learning effectiveness*, measured primarily through changes in knowledge scores. The second is *learning experience*, assessed through participant feedback, usability measures, and game-experience measures. The distinction between these domains is maintained throughout the thesis because a

positive gameplay experience does not necessarily imply effective learning, and effective learning does not necessarily imply that an intervention is practical or enjoyable to use.

3.3 Research Design

The research adopted a multi-study experimental design. Rather than treating the intervention as a fixed artifact, the research treated game-based learning as an evolving design and evaluation process. Three empirical studies form the core of the thesis. Each study addressed a related methodological question while introducing a different comparison or delivery format.

The three studies can be positioned within a common research framework:

Table 3.1: Overview of the three empirical studies

Study	**Primary intervention**	**Comparison**	**Primary methodological focus**
Study 1	Physical card-based and augmented-reality-supported playful activities	Comparison of physical and AR-supported formats	Feasibility, learning gain, gameplay experience, and technology-supported refresher learning
Study 2	Physical and digital card-based game	Conventional classroom refresher training and control condition	Immediate learning, delayed retention, and comparison of delivery formats
Study 3	Smartphone-based digital refresher game	Conventional learning/-training condition	Immediate knowledge gain, learning progression, and knowledge retention

Although the studies differ in their specific experimental configurations, they share a common methodological principle. In all of them, knowledge was assessed before and after exposure to the intervention, and the resulting change was used as the primary indicator of learning.

Where applicable, delayed assessments were conducted to examine retention.

The studies were not designed as three independent projects. Instead, they form a methodological progression. Study 1 examined the initial feasibility of transforming refresher training content into a playful, physical, and AR-supported experience. Study 2 subsequently provided a more controlled comparison among physical gameplay, digital gameplay, and conventional classroom training. Study 3 extended the digital approach to a smartphone-based intervention and examined its potential for repeated, more accessible refresher learning.

3.4 Research Context

3.4.1 Community Health Workers and the Training Context

The empirical research was conducted in the context of community-based maternal and child healthcare in India, with particular attention to ASHA workers and related CHWs. The interventions focused on knowledge areas that form part of routine maternal and child health responsibilities, including antenatal care (ANC), postnatal care (PNC), childhood immunization, and the use of medicine and nutritional supplements.

The content was organized into four broad knowledge domains, referred to in the intervention as four "silos":

1. Children below one year of age
2. Children above one year of age
3. Antenatal care (ANC) and
4. Postnatal care (PNC)

The child-related domains primarily addressed immunization and knowledge of relevant medications and supplements, whereas ANC and PNC addressed pregnancy and post-pregnancy care. The organization of the content into these domains was intended to reduce the cognitive

and navigational complexity of the learning material and to provide a structure that could be translated into game mechanics.

3.4.2 Participants

The target population consisted of CHWs, with a substantial proportion of participants being ASHA workers. Participants were recruited in connection with the local health-worker and training context in which the studies were conducted. The research involved CHWs with different educational backgrounds and years of field experience. This variation was methodologically relevant because the intervention was intended for a heterogeneous workforce rather than a narrowly defined population.

Demographic information was collected to identify potential differences between participant groups. The demographic questionnaire included variables such as age, educational qualification, and years of experience as an ASHA worker. These characteristics were considered when interpreting the results because differences in prior experience or educational background may influence baseline knowledge and learning outcomes.

The exact sample composition and group allocation for each study are reported in the corresponding publication chapters. This chapter describes the common sampling and evaluation logic, while the publication-specific chapters provide the detailed participant numbers and experimental details.

3.5 Development of the Game-Based Learning Intervention

3.5.1 Design Rationale

The intervention was developed by transforming existing refresher training content into a format that could be learned through active participation. Instead of presenting information exclusively through lectures or static written material, the intervention required participants to engage with

questions, respond to prompts, discuss alternatives, receive feedback, and progress through a game-based structure.

The design was guided by several practical requirements. First, the intervention needed to accommodate participants with different levels of formal education. Second, the content needed to remain closely aligned with the knowledge that CHWs were expected to use in their professional practice. Third, the activity needed to support group interaction because CHWs commonly work and learn within peer networks. Fourth, the intervention needed to be adaptable to settings with limited technological infrastructure. These requirements contributed to the decision to explore both physical and digital implementations rather than assuming that a digital intervention would necessarily be preferable.

3.5.2 Physical Card-Based Intervention

The physical version used a deck of cards as the primary interaction medium. The cards contained simple objective content derived from the target health-training curriculum. Participants interacted with the cards during repeated gameplay sessions, discussing questions and selecting or providing answers in accordance with the game's rules.

The physical format provided several practical advantages. It did not require smartphones, internet connectivity, or electricity during gameplay. It also allowed several CHWs to participate simultaneously and created opportunities for peer discussion. These characteristics were particularly relevant to the intended deployment context.

The physical game, therefore, served not only as an experimental intervention but also as a low-technology implementation of the same underlying game-based learning concept.

3.5.3 Digital Intervention

The digital versions translated the card-based learning interaction into a smartphone-based environment. The digital format enabled the learning content, questions, feedback, and progression to be delivered via an Android application.

The digital implementation provided additional possibilities for interaction and repetition. It also allowed the learning activity to be conducted without having to carry a physical deck of cards. At the same time, the digital format introduced dependencies on smartphone availability, familiarity with digital interfaces, and device-related constraints.

The comparison between physical and digital versions was therefore methodologically useful because it separated the broader game-based learning approach from the specific affordances of a digital platform.

3.5.4 Augmented Reality Intervention

Study 1 additionally explored an augmented-reality-supported version of the playful learning activity. AR was used to introduce digital information and interaction into a physical learning environment. The purpose of this implementation was not to test AR technology in isolation but to examine whether an additional interactive layer could support refresher learning.

The AR intervention represented an early stage of the research trajectory. It consequently served both an evaluative and exploratory purpose. Its findings contributed to the subsequent decision to investigate simpler physical and smartphone-based implementations in greater depth.

3.6 Experimental Studies

3.6.1 Study 1: Physical and Augmented Reality-Based Playful Activities

The first study, "Physical and Augmented Reality based Playful Activities for Refresher Training of ASHA Workers in India," examined the use of playful activities for refresher learning among ASHA workers.

The study compared a physical card-based activity with an AR-supported version. Participants completed a baseline knowledge assessment before participating in the intervention. Following gameplay, their knowledge and experience were evaluated through post-intervention

assessment and feedback measures.

The methodological purpose of this study was exploratory. It examined whether the transformation of refresher-training material into an interactive activity could support knowledge acquisition and whether the addition of AR produced a meaningful learning experience.

The study also provided an opportunity to identify practical issues related to gameplay, instructions, interaction, and usability. These observations informed subsequent iterations of the intervention.

3.6.2 Study 2: Physical, Digital, and Classroom-Based Refresher Training

The second study, "Refresher Training through Digital and Physical, Card-Based Game," adopted a more comparative experimental design.

Participants were exposed to one of several learning conditions. The intervention conditions used physical or digital card-based gameplay, while the comparison condition used conventional classroom-based refresher training. Where applicable, a control condition was also included.

The study consisted of a baseline assessment followed by the assigned intervention. Participants in the game-based conditions engaged in approximately four to five rounds of gameplay. Participants were encouraged to play when they met, generally for approximately one hour per day over the intervention period. Physical game decks were provided to groups of nearby CHWs, while participants assigned to the digital condition received the Android application.

The classroom condition was delivered by master trainers using the conventional training approach. This condition provided an important comparison because it represented the established mode of refresher education against which the game-based intervention could be evaluated.

A post-test was conducted after the intervention to assess immediate knowledge acquisition. A post-test was subsequently administered weeks later to assess knowledge retention.

The design, therefore, enabled three related comparisons:

1. Change in knowledge within each intervention condition

2. Differences in post-test learning between game-based and conventional training conditions and

3. Differences in delayed retention between the intervention formats

This study constituted the principal comparative evaluation of physical and digital game-based refresher learning.

3.6.3 Study 3: Smartphone-Based Refresher Training

The third study, "Replay, Revise, and Refresh: Smartphone-Based Refresher Training for Community Healthcare Workers in India," extended the digital intervention into a smartphone-based refresher-training format.

The study was motivated by the need to examine whether repeated digital gameplay could provide an effective mechanism for knowledge reinforcement. Rather than treating the game solely as a replacement for a classroom session, the intervention considered the smartphone a platform through which CHWs could repeatedly revisit learning material.

Participants completed knowledge assessments before and after the intervention. The study examined immediate learning gain and compared knowledge outcomes across the relevant intervention and comparison groups. Delayed assessment was used to investigate whether the knowledge acquired through the intervention was retained.

The study, therefore, shifted the methodological emphasis from initial feasibility and format comparison toward the potential of repeated digital interaction as a refresher-training mechanism.

3.7 Evaluation Framework

The evaluation framework was informed by the four-level Kirkpatrick model of training evaluation (Kirkpatrick, 1950). The model distinguishes between reaction, learning, behavior, and results. The present research focused primarily on the first two levels.

3.7.1 Level 1: Reaction

Level 1 concerns participants' reactions to the training intervention. In the present research, this included participants' perceptions of gameplay, enjoyment, usability, satisfaction, engagement, and overall experience.

The purpose of this evaluation was not simply to establish whether participants "liked" the game. Rather, participant reactions were considered useful for understanding whether the intervention was sufficiently accessible and engaging to function as a practical learning activity.

Usability and game-experience measures were therefore used where appropriate. Feedback from participants was also used to iteratively refine the intervention.

3.7.2 Level 2: Learning

Level 2 constituted the primary evaluation level of the thesis. Learning was measured as a measurable change in knowledge following exposure to the intervention.

A pre-test was administered before the intervention, and a post-test after the intervention. The difference between these measurements represented an immediate gain in knowledge. Where delayed post-tests were available, the post-test and delayed post-test scores were compared to assess knowledge retention.

The distinction between acquisition and retention was central to the evaluation. Immediate improvement demonstrates that participants learned during or shortly after the intervention, whereas delayed performance provides evidence about whether that knowledge persisted.

3.7.3 Levels 3 and 4

Levels 3 and 4 of the Kirkpatrick model concern behavioral transfer and broader organizational or program-level outcomes. These levels were outside the scope of the present research because they require prolonged observation of CHWs in their field practice and assessment of health-service outcomes.

The present thesis, therefore, makes no claim that improvements in questionnaire performance necessarily translate directly into improved field behavior or population-level health outcomes. Such claims would require a different research design and longer-term evaluation.

3.7.4 Conceptual Evaluation Framework

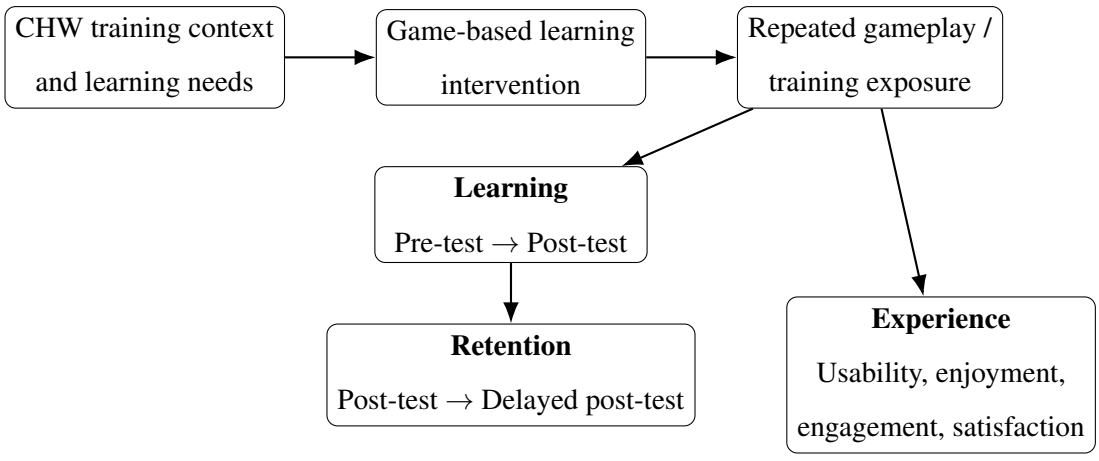


Figure 3.1: Conceptual evaluation framework linking the learning intervention, knowledge acquisition, retention, and participant experience.

3.8 Instrumentation

3.8.1 Knowledge Questionnaire

The primary learning instrument consisted of a structured questionnaire containing 40 multiple-choice questions (MCQs). The questionnaire covered all four content domains: children under 1 year, children over 1 year, ANC, and PNC.

The child-health questions focused primarily on immunization and knowledge of medications and supplements, while the maternal-health questions addressed relevant ANC and PNC practices.

The same underlying knowledge domains were assessed at different stages of the intervention to enable comparison of baseline knowledge, immediate learning, and, where applicable, retention.

The questionnaire consisted of three broad sections. The first section contained the informed-consent information. Participants read the information and provided consent before taking part in each survey or assessment. The second section collected demographic information. The third section contained the 40 knowledge questions.

The demographic section included a checklist and gap-fill items concerning characteristics such as age, educational qualification, and years of experience as an ASHA worker. These variables were collected primarily to identify potential differences between participant groups and to contextualize the learning outcomes.

A copy of the questionnaire is included in the Appendix (Figures B.1, B.2, and B.3).

3.8.2 Translation of the Knowledge Questionnaire

The questionnaire was initially developed in English and subsequently translated for participants who preferred Indian languages. Because many participants were more comfortable communicating in Hindi, Marathi, and Bengali, these languages were prioritized.

The initial translation was generated using machine-translation services and subsequently reviewed and corrected by humans. The purpose of the human review was to improve linguistic clarity, appropriateness, and comprehensibility for the intended participants.

The translation process was treated as an adaptation rather than a literal word-for-word conversion. In particular, vocabulary that was familiar to rural CHWs was preferred over highly formal or urban terminology. This was important because the objective was to measure health knowledge, not participants' ability to understand academic English.

3.8.3 Game Experience Questionnaire

The Game Experience Questionnaire (GEQ) (Ijsselsteijn et al., 2013) was considered for evaluating participants' experience of the game.

The GEQ contains items addressing dimensions including immersion, flow, competence, positive affect, negative affect, tension, and challenge. Responses are recorded using a five-point scale ranging from low to high intensity.

The original questionnaire has been used across several languages and cultural settings (Ijsselsteijn et al., 2008). Because the intended participants in the present research were primarily Indian CHWs, an adaptation and translation process was initiated.

The initial translation work focused on Marathi and Hindi. English items were translated using multiple translation sources, including Google Translate and IndiaTyping, to generate alternative linguistic formulations. Human review was then used to identify and correct wording that was unclear, overly formal, culturally inappropriate, or difficult to understand.

Particular attention was paid to the social and linguistic context of the participants. The wording was adapted to remain respectful and appropriate when addressing older female participants and to use vocabulary familiar to rural respondents. Where English terminology was commonly used in everyday communication, familiar English words were retained rather than replaced with uncommon formal equivalents.

The translation work subsequently expanded toward a broader Indian-language version, covering Bengali, Gujarati, Hindi, Kannada, Malayalam, Marathi, Tamil, Telugu, Odia, Pun-

jabi, Urdu, Assamese, Nepali, Bhojpuri, Konkani, and Maithili. These translations underwent successive rounds of human review.

The social-presence module and post-game module were excluded from the evaluation because they did not directly address the research questions of the present study.

The GEQ and documentation of the translation process are included in the Appendix (Figure B.4).

3.8.4 Usability and Participant Feedback

In addition to structured questionnaires, participant feedback was used to understand usability and the practical experience of gameplay. The feedback focused on issues such as clarity of instructions, ease of interaction, enjoyment, perceived usefulness, and difficulties encountered during play.

The System Usability Scale (SUS) was considered a standardized usability measure for the digital intervention. Where usability data were collected, these measurements were interpreted alongside the learning outcomes rather than as substitutes for them.

3.9 Data Collection Procedure

The general data collection procedure followed a repeated sequence:

1. Informed consent
2. Demographic information
3. Baseline knowledge assessment
4. Exposure to the assigned intervention or training condition
5. Post-intervention knowledge assessment
6. Gameplay or experience feedback and

7. Delayed knowledge assessment where retention was examined

Figure 3.2 presents the general methodological sequence.

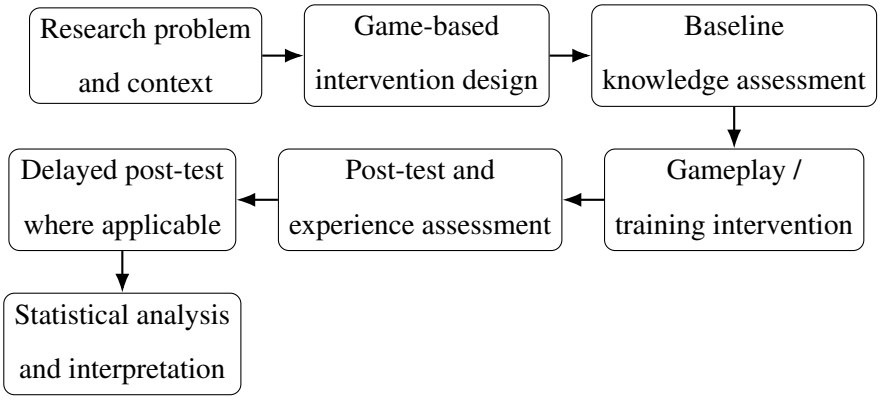


Figure 3.2: Overall methodological sequence adopted across the empirical studies.

For the game-based groups, participants were encouraged to engage with the intervention repeatedly rather than treating the game as a single exposure. In the physical and digital card-based studies, participants generally played when they met with other CHWs, for approximately 1 hour per day over 1 week. The repeated nature of the activity was intended to resemble refresher learning rather than a one-time test intervention.

The classroom comparison condition followed the established refresher-training approach, with master trainers providing conventional classroom instruction.

The delayed assessment was conducted several weeks later. The purpose of introducing a delay was to reduce the likelihood that participants would merely reproduce information still available in short-term memory.

3.10 Outcome Measures

The principal dependent variable across the studies was knowledge performance.

3.10.1 Knowledge Gain

Knowledge gain was measured as the change between baseline and post-intervention scores:

$$\Delta K = K_{post} - K_{pre} \tag{3.1}$$

where K_{pre} represents the baseline knowledge score and K_{post} represents the immediate post-intervention score.

A positive value of ΔK indicates an increase in knowledge following the intervention.

3.10.2 Knowledge Retention

For studies containing a delayed assessment, retention was examined by comparing post-test and delayed post-test performance:

$$\Delta R = K_{delayed} - K_{post} \tag{3.2}$$

where $K_{delayed}$ represents the delayed post-test score.

A smaller decline between the post-test and delayed test indicates greater retention of the acquired knowledge.

Knowledge gain and retention were treated as separate outcomes. This distinction is important because an intervention may produce a large immediate gain while producing a relatively large subsequent decline, or it may produce a more moderate initial improvement that retains over time.

3.10.3 Experience and Satisfaction

In some experiments, participants' experiences were also measured. It included usability, enjoyment, engagement, perceived competence, challenge, and other relevant dimensions of gameplay experience.

These measures were interpreted descriptively and in relation to the learning results. They were not assumed to establish causal relationships between enjoyment and learning.

3.11 Data Analysis

The statistical analysis was designed around the repeated-measures nature of the evaluation.

Descriptive statistics were first used to summarise participant characteristics and questionnaire performance. Depending on the variable, measures of central tendency and dispersion were calculated to describe the distribution of scores.

Within-group comparisons were used to determine whether participants demonstrated significant improvement from pre-test to post-test. Paired-samples t-tests were used where the assumptions of the test were considered appropriate.

Between-group comparisons were then conducted to examine whether the magnitude of learning differed across intervention conditions. In particular, the analysis compared the game-based formats with conventional classroom training and, where applicable, compared physical and digital versions of the game.

For retention analysis, post-test scores were compared with delayed post-test scores. The objective was to determine whether the amount of knowledge retained differed between intervention conditions.

The analytical strategy can therefore be summarised as:

1. Measure baseline knowledge
2. Measure within-group learning gain

3. Compare learning gain between groups

4. Assess change between immediate and delayed post-tests and

5. Compare retention across intervention formats

Where statistical assumptions permitted, parametric tests were used. The interpretation of statistical significance was complemented by examination of the direction and magnitude of observed differences. Statistical significance alone was not treated as evidence that an intervention was practically superior.

The detailed statistical procedures, sample-specific analyses, test statistics, and significance values are reported in the individual publication chapters.

3.12 Reliability, Validity, and Instrument Development

3.12.1 Content Validity

The knowledge questionnaire was constructed from the health-training content relevant to CHWs and the learning objectives represented in the intervention. The questions were therefore designed to correspond directly to the material participants were expected to learn.

This alignment between intervention content and assessment content was important because the purpose of the evaluation was to measure learning of the intended refresher material rather than general health knowledge.

3.12.2 Language Validity

Language presented a particular methodological challenge. A questionnaire that is linguistically difficult may confound knowledge with language proficiency. For this reason, the English questionnaire was adapted into languages familiar to the participants.

Machine translation was used as an initial aid, but not treated as the final linguistic authority. Human correction was incorporated to address contextual, grammatical, cultural, and vocabulary-related problems.

The same principle was followed during the development of the translated GEQ (explained above in Section 3.8.3 and Figure B.4).

3.12.3 Contextual Validity

The studies were conducted with CHWs in contexts relevant to their actual work and training practices. The intervention content was derived from their professional responsibilities, and the physical game was designed to operate without specialized infrastructure.

This approach was intended to increase ecological validity. Rather than evaluating the intervention exclusively under laboratory conditions, the research examined gameplay in contexts that approximated the circumstances in which refresher learning could realistically occur.

At the same time, this contextual approach introduces variability between participants and groups. Such variability is acknowledged as a characteristic of the field setting rather than treated automatically as experimental error.

3.13 Ethical Considerations

The study procedures were approved by the Institutional Review Board (IRB) of IIT Bombay under Proposal Number IITB-IRB/2022/051. The approval documentation is included in the Appendix (Figure C.1).

All participants provided informed consent before participating in data collection. Participants were informed about the purpose and nature of the research, and their participation was voluntary. They were free to discontinue their participation at any time without any penalty or other consequences.

The research was conducted in accordance with the applicable ethical guidelines and reg-

ulations of the Institute Ethics Committee and with the principles of the Declaration of Helsinki (WMA, 2013) developed by the World Medical Association.

The research team also recognized that CHWs contributed professional knowledge, experience, and time to the development and evaluation of the intervention. Although the study did not provide monetary or material compensation to participants, refreshments such as tea, snacks, and sweets were provided as a gesture of appreciation for their participation.

The decision not to provide monetary compensation should be understood within the institutional and ethical framework under which the studies were conducted.

3.14 Methodological Considerations and Limitations

Several methodological considerations should be taken into account when interpreting the findings of the three studies.

1. The studies primarily measure knowledge through questionnaire-based assessment. Improvements in test scores demonstrate learning within the assessed domain, but they do not necessarily establish that CHWs subsequently changed their behavior in the field. Behavioral transfer would require observation or longitudinal follow-up and was beyond the scope of this research.

2. Repeated exposure to the questionnaire may introduce a testing effect. Participants completing a post-test or delayed test may remember some questions from earlier assessments. The use of delayed assessments nevertheless provides a useful indication of whether knowledge was maintained beyond the immediate intervention period.

3. The studies were conducted with CHWs in specific Indian contexts. The findings should therefore not automatically be generalized to all CHWs, all Indian states, or other health-worker populations without considering differences in training systems, language, infrastructure, and work practices.

4. The intervention evolved across the studies. This is a methodological strength of iterative design, but it means the three studies should not be interpreted as exact replications. Each

study has some common research questions and some distinct ones, and introduces a different intervention configuration.

5. The use of digital and AR technologies introduces the possibility of a novelty effect. Participants with limited prior exposure to these technologies may initially respond positively because the technology is new. Study 1, therefore, needs to be interpreted cautiously when considering whether AR-specific effects represent durable learning benefits.

6. Game experience and learning are related but distinct constructs. A highly enjoyable intervention is not necessarily educationally effective, and an effective learning intervention may not always be highly enjoyable. The methodological framework, therefore, avoids treating subjective game experience as direct evidence of learning effectiveness.

3.15 Relationship Between the Three Studies

The three empirical studies should be understood as complementary rather than isolated experiments.

Study 1 established the feasibility of transforming CHW refresher training content into a playful activity and explored the role of augmented reality as a medium. Study 2 strengthened the comparative component by examining physical and digital card-based gameplay alongside conventional classroom training and by distinguishing immediate learning from longer-term retention. Study 3 then extended the digital approach into a smartphone-based refresher-training model and examined whether repeated digital interaction could support knowledge acquisition and retention.

The methodological progression can be represented as follows:

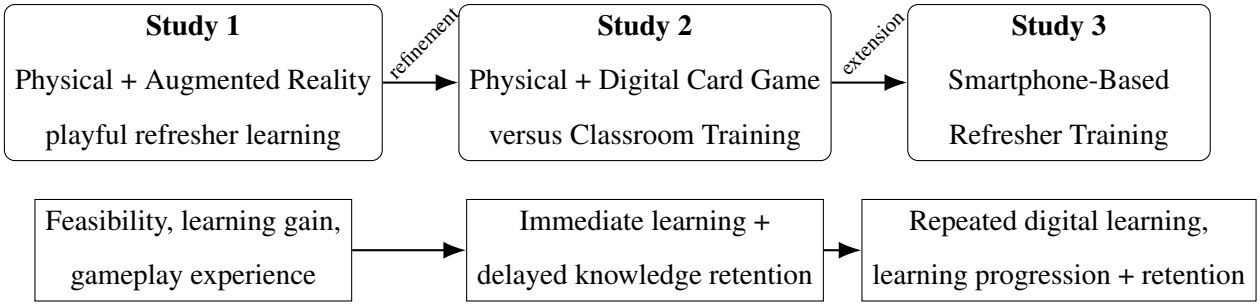


Figure 3.3: Methodological progression across the three empirical studies.

The progression from physical and AR-supported activities to physical and digital card games and finally to smartphone-based gameplay reflects an increasing focus on scalability, repeatability, and sustained learning. At the same time, the studies share an emphasis on learner-centered interaction, active participation, and measurable knowledge outcomes.

This progression also provides the methodological bridge between the individual publications. The publications address different research questions, but all examine the broader proposition that gameful and interactive learning can provide an alternative to passive refresher training for CHWs.

3.16 Methodological Contribution of the Thesis

The methodological contribution of this research lies not in the use of a single experimental technique but in the integration of several approaches appropriate to the complexity of the research problem.

1. The research evaluates game-based learning using measurable learning outcomes rather than relying exclusively on participant perceptions.

2. The research distinguishes immediate knowledge acquisition from knowledge retention. This distinction is particularly relevant to refresher training because the practical value of an intervention depends on knowledge being available beyond the training session itself.

3. The research compares physical and digital implementations of a common game-based

learning concept. This makes it possible to examine whether the observed benefits are associated more broadly with game-based interaction or with specific technological affordances.

4. The research incorporates usability and game-experience measures alongside knowledge assessment. This acknowledges that an intervention intended for real-world deployment must be both educationally useful and practically usable.

5. The research adopts an iterative approach in which findings from earlier intervention formats informed subsequent designs. This approach is consistent with the HCI tradition of designing, evaluating, and refining interactive systems within their intended social and technological context.

3.17 Chapter Summary

This chapter presented the methodological framework underlying the thesis's three empirical studies. The research was motivated by the need to investigate alternatives to conventional refresher training for CHWs, specifically whether interactive and game-based approaches could improve knowledge acquisition and retention.

A multi-study experimental strategy was adopted. Study 1 explored physical and augmented-reality-supported playful learning. Study 2 compared physical and digital card-based gameplay with conventional classroom training and introduced delayed assessment of knowledge retention. Study 3 extended the digital intervention into a smartphone-based refresher-training format.

Across the studies, the principal learning outcome was a change in knowledge, measured using a 40-item MCQ questionnaire covering childhood immunization, medicines and supplements, and ANC and PNC. Participant experience was examined through feedback, usability measures, and the GEQ where appropriate. The evaluation was conceptually aligned with Levels 1 and 2 of the Kirkpatrick model, focusing on participant reaction and learning rather than longer-term behavioral or organizational outcomes.

The chapter also described the intervention development process, participant context,

translation procedures, data collection sequence, statistical analysis strategy, and ethical safeguards. The methodological framework provides the common foundation for the three publication chapters that follow. The individual chapters subsequently provide the study-specific experimental details, results, and interpretation. The game development chapter discusses the iterative design and development process. The Discussion chapter brings the findings together and synthesizes to examine what the three studies collectively suggest about game-based refresher learning for CHWs.

Chapter 4

Study 1: Comparing Physical and Augmented Reality-based Playful Activities for Refresher Training of ASHA Workers

This study seeks to design and evaluate playful refresher training activities for CHWs on immunization schedules and maternal-child healthcare practices. It compares a physical card game and an Augmented Reality (AR)-enabled smartphone application. Hypothesis (**H1**): ASHAs playing the AR-based digital version would show a higher improvement in knowledge of immunization compared to the group using the physical card-based game.

This study aims to empirically validate the pedagogical potential of immersive technologies like AR in public health training, address gaps in CHW knowledge through scalable learning tools, and contribute a novel perspective to digital education research in low-resource settings. The study proposes an inclusive training methodology easily adapted across diverse public health topics and geographies by making learning interactive, language-independent, and

play-based.

4.1 Study-1 Details

Parts of this section are from a published research Study [1]

1. **Introduction**

   Research conducted on training CHWs using technology through means of texts, voice messages, videos, and quizzes has shown mixed results (Section 2.4). While discussing with CHWs, we found that most of them prefer to learn when they are doing something tangible with their hands or receive instant feedback. Augmented Reality can address this gap (Billinghurst and Dünser, 2012; Dunleavy et al., 2009; Wu et al., 2013) by layering a virtual world over the physical world through a smartphone screen, which enables learning by doing (Azuma, 1997). With the increasing use of mobile devices, the adoption of interactive and gaming-style learning strategies (Peek, 2017) presents new opportunities for training in AR (Santos et al., 2014; Sarkar et al., 2020). AR provides an opportunity to transform an average room into an interactive learning lab, where child profiles and vaccine cards come to life, and choices made by the player have consequences within that world.

[1] **Majhi, A.**, Agnihotri, S. B., Mondal, A., 2022. Physical and Augmented Reality-based Playful Activities for Refresher Training of ASHA Workers in India ", *CHI 2022: Asian CHI Symposium 2022* (Best Paper and Best Presentation award in Long Paper category) https://dl.acm.org/doi/10.1145/3516492.3558788

2. **Methods**

We conducted the study in multiple stages, i.e., conceptualization of a training game, content validation, game design, development, field testing, and comparative assessment.

(a) **Focused Group Discussion with CHWs**

To complement the quantitative findings, a qualitative component was incorporated into the study through a Focus Group Discussion (FGD). While the quantitative data assessed measurable outcomes related to immunization training, the FGD was conducted to gain a deeper understanding of Accredited Social Health Activists' (ASHAs) lived experiences with existing training practices, the challenges they encountered, and their perceptions of the effectiveness of current training approaches. These qualitative insights provided contextual explanations for the quantitative findings and informed the interpretation of the results and the design recommendations presented in this thesis.

A FGD was conducted with (n=14) ASHAs who were available and willing to participate at the time of the study. Participants were recruited using convenience sampling from the study area. The discussion was facilitated using a semi-structured discussion guide consisting of open-ended questions related to participants' experiences with immunization training, perceived barriers to learning, confidence in performing immunization-related tasks, and suggestions for improving future training programmes. The discussion encouraged participants to share their experiences freely while allowing the facilitator to probe emerging topics where appropriate.

As audio recording was not feasible during the session, detailed written field notes were taken throughout the discussion to document participants' responses, examples, interactions, and key observations. Immediately after the session, these field notes were expanded to preserve contextual details and improve the completeness of the qualitative dataset.

The qualitative data were analyzed manually using reflexive thematic analysis following the approach proposed by Braun and Clarke (Braun and Clarke, 2006, 2019). The thematic analysis was conducted on the detailed field notes. The analysis began with repeated reading of the field notes to achieve familiarity with the data. Meaningful segments of text were then coded inductively, allowing codes to emerge from

participants' responses rather than being predetermined. Related codes were iteratively compared and grouped into broader candidate themes. These themes were subsequently reviewed and refined to ensure they accurately represented recurring patterns across participants' experiences. Throughout the analysis, principles of grounded theory, particularly inductive coding and constant comparison, informed the development and refinement of themes (Charmaz, 2014). The qualitative analysis was performed manually, and no qualitative data analysis software was used.

The themes identified from the FGD were used to contextualize and interpret the quantitative findings reported in this thesis. Specifically, the qualitative findings helped explain participants' experiences underlying the observed quantitative results, identified practical challenges encountered during immunization training, and informed the discussion and design implications for future game-based training interventions for ASHAs.

An important theme that emerged is the knowledge gap among participants, particularly in recalling key aspects of the immunization schedule. Many ASHAs reported difficulty remembering the names of vaccines, the sequence in which they are administered, and the specific timing of each dose. This gap was often described in relation to their day-to-day work, where accurate recall is essential for effective service delivery.

Participants also reflected on their prior training experiences, noting that these sessions were predominantly lecture-based and, in many cases, delivered within limited timeframes. As a result, the training was perceived as rushed and difficult to comprehend, with limited opportunities for interaction, clarification, or reinforcement of key concepts. Several participants indicated that this format made it challenging to retain information over time.

These observations suggest that the issue extends beyond the complexity of the content itself and is closely tied to how the information is presented and reinforced. The lack of engaging, accessible, and contextually relevant learning formats appears to contribute to reduced retention. These insights informed our subsequent design direction, motivating us to explore more visual, interactive, and practice-oriented approaches aimed at improving both understanding and long-term retention among ASHAs.

(b) **Co-Design with CHWs**

We chose to design along with the ASHAs. We hosted small co-design sessions where ASHAs reviewed sample illustrations, suggested modifications, and made them draw. Their inputs directly influenced the choice of imagery, terminology, and the flow of the physical and AR-based games. This collaborative approach ensured that the final tools were intuitive and familiar. It also helped build a sense of ownership among the ASHAs, making them more invested in learning.

(c) **Gamified Activity - Physical Card Game**

We developed a deck of 20 cards, covering the administration of age-specific vaccines, which was used across both the physical and AR-based interventions to maintain consistency and comparability of content. The detailed process of designing the card deck is discussed in Section 8.2.2 and 8.4.2. Players received a random mix of cards in the Physical Card Game version. The task in the game is to simulate the experience of examining child profiles and administering age-appropriate vaccines. Players receive points for correct matches and can challenge each other's combinations. The activity encouraged peer validation, discussion, and friendly competition. Each round took about 15 minutes and was played by 2–5 CHWs.

(d) **Gamified Activity - AR-based Digital Game**

The AR-based game or digital version added an interactive twist. CHWs placed printed stickers on the walls representing children and mothers. When scanned with a smartphone using the AR app, a 3D model appeared, showing age and other key details. Players then scanned the appropriate vaccine cards to administer them virtually. The game rewarded correct matches with laughter and positive reinforcement. Mistakes led to crying sounds and point deductions. This instant feedback made the experience memorable and emotionally engaging. Players can play the AR game solo or in teams. In multiplayer settings, we observed role-playing dynamics, with some participants taking the lead in scanning while others handled cards or assisted with strategy.

(e) **Evaluation**

To test both tools, we collaborated with an NGO involved in ASHA training. We aligned our interventions with their existing schedule and conducted five sessions. Each group of participants went through either the physical or digital version of the

training. ASHAs were informed of the purpose of the study and agreed to it through SMS or WhatsApp. The participation was not mandatory, and they were free to exit anytime. Of the initial 100 ASHAs recruited for the study, 86 went through all phases. Most were between 30 and 50 years old and had a secondary school level of education. They were all proficient in Bengali, although some had difficulty reading or writing it. The demographic data is tabulated in Table 4.1.

We used a questionnaire consisting of 10 questions focused on the immunization schedule to measure knowledge gains. We conducted the questionnaire survey both before and after the training. Questions covered topics such as the timing of vaccines, matching vaccines to ages, and interpreting information from the MCP card. We divided the participants into two groups: one experienced the card game (n=42), and the other used the AR app (n=44). We analyzed the change in their scores using paired t-tests within groups and two-sample t-tests between groups (Dimitrov and Rumrill, 2003).

We divided the participants into two groups: one used the physical card game, and the other used the AR app. Before starting, we asked each ASHA to complete a short quiz (pre-test) covering key points of the immunization schedule. We repeated the same quiz right after the play session (post-test).

Table 4.1: Study-1 Participants Demographics

Parameter	Number of ASHAs (%)
Age	
<30	4 (4.65%)
30–40	41 (47.67%)
40–50	37 (43.02%)
No information	4 (4.65%)
Education (Grade)	
Below 8th	8 (9.30%)
8th–10th	12 (13.95%)
10th–12th	56 (65.12%)
Graduation and above	8 (9.30%)
No information	2 (2.33%)
Experience (Years)	
0–5	4 (4.65%)
5–10	32 (37.21%)
10–15	34 (39.53%)
Above 15	11 (12.79%)
No information	5 (5.81%)

3. **Results**

On average, ASHAs who played the physical card game improved their scores by about 3.4 points. Users of the AR app improved by over 6.5 points. We observed a statistical difference between the two groups. The results showed that the AR-based and physical games significantly increased the ASHA workers' knowledge of child immunization schedules. The intuitive and interactive nature of the interface of the AR game enhanced its appeal among the participants, making learning more effective. We also found that AR-based gaming was effective for visual learners. [2] However, physical card-based gameplay was liked by auditory learners or those who support verbal cues during playtime.

[2] "Visual learners" prefer to understand and remember information they can see, such as images, diagrams, or spatial layouts, whereas "auditory learners" tend to learn more effectively from spoken explanations, discussions, and other sound-based inputs.

Intervention	Pre-test score	Post-test score	Difference in Score
IG-1 (n=44) (AR)	2.7 (SD=1.1)	9.2 (SD=2.1)	6.5
IG-2 (n=42) (Physical Cards)	2.8 (SD=1.4)	6.2 (SD=1.8)	3.4
Difference of difference			Difference in Score
(IG1post - IG1pre) - (IG2post - IG2pre)			3.1

Table 4.2: Study 1: Summary of means of pre-test and post-test scores and their difference. Learning through Mobile App (Intervention Group 1 or IG 1). Learning through Physical Card (Intervention Group 2 or IG 2)

4. **Discussion**

(a) **Findings in Playful Learning**

Both modalities showed overall effectiveness, but the study shows a few counterintuitive findings.

i. **Short-term vs Long-term knowledge retention**

The AR-based game showed effective, immediate knowledge gains compared to the physical card group. This reflects Kahneman's dual-process theory (Kahneman, 2011). AR-based game engaged System-1 through fast, intuitive cues like the crying or laughing baby, signalling a right or wrong move. Whereas playing with physical playing cards demanded carefully weighing arguments and perspectives to make a thoughtful decision through taking turns, discussion, and rule-based negotiations. System-1 supported immediacy and memorability, but System-2 supported deeper understanding and codifying the content. An ASHA who had been in the physical card play group stayed silent through the first play round, later started correcting her peers mid-game, saying, "No, not at six months, the vaccine comes later—let me explain properly," suggesting that peer interaction and slow reasoning supported knowledge retention.

ii. **Low literate CHWs performed better with AR than physical cards**

Another counterintuitive finding was that participants with lower literacy or limited smartphone exposure often showed higher comfort with AR-based gameplay than with physical cards. The conjecture is that in the digital version, the

multimodal cues reduced the cognitive burden of textual recall. This is consistent with embodied cognition (Wilson, 2002). During playtesting, an ASHA burst out laughing at the crying baby sound and remarked, "Now I will never forget this mistake—the baby scolded me," while making a sarcastical facial expression. It shows that sensory feedback can act as a strong anchor for memory. However, literate participants often preferred the tactile and visual-symbolic reasoning of the cards, where they could compare options.

iii. **Constructive and Social Learning pathways**

While the AR-based game offered a learning loop through trial and error, the physical card game offered social play, elements of collaboration, competition, and correction. This exhibits the factor of relatedness in Self-Determination Theory (SDT) (Miller et al., 1985). ASHAs frequently joked and teased one another and took the game in good spirits. One of them said, "In this game, mistakes are fun, not shameful." This environment of safe social space aligns with Vygotskian ideas (Vygotsky, 1978) where scaffolding from colearners turns mistakes into a shared learning opportunity. Facilitators of the experiment observed that the physical card gameplay group had more laughter and intense debates, as compared to the digital group. This suggests that a technologically complex method or solution does not necessarily translate linearly into engagement or effectiveness. Instead, each form of gameplay was supported by different motivational and cognitive pathways. AR-based gameplay supported competence through feedback, whereas the physical card version supported relatedness through collaboration, and both were complementary to each other.

(b) **Knowledge Graph**

By about 10-15 rounds, the novelty and challenge dipped since the scenarios were stereotypical. It suggests incorporating more realistic, diversified scenarios and adaptive gameplay for long-term interest. Peer collaboration in both versions also allowed for discussion, dispelling misconceptions, and learning together, pointing out the social facet of learning through play.

(c) **Co-Design**

The intervention became contextually grounded and language-independent by involving ASHAs in the participatory co-design process, by selecting illustrations and

developing rules.

(d) **Institutional Support**

The co-design and field-testing were supported by the Child in Need Institute (CINI), which also serves as the State Training Centre for ASHAs in West Bengal. CINI's long presence in communities and the trust it has built through its social work enabled us to reach remote areas and work closely with people who might otherwise have been difficult to engage. This experience shows how important institutional support is for enabling access and ensuring interventions are welcomed in community settings.

5. **Conclusion** This study began with the conjecture that training can be more effective and enjoyable if it is more game-like. Based on our observations and measurements, we found that both tools, the physical card game and the AR app, made learning more interactive, memorable, and enjoyable for ASHAs. The AR app, in particular, showed greater improvement in test scores. These findings matter because ASHAs often have limited time and resources for refresher training. If we can make those brief windows more impactful using tools they enjoy, we can improve their knowledge and empower them in their vital roles.

This study demonstrated the potential of AR-supported learning tools to enhance healthcare training for CHWs by providing interactive, immersive, and visually rich learning experiences. The findings suggest that AR-supported learning tools can overcome the limitations of traditional training methods. Future efforts should prioritize refining AR game design for low-literacy users and scaling up this intervention across various healthcare settings. The current game's content is limited to a fixed number of scenarios. However, in reality, there can be infinite possibilities. Adding more complex narratives might make the game's content more comprehensive, but it also makes the gameplay more challenging. The long-term impact is the actual test of any training tool. With further refinement and support, tools like these can become part of mainstream ASHA training across states, translated into different languages, and used in both online and offline formats.

4.2 Study-1 Reflections

This study reaffirmed our initial conjecture that experiential, game-based refresher learning tools can contribute to capacity-building among frontline health workers. By shifting the training mode from predominantly passive, lecture-based instruction to active participation, interaction, and learning-by-doing, the interventions created opportunities for CHWs to engage with immunization content in a more memorable and enjoyable manner. The findings also demonstrated that different forms of play can support learning through different pathways. The AR-based intervention provided immediate visual and auditory feedback, while the physical card game encouraged peer discussion, correction, and collaborative reasoning. Thus, the study shifted our understanding of game-based learning from a question of which technology is more effective to a broader consideration of how different forms of play support different learning processes. The study found that both interventions were effective, although the AR-based group showed greater immediate gains in knowledge.

At the same time, the study raised an important question concerning the role of novelty in technology-mediated learning. Critics could reasonably argue that the use of Augmented Reality (AR), as a relatively unfamiliar technology for many participants, may have introduced a novelty effect that influenced engagement and learning outcomes. The increased attention and excitement associated with scanning cards, seeing virtual characters, and receiving audiovisual feedback may have contributed to the observed improvement. This leaves an important question unresolved: whether the higher learning gains were primarily attributable to the pedagogical affordances of AR or, at least partly, to the novelty of interacting with a new technology. A longer-term evaluation would be necessary to distinguish sustained pedagogical value from short-term novelty effects.

The study also highlighted the challenge of balancing realism, complexity, and accessibility. While increasingly realistic scenarios and more sophisticated gameplay could potentially provide richer opportunities for skill development, they also require greater cognitive, technical, and procedural proficiency from players. This becomes particularly important when designing for CHWs with varying levels of literacy, smartphone familiarity, and technical confidence. During the study, it became evident that an intervention designed to be technologically engag-

ing could simultaneously introduce additional barriers to participation. Simplified instructions, visual cues, intuitive interaction, and minimal dependence on textual input can help address some of these barriers, but they cannot entirely overcome limitations related to device access, infrastructure, or digital familiarity.

These reflections influenced an important design decision in the subsequent studies. Although the AR intervention produced promising immediate learning outcomes, we decided not to pursue AR as the primary direction for the subsequent studies. This was not because AR was found to be ineffective, but because its potential advantages had to be considered alongside the practical realities of scalability, infrastructure requirements, device availability, and adoption among less digitally experienced and older CHWs. In comparison, the physical card game required substantially fewer technological resources and was easier to introduce within existing training environments. The card-based approach, therefore, appeared more compatible with the contextual constraints of the target population. Consequently, the subsequent studies focus on card-based games and investigate how the same underlying game-based learning approach can be further developed across physical and digital forms.

Another limitation of this study was the absence of a placebo or no-intervention control condition. Although the comparison between the AR and physical card groups allowed us to examine differences between two playful learning modalities, it did not allow us to isolate the effect of game-based training from the effects of participating in any training activity itself. Improvements in both groups could therefore potentially reflect factors such as repeated exposure to the content, interaction with peers, attention from facilitators, or participation in the assessment process. This limitation also restricts the extent to which the observed improvements can be attributed specifically to the game-based interventions.

These reflections directly informed the methodological and design decisions made in the subsequent study in which a placebo or control group was introduced to provide a more controlled comparison and better isolate the intervention's contribution from usual training or repeated exposure.

Chapter 5

Study-2: Refresher Training through Digital and Physical, Card-Based Game for Accredited Social Health Activists (ASHAs) and Anganwadi Workers (AWWs) in India

This study was built on the findings of the previous research by expanding the comparison to include both digital and physical versions of the same card game, following the same rules of play for refresher training in immunization knowledge. The objective was to evaluate the comparative effectiveness of these two game modalities in improving knowledge retention and engagement over time.

A significant methodological limitation of the previous study is its lack of an appropriate control group. Educational research demonstrates that most instructional activities, regardless of their novelty, tend to yield positive learning gains. Assessing a new teaching practice against

a condition of "no teaching" only provides a weak basis for causal inference. A more rigorous approach would evaluate the intervention in relation to traditional or established teaching methods, thereby providing a more valid and interpretable measure of its effectiveness (Hattie, 2008).

Building upon the learnings from the previous experiment in Study 1, this experiment is four-armed and compares the impact of four training groups: (i) A smartphone-based digital card game, (ii) A physical card-based game mimicking similar mechanics, (iii) Traditional classroom teaching, (iv) A control group receiving no training.

The intervention is designed around the ANC, PNC, and mother and child immunization schedule and medications, integrating age-appropriate vaccination logic into the game mechanics. We categorized vaccines into children's age groups: Below 1 year, above 1 year, and mothers' phases: Antenatal care, and Postnatal Care. The research questions are as follows: **RQ1:** Does knowledge gain and retention differ significantly among CHWs trained using digital games, physical card games, and traditional classroom methods? **RQ2:** Is there evidence of a novelty effect that fades over time?

5.1 Study-2 Details

Parts of this section are from a published research Study [1]

1. **Introduction** The prevailing concerns regarding training CHWs have already been discussed in the previous Chapter. This study examines the effectiveness of card-based games—both digital and physical—as refresher training tools for CHWs. We also wanted to find whether these games helped participants retain information over time and how they compared with conventional classroom instruction. Throughout the design process, we involved CHWs to ensure the games reflected their learning needs and constraints. We prototyped and piloted a card game-based refresher training solution for CHWs, particularly focusing on child immunization. This game tried to make learning more interactive

[1] **Majhi, A.**, Agnihotri, S. B., Mondal, A., 2024. Refresher Training through Digital and Physical, Card-Based Games for Accredited Social Health Activists (ASHAs) and Anganwadi Workers (AWWs) in India. CHI PLAY Companion '24, October 14–17, 2024, Tampere, Finland (https://dl.acm.org/doi/10.1145/3665463.3678819)

and engaging, allowing CHWs to practice decision-making in a social and dynamic setting.

2. **Methods**

We contacted the CHW supervisors and recruited about 400 CHWs (almost equal number of ASHAs and AWWs) through convenient sampling to account for possible dropouts. By the end, 368 from those who retained,were selected. They paired up as ASHA AWW groups as per their areas where they live and work, forming 181 groups. Then they were randomly alloted into four experimental groups of 92 each. Group 1 (IG-1) (45 pairs of ASHAs and AWWs): Trained with the digital card game on smartphones, Group 2 (IG-2) (45 pairs of ASHAs and AWWs): Trained with physical cards, Group 3 (IG-3) (45 pairs of ASHAs and AWWs): Received a traditional classroom session with the same content, and Group 4 (CG) (46 pairs of ASHAs and AWWs): Control group with no training during the study period. The demographic details of the CHWs in this study are tabulated in Table 5.1 on the next page. We tried our best to balance age groups, educational levels, and years of experience among CHWs.

All groups underwent a baseline test. The three intervention groups received their respective training formats. A post-test followed. After three weeks, we conducted a final delayed retention test (DRT) to measure how much they remembered. We trained the control group only after completing all evaluations to ensure fairness. Before participating, we gave all CHWs verbal and written explanations about the study's purpose and process. We designed digital and physical card games to help CHWs memorize the sequence and dosage of immunizations and supplements required for children and mothers. We used questionnaire surveys to check knowledge gain and retention. Focus group discussions with CHWs captured engagement levels, user satisfaction, and feedback on the training experience. Data analysis included paired t-tests for knowledge score comparisons and thematic analysis for qualitative data.

We built the training game using 60 cards. The process of designing the card deck is discussed in Section 8.2.2 and 8.4.2. We printed the playing cards on card paper for the physical version, and the digital game mimicked the same format. The rules of play are discussed in Appendix A.6. The players played the game in teams of four. Each player received 15 shuffled cards. The goal was to place cards in the correct chronological order

Table 5.1: Study-2 Participants' Demography

	Intervention Group-1 Digital Card Game		Intervention Group-2 Physical Card Game		Intervention Group-3 Classroom training		Control Group No training	
Parameters	ASHAs(%)	AWWs(%)	ASHAs(%)	AWWs(%))	ASHAs(%)	AWWs(%)	ASHAs(%)	AWWs(%)
Age								
Less than 30	2(4.35%)	1(2.17%)	2(4.35%)	2(4.35%)	1(2.17%)	1(2.17%)	2(4.35%)	1(2.17%)
30-40	24(52.17%)	22(47.83%)	25(54.35%)	20(43.48%)	24(52.17%)	21(45.65%)	24(52.17%)	20(43.48%)
40-50	19(41.3%)	22(47.83%)	18(39.13%)	23(50%)	20(43.48%)	23(50%)	18(39.13%)	24(52.17%)
No information	1(2.17%)	1(2.17%)	1(2.17%)	1(2.17%)	1(2.17%)	1(2.17%)	2(4.35%)	1(2.17%)
Education (grade)								
$Below 8^{th}$	2(4.35%)	2(4.35%)	2(4.35%)	2(4.35%)	2(4.35%)	3(6.52%)	2(4.35%)	2(4.35%)
$8^{th} - 10^{th}$	10(21.74%)	15(32.61%)	10(21.74%)	12(26.09%)	9(19.57%)	11(23.91%)	8(17.39%)	13(28.26%)
$10^{th} - 12^{th}$	30(65.22%)	24(52.17%)	29(63.04%)	28(60.87%)	30(65.22%)	28(60.87%)	31(67.39%)	27(58.7%)
Graduate and above	3(6.52%)	4(8.7%)	4(8.7%)	3(6.52%)	4(8.7%)	3(6.52%)	4(8.7%)	3(6.52%)
No information	1(2.17%)	1(2.17%)	1(2.17%)	1(2.17%)	1(2.17%)	1(2.17%)	1(2.17%)	1(2.17%)
Experience as CHWs (years)								
0-5	5(10.87%)	4(8.7%)	4(8.7%)	4(8.7%)	4(8.7%)	3(6.52%)	4(8.7%)	4(8.7%)
5-10	19(41.3%)	16(34.78%)	20(43.48%)	16(34.78%)	19(41.3%)	16(34.78%)	17(36.96%)	17(36.96%)
10-15	17(36.96%)	19(41.3%)	16(34.78%)	18(39.13%)	16(34.78%)	19(41.3%)	19(41.3%)	18(39.13%)
Above 15	4(8.7%)	5(10.87%)	5(10.87%)	7(15.22%)	5(10.87%)	7(15.22%)	5(10.87%)	6(13.04%)
No information	1(2.17%)	2(4.35%)	1(2.17%)	1(2.17%)	2(4.35%)	1(2.17%)	1(2.17%)	1(2.17%)

based on the child's or mother's age and stage of care. Players took turns placing cards, and correct moves awarded bonus turns. The game ends when one player finishes their cards in their hand. The digital game automated the process of shuffling, distribution, and ensuring rules are followed. This reduces the cognitive load on the players. Instant feedback gives them the opportunity to learn and reflect.

We used a pen-and-paper questionnaire (Appendix B) comprising 40 multiple-choice questions covering the MCP card's contents. Each question had a single correct answer. Public health experts reviewed the questions and pilot-tested them with a small CHW group. Each survey phase—pre-test, post-test, and DRT—used the same question pool but in different orders to reduce memory bias. We translated the questionnaire into local languages to include the diversity of participants across India in the evaluation. We conducted interviews and informal discussions with CHWs to obtain qualitative feedback on their gameplay experience.

3. **Results** A mixed-design ANOVA was conducted with training condition (digital card game on smartphones, physical cards, traditional classroom session, and control with no training) as a between-subjects factor and stage (Pre, Post, Long Post) as a within-subjects factor.

   Figure 5.1 visualizes the dispersion of the scores of each group through a series of box plots.

   The analysis revealed a significant main effect of stage, $F(2, 648) = 85.58$, $p < .001$, $\eta^2 = .174$, indicating significant changes in scores over time. A significant main effect of training condition was also observed, $F(3, 648) = 37.16$, $p < .001$, $\eta^2 = .113$, suggesting overall differences between instructional approaches. A significant interaction between training condition and stage was found, $F(6, 648) = 8.75$, $p < .001$, $\eta^2 = .053$, indicating that learning trajectories differed across conditions.

   Post-hoc analyses showed that participants trained using the digital card game on smartphones exhibited the largest improvement from Pre to Post ($d = 1.83$), followed by those trained with physical cards ($d = 1.66$). The traditional classroom condition showed a smaller improvement ($d = 0.65$), while the control group exhibited only modest gains ($d = 0.65$), likely reflecting practice effects.

   At Long Post, both card-based training conditions retained significant improvements rel-

	Intervention Group-1 (IG-1) Digital Card Game		Intervention Group-2 (IG-2) Physical Card Game	
Test phase	Mean (SD)	Median (min-max)	Mean (SD)	Median (min-max)
Pre-test	25.80 (4.81)	26 (15-36)	24.35 (5.02)	24 (16-35)
Post-test	33.83 (4.03)	34 (25-40)	32.12 (4.36)	32 (21-40)
Late Post-test (After 3 weeks)	30.55 (4.04)	31 (23-39)	30.43 (4.20)	30 (23-39)
	Intervention Group-3 (IG-3) Digital Card Game		Control Group (CG) Physical Card Game	
Test phase	Mean (SD)	Median (min-max)	Mean (SD)	Median (min-max)
Pre-test	25.35 (4.30)	25 (15-34)	24.61 (5.65)	24 (10-35)
Post-test	28.05 (3.73)	28 (19-36)	27.64 (3.69)	28 (19-36)
Late Post-test (After 3 weeks)	26.25 (3.32)	26 (19-33)	24.67 (4.10)	25 (15-32)

Table 5.2: Study-2: Test Scores

Table 5.3: Mixed ANOVA Results

Effect	df	F	p	η^2
Group	3	37.16	<.001	.113
Stage	2	85.58	<.001	.174
Group × Stage	6	8.75	<.001	.053

ative to baseline (both $p < .001$), although scores declined slightly from Post. In contrast, participants in the traditional classroom condition did not show significant retention ($p = .267$), performing similarly to baseline. The control group likewise showed no sustained improvement ($p = .939$).

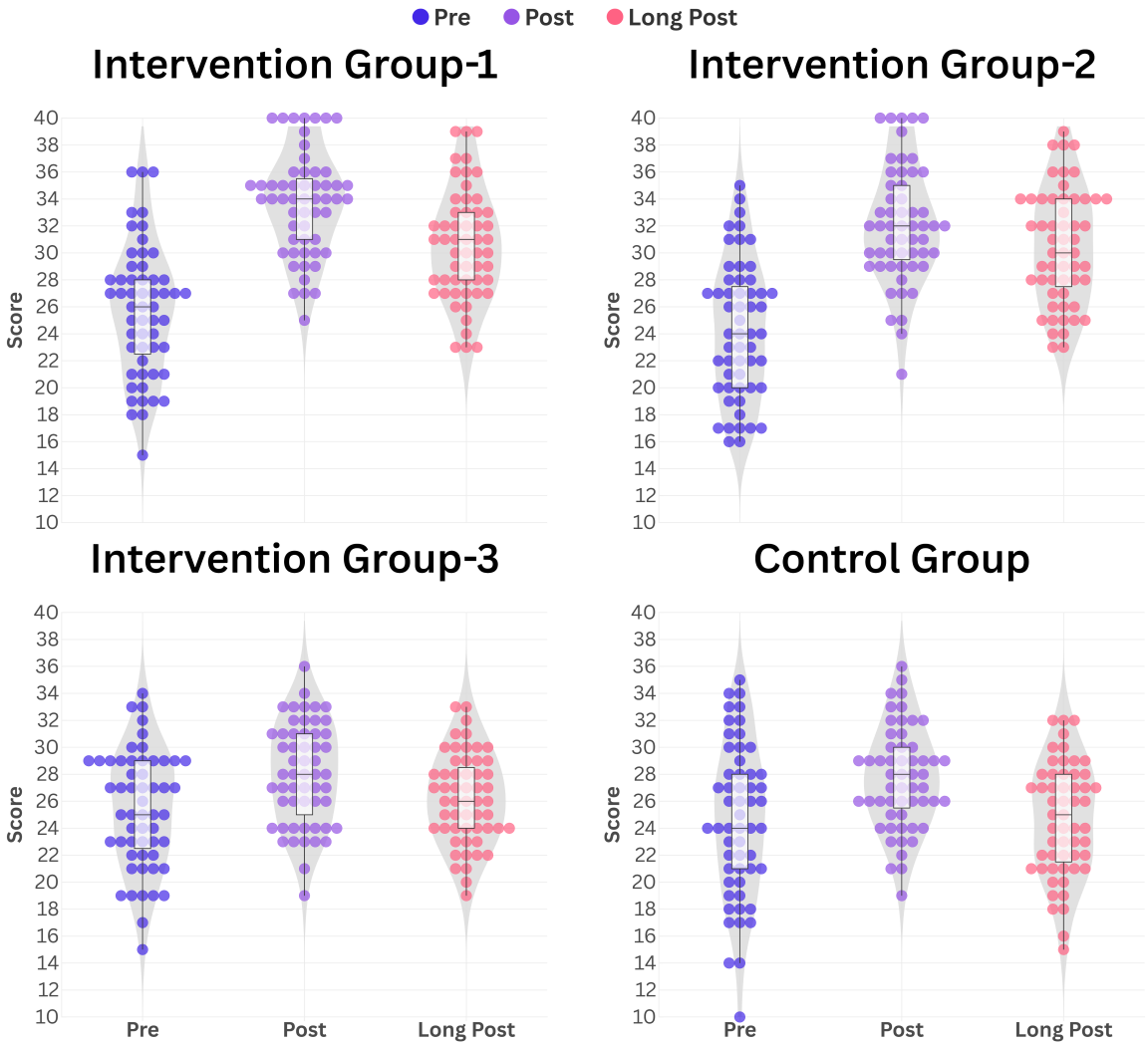


Figure 5.1: Study 2 : Results from comparing the scores of IG1 (Digital Card Games group), IG2 (Physical Card Game group), IG3 (Classroom group) and CG (Control Group)

Between-condition comparisons confirmed that, while all groups were equivalent at baseline, both the digital card game and the physical card conditions significantly outperformed the control group at both Post and Long Post. The digital card game condition consistently outperformed the traditional classroom condition, and the latter did not differ significantly from the control group at Long Post.

These findings indicate that card-based interactive training, particularly when delivered digitally, produces stronger and more durable learning outcomes than traditional classroom instruction.

4. **Discussion**

Table 5.4: Within-Group Comparisons Across Training Conditions

Condition	Comparison	*t*	*p*	*d*	95% CI
Digital card game (smartphone)	Pre–Post	-9.57	<.001	-1.83	[-2.25, -1.41]
	Post–Long	4.17	<.001	0.79	[0.40, 1.18]
	Pre–Long	-5.66	<.001	-1.08	[-1.47, -0.69]
Physical cards	Pre–Post	-8.72	<.001	-1.66	[-2.05, -1.27]
	Post–Long	2.03	.044	0.39	[0.01, 0.77]
	Pre–Long	-6.91	<.001	-1.32	[-1.70, -0.94]
Traditional classroom	Pre–Post	-3.42	<.001	-0.65	[-1.02, -0.28]
	Post–Long	2.70	.008	0.51	[0.14, 0.88]
	Pre–Long	-1.12	.267	-0.21	[-0.57, 0.15]
Control (no training)	Pre–Post	-3.39	<.001	-0.65	[-1.01, -0.28]
	Post–Long	4.03	<.001	0.77	[0.40, 1.14]
	Pre–Long	-0.08	.939	-0.01	[-0.36, 0.34]

Table 5.5: Between-Condition Comparisons (Tukey HSD)

Stage	Comparison	Result
Pre	All conditions	Not Significant
Post	Digital game > Control	***
	Physical cards > Control	***
	Digital game > Classroom	***
	Physical cards > Classroom	*
Long Post	Digital game > Control	***
	Physical cards > Control	***
	Classroom vs Control	Not Significant
	Digital game > Classroom	**

The results support both H1 and H2. Consistent with H1, knowledge gain and retention differed significantly across training modalities, with both the digital card game on

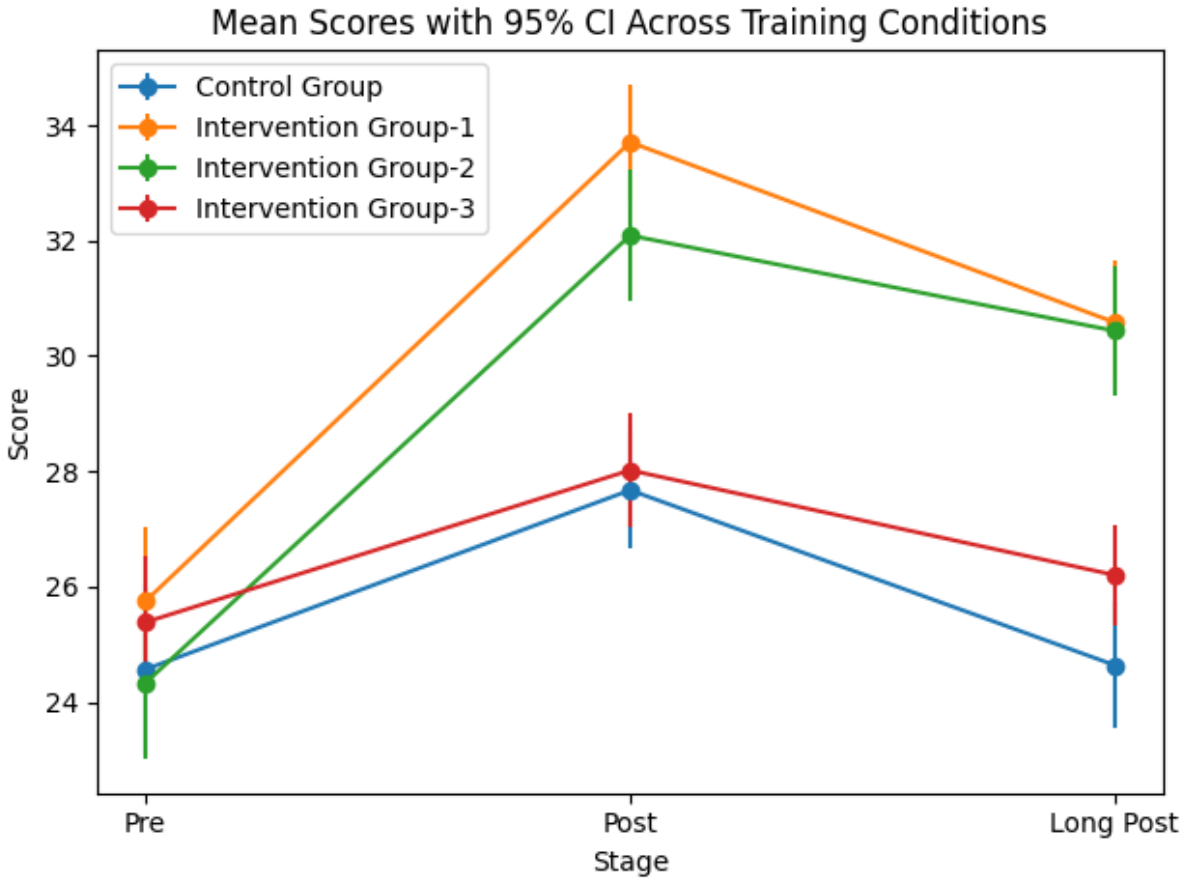


Figure 5.2: Mean scores with 95% confidence intervals across training conditions and time. Participants trained using the digital card game on smartphones showed the largest improvement, followed by those trained with physical cards.

smartphones and the physical card game producing substantially greater improvements than traditional classroom instruction and the control condition. These gains persisted at the delayed assessment, indicating sustained learning. Supporting H2, although some decline from Post to Long Post was observed, the continued advantage of the card-based conditions over baseline suggests that improvements cannot be attributed solely to a novelty effect. While the digital intervention showed the largest immediate gains—likely driven by engagement—the physical card condition demonstrated comparable retention, suggesting that interactive, game-based approaches enable deeper, more durable learning than traditional methods.

Findings in Playful Learning

Although digital and physical card games both significantly improved knowledge, several observations challenged conventional assumptions about game-based learning for CHWs.

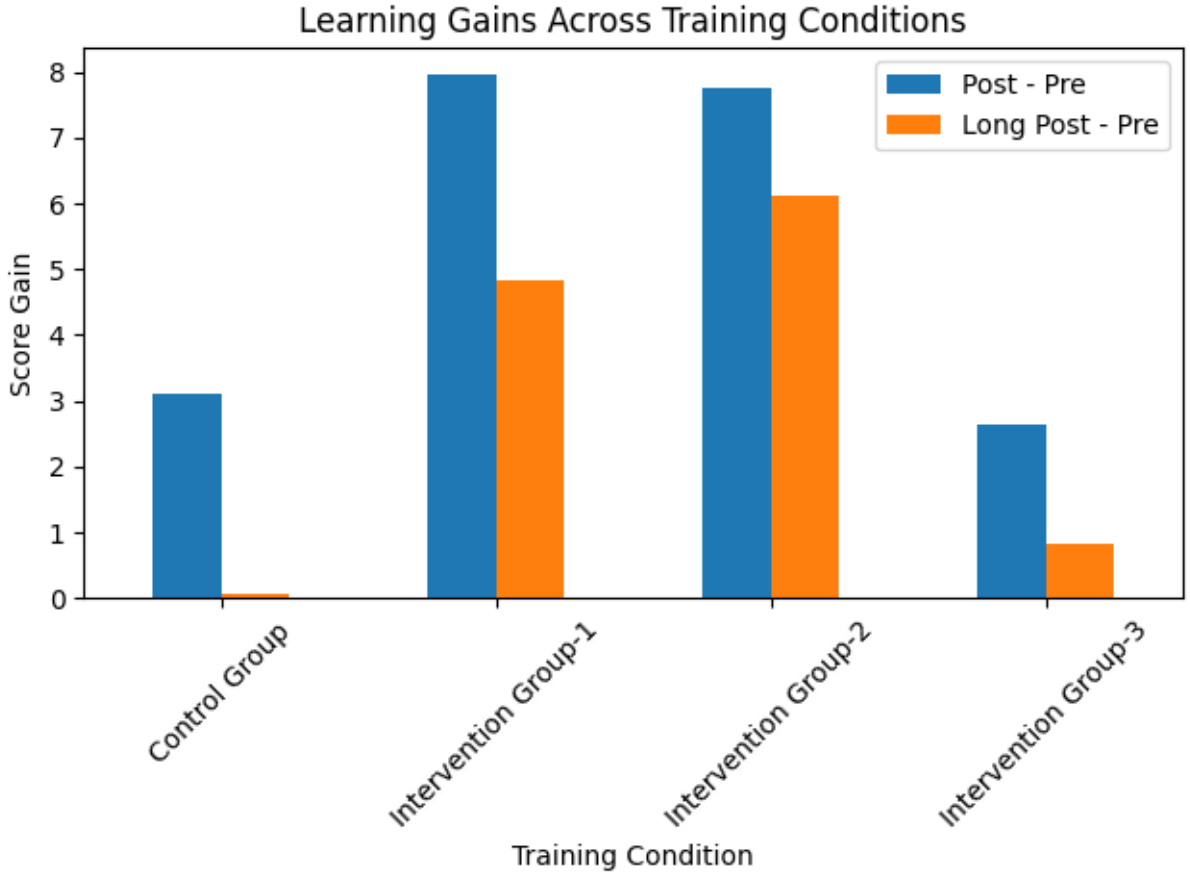


Figure 5.3: Learning gains relative to baseline. Both card-based interventions demonstrate substantially larger gains than the traditional classroom and control conditions. While some decay is observed at Long Post, card-based training retains meaningful gains, whereas classroom and control conditions show minimal sustained improvement.

(a) **Familiarity with Technology reduced Cognitive Engagement**

Some CHWs with prior experience with smartphones performed poorly in the digital game compared to expectations. Discussions with the participants later revealed that they relied on rapid chances and mindless tapping of cards, rather than carefully evaluating the sequence of the immunization schedule. A CHW remarked, “I thought I knew the order because I use phones every day, but I kept placing cards wrong because I was rushing.” This is in line with "Cognitive Load Theory" [2] (Sweller, 2022), which suggests that familiarity with technology often leads to

[2]Cognitive Load Theory (Sweller, 1988) : Technology familiarity creates overconfidence, leading to surface-level engagement where interface cues overload working memory instead of supporting deliberate content processing.

surface-level engagement and overreliance on interface cues. However, CHWs with minimal digital experience approached the game more purposefully, taking extra time to reason through each card placement, which improved knowledge retention. This is supported by dual-process theory (System-2 thinking) [3] (Kahneman, 2011), highlighting the benefits of conscious or thoughtful learning.

(b) **Highly competitive groups bargain Accuracy for Speed**

Another counterintuitive finding was that highly competitive teams of CHWs sometimes showed lower overall accuracy during early rounds of digital-based card play. While competition increased engagement and motivation, it occasionally triggered speed-over-accuracy strategies. CHWs prioritized finishing first over correctly sequencing cards. Meanwhile, we observed that CHWs playing with physical cards were less competitively driven and more collaborative in their approach. They achieved higher correctness by playing at a slower pace. This is in line with the principles of Self-Determination Theory [4] (Miller et al., 1985).

Learning - A Social Experience

The results suggest that when CHWs engage in game-based learning, they are not just having fun but learning in a way that retains them. The interactive and hands-on nature of the digital and tangible card games increased the interactivity and engagement of the training. It aligns with the Social Learning Theory [5] (Bandura, 1977).

Number Literacy and Content Standardization

Several issues came up during testing. A few CHWs struggled with numeric literacy, especially those who had studied in madrasas, where English numerals were more familiar than Hindi. We responded by changing card numbers to English digits. We also found that some vaccines mentioned in our materials were not administered in all districts, which confused participants during the evaluation. A handful of participants attempted to

[3] Kahneman's dual-process theory: System 1-driven quick moves in the game can result in immediate gains through game scores, but shallow knowledge retention without slow reflecting and reasoning driven by System 2. (Explained in detail in Theoretical Framework In the chapter Background and Literature Review)

[4] Self-Determination Theory : Intrinsic motivation and social relatedness can override extrinsic incentives like competition in supporting effective learning.

[5] Social Learning Theory: Learning is a social activity where we learn by observing, imitating, and modeling. We learn better when peers are actively engaged in the learning process.

refer to printed job aids during testing. We removed these cases from the final analysis. To prevent bias, we randomized the order of questions in later survey rounds.

Learning Experience through Games

During informal feedback sessions, most of the CHWs indicated that they enjoyed playing the game. They found it better than classroom sessions and more relevant to their work. Some teams were very competitive, which helped reinforce memorization of the order of the cards. The findings point to a broader potential shift in how we might train CHWs. Physical card games were easy to adopt as they did not require high-tech hardware or advanced literacy. They can be designed locally for local health agendas and scaled up with little technology. They offer a practical, scalable solution to this problem, even at their simplest.

5. **Conclusion**

   This study shows that learning through games with smartphones or paper cards can be a robust and feasible method for training community health workers. Both formats helped CHWs learn and retain key health information better than classroom sessions. With rising smartphone access, digital tools offer slightly better outcomes and added flexibility, which has the advantages of scalability and long-term retention. This strategy can help bridge long-standing training gaps and improve public health outcomes across India by scaling the solution. Physical card-based games can be an alternative at places where access to digital infrastructure is scarce. The findings also suggest that training approaches combining physical and digital game-based learning can be more effective.

5.2 Study-2 Reflections

The findings of this study provided a more nuanced understanding of how game-based refresher training can support knowledge acquisition and retention among CHWs. Both the digital and physical card-based game interventions produced substantially greater immediate knowledge gains than traditional classroom training and the control condition. More importantly, the card-based interventions retained significant improvements at the delayed assessment, whereas the classroom condition did not show a significant improvement over baseline after three weeks.

Although the digital version produced the largest immediate learning gain and showed somewhat stronger retention, the difference between the digital and physical modalities was not large enough to suggest that the technology itself was the primary factor behind better learning. This finding was particularly important for our understanding of game-based learning in resource-constrained settings. It suggested that the underlying interactive structure of the game may be more important than whether it is delivered via a smartphone or physical cards.

The study also challenged some of our initial assumptions about engagement. We expected familiarity with smartphones to facilitate interaction with the digital game. However, observations during gameplay suggested that familiarity could sometimes work in the opposite direction. Some participants interacted with the digital game very quickly, relying on interface cues and attempting to progress through the game rather than carefully reasoning about the immunization sequence. Similarly, highly competitive groups occasionally prioritized speed over accuracy. In contrast, participants using the physical cards often adopted a slower and more collaborative approach, discussing card placement and reasoning through disagreements. These observations suggest that engagement should not be equated with effective learning. A participant who is highly active within a game is not necessarily processing the learning content more deeply. This became an important design consideration for us: game mechanics need to encourage reflection and accuracy, rather than simply rewarding speed or successful completion.

The competitive and collaborative elements of the game created valuable opportunities for peer learning and social reinforcement. CHWs worked together, observed one another's decisions, negotiated card placements, and discussed the correct sequence. Such interactions are particularly relevant to community health work, where knowledge is frequently applied through interaction with colleagues and communities rather than in isolation. The game, therefore, provided not only a mechanism for knowledge retrieval but also a social setting in which participants could explain, challenge, and reinforce one another's understanding. This observation strengthened our view that playful learning for CHWs can benefit from being designed as a social experience rather than solely as an individual digital learning activity.

The study also raised questions about the role of novelty in game-based learning. The initial exposure to the digital game generated considerable interest, partly because of the introduction of a new technology and the availability of immediate feedback. However, as participants

became familiar with the mechanics and encountered the limited set of scenarios repeatedly, this interest appeared to level off. While the delayed-test results suggest that the learning gains cannot be explained entirely by a novelty effect, the observations indicate that novelty may still influence early engagement. This led us to reconsider how game-based refresher training should be sustained over longer periods. Rather than treating a game as a fixed intervention, future systems may need content that is periodically refreshed, more varied scenarios, and increasingly context-dependent challenges. Such variation could help maintain interest while encouraging CHWs to apply their knowledge in situations that more closely reflect the complexity of their actual work.

Another important reflection concerned the intervention's accessibility and practicality. The physical card game performed surprisingly well despite requiring no technological infrastructure, unlike the digital version. This demonstrated that effective game-based learning does not necessarily require sophisticated technology. In settings where smartphones, connectivity, or technical support may be limited, physical game-based interventions can provide a relatively simple and scalable alternative. At the same time, the digital version offers advantages in automated gameplay, immediate feedback, content distribution, and potentially easier future updates. Rather than identifying one modality as universally superior, the study therefore encouraged us to think of digital and physical interventions as complementary approaches that can be selected according to the context in which training takes place.

The study discussed several methodological and practical challenges that influenced the design of subsequent experiments. Although considerable effort was made to balance the experimental groups, the sample was recruited from a relatively constrained geographical and contextual setting. Consequently, the participants were unlikely to represent the full demographic, educational, technological, and occupational diversity of CHWs across India. The findings therefore provide strong evidence of the intervention's effectiveness within the study context, but their generalizability to other regions and populations remains limited. This limitation directly informed the design of the subsequent experiment, in which we sought to broaden the intervention's geographical and contextual reach by implementing it across more demographically diverse sites.

The four-arm experimental design also highlighted the practical difficulties of conducting controlled studies in community-based settings. Maintaining clear separation between inter-

vention groups was challenging, particularly where participants lived and worked within the same communities. There was a risk of intervention spillover, whereby participants from different conditions could discuss or share elements of the training with one another. In addition, implementing multiple conditions increased the logistical complexity of recruitment, training, coordination, and data collection. These experiences influenced the methodological decisions in the subsequent study. We condensed the design to three intervention arms and removed the no-intervention control condition. This made the study more feasible in geographically distributed community settings while allowing us to focus directly on comparisons between the active training approaches. Although this reduced the ability to compare the interventions against a no-training baseline, it provided a more practical design for examining differences between alternative training modalities under real-world conditions.

Chapter 6

Study-3: Replay, Revise, and Refresh: Smartphone-Based Refresher Training for Community Healthcare Workers in India

This study is similar to the previous study. However, in this study, the experiment was conducted in various locations in Madhya Pradesh, improving the representativeness of the population. We chose rural, tribal, and urban areas of Madhya Pradesh and Maharashtra. Mostly Hindi versions of digital and physical cards and questionnaires were used; however, in some areas bordering Maharashtra, we also used the Marathi version of the questionnaire.

6.1 Study-3 Details

Parts of this section are from a published research Study [1]

1. **Introduction**

   The prevailing concerns on training CHWs are already discussed in the chapter 1.

   This study investigates the comparative effectiveness of three refresher training approaches: standard classroom training, a physical card game, and a smartphone digital game.

   This study particularly wants to address the following research questions:

   - RQ1: Is gaming an effective replacement for standard classroom-based refresher training in enhancing CHW knowledge?
   - RQ2: Do digital and physical gameplay modalities differ significantly in terms of the immediate gains in knowledge?
   - RQ3: Do the two modalities differ significantly regarding long-term knowledge retention?

2. **Methods**

   We built the training game using 60 cards (Chapter 8.4.2). The process of designing the card deck is discussed in Appendix A.3. We printed the playing cards on card paper for the physical version, and the digital game mimicked the same format and rules of play, discussed in Appendix A.6. We developed a digital version of the card game as a smartphone application with the same content and rules of play as the physical game. We calculated the sample size using G*Power software (d = 0.5, 1-β = 0.95, α = 0.05), leading to a required sample of 88 participants per group. Considering attrition, we initially recruited 95 participants per group. 90 participants per group retained, totaling 270 participants. Each group had an equal distribution of ASHAs and AWWs (45 each).

[1] **Majhi, A.**, Agnihotri, S. B., Mondal, A., 2024. Replay, Revise and Refresh: Smartphone-Based Refresher Training for Community Healthcare Workers in India ", *HCI International 2024, Washington DC, USA* https://link.springer.com/chapter/10.1007/978-3-031-61966-3_34)

A quasi-experimental design was conducted. We divided the participants into three groups:

- IG1: Digital card game (smartphone-based)
- IG2: Physical card game
- IG3: Traditional classroom training (control group)

We conducted three assessments for each group: a baseline pre-test, a post-intervention test, and a delayed post-test after three weeks. All tests included 45 multiple-choice questions (Appendix B) focused on immunization and healthcare scheduling, shuffled across instances to reduce recall bias. We performed quantitative analysis to evaluate intra- and inter-group differences.

3. **Results**

Immediate knowledge gain was significantly greater in the digital card game group, while there was no significant difference in knowledge retention between physical and digital card game-based groups. Traditional classroom training showed significantly low retention of knowledge. The qualitative feedback indicated that CHWs liked the digital version of the card game due to its convenience and interactivity, but had reservations about distractions and the requirement of printed versions for reference.

A. Participants' Demography						
	Intervention Group-1 Digital Card Game (n=90)		**Intervention Group-2 Physical Card Game** (n=90)		**Intervention Group-3 Classroom In-Person training** (n=90)	
Parameters	ASHAs(%) (n=45)	AWWs(%) (n=45)	ASHAs(%) (n=45)	AWWs(%)) (n=45)	ASHAs(%) (n=45)	AWWs(%) (n=45)
Age group (years)						
Less than 30	1 (2.22%)	0 (0%)	1 (2.22%)	1 (2.22%)	2 (4.44%)	2 (4.44%)
30-40	22 (48.89%)	19 (42.22%)	22 (48.89%)	20 (44.44%)	23 (51.11%)	19 (42.22%)
40-50	21 (46.67%)	25 (55.56%)	20 (44.44%)	24 (53.33%)	20 (44.44%)	24 (53.33%)
No information	1 (2.22%)	1 (2.22%)	2 (4.44%)	0 (0%)	0 (0%)	0 (0%)
Education (schooling grade)						
Below 8^{th}	3 (6.67%)	2 (4.44%)	1 (2.27%)	2 (4.44%)	1 (2.22%)	2 (4.44%)
$8^{th} - 10^{th}$	9 (20%)	11 (24.44%)	9 (20.45%)	14 (31.11%)	10 (22.22%)	12 (26.67%)
$10^{th} - 12^{th}$	29 (64.44%)	28 (62.22%)	29 (65.91%)	23 (51.11%)	29 (64.44%)	27 (60%)
Graduate and above	4 (8.89%)	4 (8.89%)	4 (9.09%)	5 (11.11%)	5 (11.11%)	4 (8.89%)
No information	0 (0%)	0 (0%)	1 (2.27%)	1 (2.22%)	0 (0%)	0 (0%)
Experience as CHWs (years)						
0-5	2 (4.44%)	2 (4.44%)	4 (8.89%)	3 (6.67%)	4 (8.89%)	2 (4.35%)
5-10	18 (40%)	17 (37.78%)	16 (35.56%)	16 (35.56%)	19 (42.22%)	17 (36.96%)
10-15	18 (40%)	18 (40%)	18 (40%)	19 (42.22%)	17 (37.78%)	18 (39.13%)
Above 15	7 (15.56%)	8 (17.78%)	7 (15.56%)	6 (13.33%)	4 (8.89%)	8 (17.39%)
No information	0 (0%)	0 (0%)	0 (0%)	1 (2.22%)	1 (2.22%)	1 (2.17%)

Table 6.1: Study-3: ParticipantsDemography

Experimental Study Results	**Intervention Group-1 Test Scores**		**Intervention Group-2 Test Scores**		**Intervention Group-3 Test Scores**	
Test phase	Mean (SD)	Median (min-max)	Mean (SD)	Median (min-max)	Mean (SD)	Median (min-max)
Pre-test scores	24.85 (3.34)	24.16 (18-33)	24.28 (2.88)	24.13 (19-30)	25.02 (3.19)	24.69 (19-32)
Post-test scores	35.62 (3.57)	35.56 (28-42)	31.86 (4.1)	31.48 (24-42)	30.01 (4.32)	29.71 (19-39)
Delayed Post-test scores (After 3 weeks)	31.75 (4.69)	32.23 (22-41)	30.62 (4.1)	30.82 (21-40)	26.46 (3.54)	26.75 (19-34)

Table 6.2: Study-3: Study Results

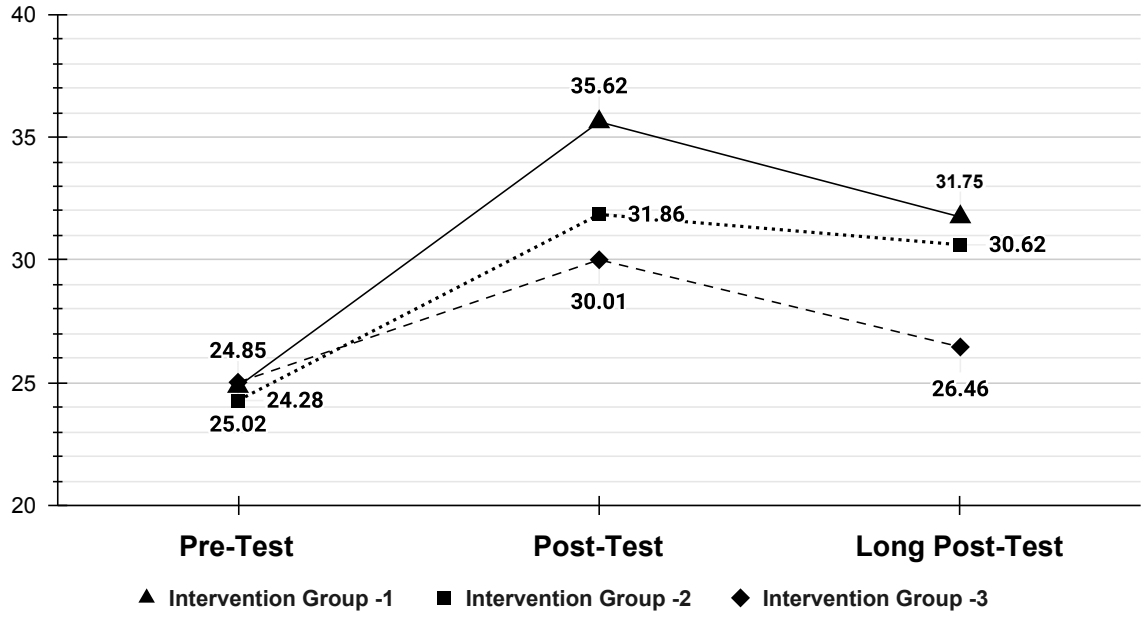


Figure 6.1: Study 3: Line chart showing trends of change of mean value in the pre-test, post-test, and delayed post-test

Pre-test scores did not differ significantly across the three groups ($F = 0.84$, $p = 0.44$), testifying to an equal baseline. Post-intervention scores differed significantly ($F = 22.96$, $p < 0.00001$). Tukey's HSD test revealed that IG1 performed significantly better than both IG2 ($F = 6.36$, $p = 0.00004$) and IG3 ($F = 9.39$, $p < 0.000001$). We did not observe

a statistically significant difference between IG2 and IG3 ($F = 3.03$, $p = 0.085$).

After three weeks, delayed post-test scores varied significantly ($F = 20.62$, $p < 0.00001$). IG1 and IG2 scored significantly higher than IG3 ($p < 0.05$). No significant difference existed between IG1 and IG2 ($F = 1.84$, $p = 0.4$), indicating equivalent retention. There were significant gains from the pre-test to the post-test for all groups. IG1 gained the most in scores, followed by IG2, then IG3. While IG1 gained the most immediately, its delayed retention dropped more significantly than IG2, which retained more steadily. IG3 consistently had the lowest scores at every phase.

4. Discussion

This study validates the hypothesis that gamified training methods, particularly smartphone-based interventions, can outperform traditional classroom training in immediate knowledge acquisition. However, the performance of digital and physical card games converged when we checked knowledge retention over three weeks. This insight underscores the potential of both gamified methods as viable tools for CHW refresher training.

Findings in Playful Learning

Although digital and physical card games both significantly improved knowledge, several observations challenged conventional assumptions and revealed nuanced interactions between digital literacy, local context, and learning behavior.

(a) **Technical Constraints hinders Cognitive Processing**

CHWS from rural and tribal areas who were in the intervention group, which used the digital card game, sometimes performed more slowly than those who were intervened with the physical card game, despite having prior experience with smartphones. Interviews from the feedback session and careful introspection revealed that limited network connectivity and small smartphone screens caused "cognitive friction", reducing engagement. This behavior is in line with Cognitive Load Theory (Sweller, 2022). [2]

One ASHA from a tribal block remarked, “I know how to use a phone, but tapping through this many screens was confusing. I liked spreading the cards on the table better.” Physical cards enabled better reasoning and collaborative discussions, which led to a reduction in extrinsic cognitive load and supported "distributed cognition" [3] (Hutchins, 1995) through visible sequencing and scaffolding from coparticipants.

(b) **Switching to Physical Cards**

There were instances where CHWs in the intervention group, who had access to the digital card game, voluntarily opted to switch to physical cards for peer learning or practicing at home. One CHW commented, “I liked the app, but with the real cards, I could show my friends and explain everything. It stays in my memory better.”

[2]Cognitive Load Theory (Sweller, 1988): Learning suffers when extraneous demands (e.g., poor interfaces) overload limited working memory alongside essential content.

[3]Distributed Cognition (Hutchins, 1995): Intelligence emerges across people/tools; physical cards externalize thought processes for group reasoning.

This behavior is in line with the principles of "Social Constructivism" [4] (Vygotsky, 1978). Collaborative dialogue between co-learners and tangible artifacts facilitates the co-creation of knowledge. These findings have important implications for public health training programs. Gamified learning can offer a scalable, interactive, and efficient approach to training. However, hybrid models that combine digital and printed materials may more appropriately address the varied needs and constraints of CHWs.

(c) **Drop in Retention of Knowledge among Digital Play Group**

CHWS in the intervention group who played the digital card game sometimes showed a sharper drop in retention compared to those in the intervention group who played the physical card game. The novelty and interactivity of the app generated intrinsic motivation (Miller et al., 1985), but in some cases also promoted overconfidence and choosing quick strategies, reducing careful thinking. A participant reflected, "I wanted to finish first, but then I realized I had misplaced some cards—I remembered better after playing slowly with my friends." This gels with Kahneman's dual-process theory [5] (Kahneman, 2011).

(d) **Context-Dependent Perception of Games**

We observed that urban CHWs sometimes treated the digital game as a leisure activity. However, rural and tribal CHWs engaged with it as a formal learning tool. One rural CHW explained, "Playing on the phone with my community CHWs felt serious here, but when I go for training and play it with the city CHWs, it seemed like a game to pass the time." This underscores that learning outcomes are co-determined by social, cultural, and environmental factors, rather than being solely determined by the modality itself. This is in line with Situated Learning Theory [6] (Lave and Wenger, 1991).

[4] Social Constructivism: Knowledge builds through social interaction within the zone of proximal development, where peers scaffold understanding via shared tools and dialogue.

[5] Kahneman's dual-process theory: System 1-driven quick moves in the game can result in immediate gains through game scores, but shallow knowledge retention without slow reflecting and reasoning driven by System 2. (Explained in detail in Theoretical Framework In the chapter Background and Literature Review)

[6] Situated Learning Theory (Lave & Wenger, 1991): Knowledge emerges through participation in "communities of practice," where context shapes meaning-making. Rural CHWs' serious community setting fostered legitimate peripheral participation; urban mixed groups diluted it into casual play.

5. **Conclusion**

   This study shows that training using smartphones could be an effective method for refresher training, by enhancing the knowledge retention of CHWs. Though the digital game showed a higher short-term gain in knowledge, long-term retention was almost similar in both digital and physical card games. Evidence also shows that game-based training can complement conventional training programs in low-resource contexts. Future research can investigate long-term effects, motivational determinants of the use of games, and integration of gamified learning into routine CHW workflows.

6.2 Study-3 Reflections

The results of this study strengthened our belief that game-based learning can provide a meaningful alternative to conventional classroom-based refresher training for CHWs. Both the physical and digital game groups showed greater improvements in knowledge than the classroom group, with the digital gameplay group demonstrating the highest immediate gain. These findings support our initial assumption that interactive and engaging learning formats can be particularly useful when the objective is immediate knowledge enhancement. At the same time, the results also made us more cautious about treating immediate performance as the primary indicator of learning.

The three-week delayed post-test provided an important perspective. Although both game-based groups performed significantly better than the classroom group, there was no significant difference between the digital and physical groups in long-term retention. This was somewhat unexpected, particularly because the digital version produced the strongest immediate improvement. It made us reflect on the distinction between being engaged in an activity and actually retaining what has been learned. The digital game may create a more stimulating and convenient learning experience, but stimulation alone does not guarantee stronger long-term learning. In this sense, the study encouraged us to look beyond a game's initial effectiveness and consider how its design supports reflection, repetition, and continued engagement after the intervention.

The focus group discussions further complicated our understanding of what makes a game-based intervention effective. We analyzed the written notes from the discussions manually us-

ing thematic analysis (Braun and Clarke, 2006, 2019), informed by grounded theory guidelines (Charmaz, 2014). Through repeated reading, initial coding, and identification of broader patterns, we found that participants' experiences were shaped by several sociocultural and contextual factors. Some CHWs appreciated the smartphone game's independence, interactivity, and convenience. Others preferred physical cards because they were tangible, easier to handle, and could be used as counseling materials during field visits. These responses reminded us that the value of a learning intervention is not determined only by its learning outcomes. Its usefulness also depends on whether it fits naturally into the everyday practices of those expected to use it.

The issue of smartphone distraction was particularly revealing. We initially viewed the smartphone primarily as a convenient platform for delivering the game. However, in practice, the same device that enabled access to the intervention could also introduce distractions, particularly when smartphones were shared among CHWs. This made us reconsider the common assumption that moving a learning intervention to a digital platform automatically makes it more accessible or scalable. Digital delivery can remove some barriers, but it can also introduce new ones. The experience reinforced the importance of considering the complete learning environment rather than evaluating the technology in isolation.

The physical card game also changed our perception of what constitutes a "low-tech" intervention. Before conducting the study, it could have been easy to view the physical version as simply a less sophisticated alternative to the smartphone application. However, the experiences of the CHWs showed that its simplicity was also one of its strengths. Physical cards could be handled collectively, discussed with peers, and taken into the field as a tangible resource. Their lack of technological requirements meant that they could function even when connectivity, device availability, or digital familiarity became concerns. This led us to think differently about innovation in low-resource contexts. Sometimes, an intervention becomes more useful not by adding technology, but by removing unnecessary technological barriers.

At the same time, we do not see these findings as an argument against digital game-based learning. The smartphone version has clear advantages in terms of scalability, automated feedback, progression tracking, and the potential for wider dissemination through public health platforms or app-based systems. These advantages become particularly important when training needs to reach large numbers of CHWs across geographically dispersed locations. What the study made clearer to us is that scalability and contextual appropriateness are not always the

same thing. A digitally scalable intervention may still require physical or social components to work effectively within particular communities.

This study, therefore, shifted our thinking from choosing between digital and physical learning toward considering how the two can complement each other. A hybrid model may allow the digital game to provide convenient, repeatable refresher learning, while physical cards support peer learning, discussion, and field-based use. Rather than replacing classroom training entirely, these interventions could also function as reinforcement mechanisms that help CHWs revisit and consolidate knowledge after formal training. Such an approach may be particularly relevant for CHWs whose learning does not occur solely in training rooms but is continuously shaped by their interactions with colleagues and communities.

At a broader level, the study also made us reflect on the potential role of these interventions within public health systems. Low-cost, culturally and contextually targeted game-based training could provide a practical way of reinforcing knowledge and confidence among CHWs between formal training sessions. If integrated thoughtfully into existing training structures, such interventions may contribute to greater consistency in knowledge and, ultimately, better delivery of community health services. However, our findings also caution against assuming that a single technological solution can be applied uniformly across communities.

Chapter 7

Experimental Studies

7.1 From Knowledge Acquisition to Field Performance

The first three experimental studies established the potential of game-based interventions to improve knowledge acquisition, retention, motivation, and engagement among Community Health Workers (CHWs). Across different game formats and learning contexts, the findings consistently suggested that active and interactive learning could provide advantages over conventional training approaches. However, these studies also revealed an important limitation in evaluating the effectiveness of such interventions. Improvements in knowledge and test performance do not necessarily indicate that the acquired knowledge is transferred to routine healthcare practice.

This distinction is particularly important in the context of CHWs. Their work extends beyond learning health-related information and includes the regular application of that knowledge during household visits, data collection, counseling, and community-level health activities. The question remained unanswered whether improved learning through game-based training translates into better performance in actual field activities.

This question motivated the next stage of the research. Rather than evaluating game-

based interventions solely through conventional pre- and post-test scores, the subsequent study examined whether such an intervention could influence measurable aspects of CHW field performance. In particular, the study explored the use of location-based game mechanics to support and monitor routine data-collection activities. This provided an opportunity to examine whether the motivational and engagement mechanisms identified in the earlier studies could extend beyond the training environment and contribute to sustained field-level performance.

At the same time, the earlier studies raised another question concerning the applicability of game-based refresher training to learners at different stages of healthcare education. The effectiveness of a refresher intervention may depend substantially on whether learners already possess foundational knowledge of the subject. To examine this issue, we subsequently explored the intervention with pre-service healthcare trainees who had received different levels of formal instruction. This allowed us to distinguish between the intervention's effectiveness for initial learning and its potential value as a refresher-learning mechanism.

Thus, the subsequent studies extended the research in two related directions. The first examined the transfer of game-based learning from training to field performance, while the second examined the intervention's applicability across different levels of healthcare training and varying levels of prior knowledge. Together, these studies moved the research beyond the question of whether games can improve test scores toward the broader question of whether game-based learning can support meaningful and sustained performance in healthcare practice.

7.2 Experimental Study - 1: Mapping Child Malnutrition and Measuring Efficiency of Community Healthcare Workers through Location-Based Games in India

The preceding studies demonstrated that game-based learning could improve knowledge acquisition and, in some cases, knowledge retention among CHWs. However, the practical value of such interventions ultimately depends on whether these learning gains translate into improved performance in routine field activities. This study, therefore, shifted the evaluation from primarily educational outcomes to field-level performance.

CHWs routinely collect health information during household visits, and the quality, coverage, and timeliness of this data can directly influence community-level health planning. Anthropometric measurements of children are particularly important for identifying conditions such as stunting, wasting, and underweight. Yet these activities can be difficult to perform consistently, particularly in geographically dispersed or hard-to-reach communities. Conventional data-entry applications may facilitate recording information, but they do not necessarily provide mechanisms that encourage CHWs to visit households consistently, monitor their progress, or sustain engagement with data-collection activities.

The availability of smartphones and location-based technologies provides an opportunity to address this limitation. Rather than treating the mobile device solely as a data-entry tool, a location-based game can connect data collection with navigation, task completion, exploration, feedback, and achievement. Such mechanisms may encourage CHWs to reach assigned households and complete measurements, while simultaneously generating geographically referenced information to support health surveillance.

The study, therefore, examined whether transforming routine anthropometric data collection into a location-based game could improve CHWs' measurement efficiency compared with a conventional data-entry application. It also examined whether any improvement could be sustained over time. The study addressed the following research questions:

RQ1: Does implementing a location-based game improve the measurement efficiency of CHWs more than a conventional data entry application?

RQ2: Is the efficiency gained through the game-based intervention maintained in the long term?

The Measurement Efficiency Index (MEI) is a composite score ranging from 1 to 100 that measures the effectiveness of Community Health Workers (CHWs) in performing anthropometric data-collection tasks, such as measuring height, weight, and mid-upper arm circumference (MUAC) of newborns during household visits. It combines three main parts, each scored from 0 to 100, with equal weight: MEI = (Accuracy + Coverage + Timeliness)/3. Accuracy checks the precision of measurements by comparing CHW readings to re-measurements performed by a local supervisor who shadows the CHW and verifies random cases on-site (for example, full points are awarded if the weight is within measurement tolerance, with partial scores given

for minor errors). Coverage measures how completely CHWs visit assigned households, using GPS data from the game to calculate the percentage of eligible children reached (with extra points for hard-to-reach rural or tribal areas). Timeliness scores are based on app timestamps, including the average time per child (target under 5 minutes) and visits completed within due dates, benchmarked against the group average. This index offers a clear and objective way to compare game-based training with traditional methods.

Data collection happens through the Android geolocation game app. CHWs receive a list of tasks to perform daily. They use GPS to navigate to homes, pin locations, scan child IDs, enter measurements with photos of tools, and get instant feedback. The app syncs data to a central dashboard when online, building community heatmaps of stunting, wasting, and underweight rates, while awarding game points for streaks and exploration. Supervisors use a linked app to log re-checks anonymously.

We recruited around 200 CHWs across multiple Indian states following convenience sampling, and 94 participants per group completed the study. We conducted a quasi-experiment with 188 CHWs in two groups: one using the game-based app and the other using a typical non-game data entry app. There were pre- and post-intervention tests, and a delayed retention test (DRT) three months after the intervention to assess long-term engagement and data-collection efficiency.

The intervention group (geolocation game app) demonstrated statistically significant improvements in measurement efficiency compared to the control group (regular data collection app) ($p = 0.00004$; Cohen's $d = 1.6$). While both groups showed a reduction in scores on the delayed post-test, the IG maintained a significantly higher performance level than the CG, supporting both RQ1 and RQ2. Mean scores for the IG improved from 49.04 at baseline to 73.9 post-test and slightly declined to 69.14 at DRT. In contrast, the CG scores moved from 51.46 to 54.84 post-test and returned to 52.58 in DRT.

The findings indicated that the game-based application significantly improved the efficiency of data measurement among CHWs compared to the traditional non-game app. The game-based model showed better engagement and retention. CHWs reported higher motivation to gather data in untapped regions. The study found that initial motivation for the game-based application gradually declined, indicating a need for sustained reinforcement and support to achieve long-term engagement.

The findings also prompted a further question about the scope of the intervention. If game-based learning is intended not only as a field-support tool but also as a training mechanism, its applicability should be examined across different stages of healthcare education. In particular, it was necessary to determine whether the intervention functioned primarily as an initial learning tool or whether its value was greater when learners already possessed foundational knowledge and could use the game to reinforce previously acquired information. This motivated the subsequent study with pre-service healthcare trainees.

7.3 Experimental Study – 2: Pilot Testing Game-Based Training with Pre-Service Health Trainees

The preceding field study demonstrated the potential of game-based mechanisms to support healthcare-related activities beyond conventional classroom learning. However, it primarily focused on practicing CHWs. A complementary question was whether similar game-based approaches could be incorporated into the formal education of future healthcare workers. Exploring this question was important for two reasons. First, pre-service training provides an opportunity to introduce interactive learning before trainees enter routine field practice. Second, comparing learners with varying levels of prior knowledge can help clarify whether the intervention is better suited to initial learning or to reinforcing knowledge that has already been acquired.

The study, therefore, focused on nursing trainees at different stages of their formal education. Particular attention was given to learners who had already received immunization instruction, since the broader purpose of the intervention was to examine its potential as a refresher-training tool rather than as a substitute for foundational instruction. Including students with different levels of prior exposure also provided an opportunity to identify whether baseline knowledge influenced the magnitude of learning gains.

The first-year ANM batch (n=39, Average Age=19 years, SD=2.28) had recently joined the course in the last two months. They had little or no prior formal knowledge about immunization and consequently relied heavily on guesswork while completing the pre-test questionnaire. Conducting t-tests, we found no significant difference between the pre- and post-test scores

of the physical card group [p=0.197, average improvement in scores = 4.6], but a significant difference was observed in the digital group [p=0.0009 ($p < 0.001$), average improvement in scores = 16.5]. Although the improvement in the digital group was promising, this cohort did not closely align with the intended purpose of evaluating the intervention as a refresher-training tool, as participants had not yet developed sufficient foundational knowledge.

The second-year ANM batch (n=29, Average Age=20.25 years, SD=2.25), who had prior introductory coursework in immunization, showed significant differences in both the physical card group [n=19, p=0.003, average improvement in scores = 3.63, SD=4.7] and the digital app group [n=10, p=0.002, average improvement in scores = 8.5, SD=6.4]. The digital app group showed higher gains, but with larger variation. A t-test comparing the card and digital app groups showed no statistically significant difference (p=0.051), indicating a marginal trend but insufficient evidence to establish superiority of one intervention over the other.

The first- and second-year B.Sc. Nursing batches were excluded as they had not yet been taught immunization in detail, and fourth-year students were unavailable due to clinical placements. Therefore, the study focused on the third-year B.Sc. Nursing batch (n=35, Average Age=21 years, SD=1.14), who had prior exposure to immunization coursework. Significant differences between pre- and post-tests were observed in both the physical card group [n=25, p=0.0000007 ($p < 0.001$), average improvement in scores = 9.16, SD=6.88] and the digital app group [n=10, p=0.00005 ($p < 0.001$), average improvement in scores = 12.3, SD=5.38]. Although the digital app group showed higher average gains, a t-test comparing the differences between groups indicated no statistically significant difference (p=0.17). Thus, while both interventions appeared effective as refresher-training tools, the results did not provide sufficient evidence to conclude that the digital intervention was superior to the physical card-based intervention.

7.4 Reflections on Experimental Studies 1 and 2

These two studies marked an important shift in the direction of this research. The earlier experimental studies had primarily examined whether game-based interventions could improve knowledge acquisition, retention, motivation, and engagement. Studies 4 and 5 extended this

Figure 7.1: Gameplay session with nursing students

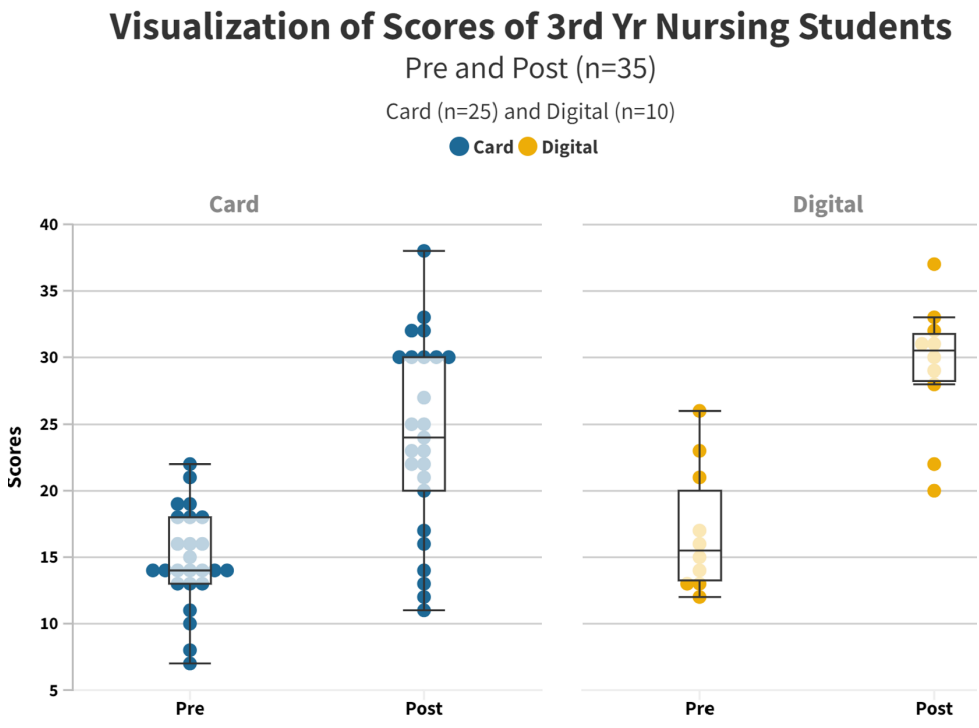
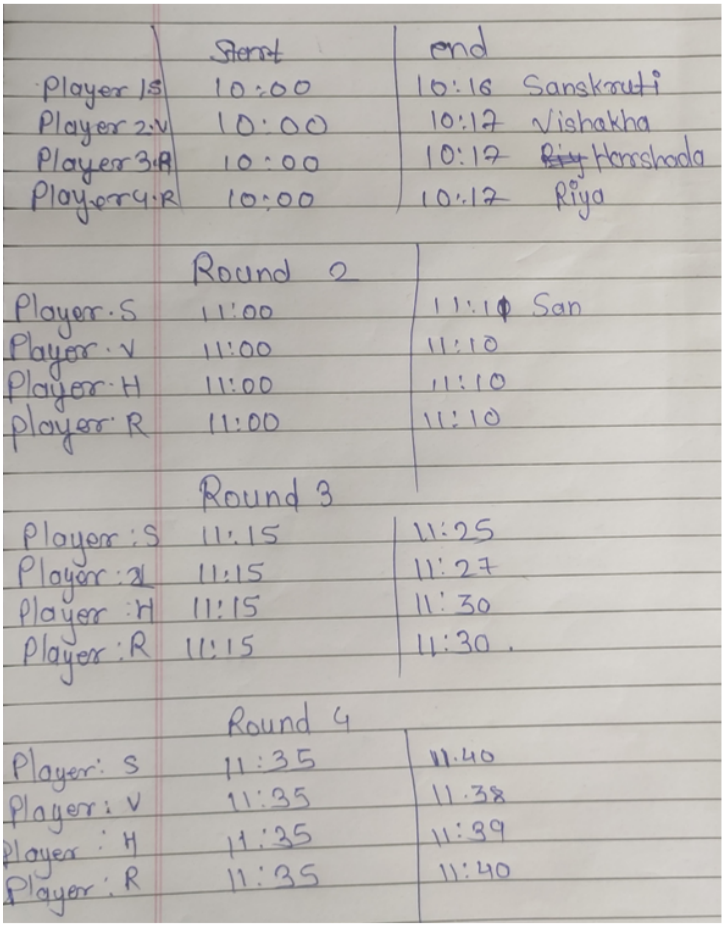


Figure 7.2: Results from comparing the scores of nursing students



	Start	end	
Player 1:S	10:00	10:16	Sanskruti
Player 2:V	10:00	10:17	Vishakha
Player 3:H	10:00	10:17	~~Riya~~ Harshada
Player 4:R	10:00	10:17	Riya

Round 2		
Player:S	11:00	11:10 San
Player:V	11:00	11:10
Player:H	11:00	11:10
player:R	11:00	11:10

Round 3		
Player:S	11:15	11:25
Player:V	11:15	11:27
Player:H	11:15	11:30
Player:R	11:15	11:30

Round 4		
Player:S	11:35	11:40
Player:V	11:35	11:38
Player:H	11:35	11:39
Player:R	11:35	11:40

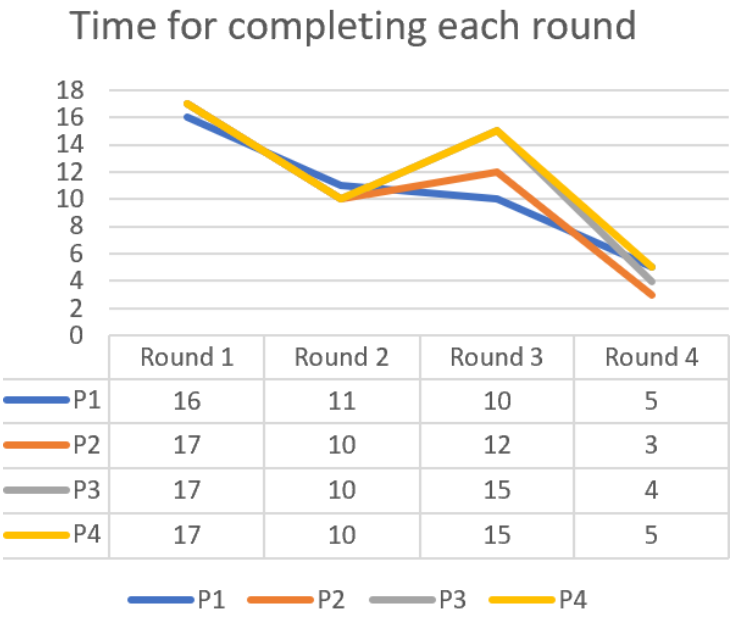


Figure 7.3: Decrease in time taken to complete a round showing mastery

inquiry by asking two related questions: whether game-based mechanisms could support performance beyond the classroom and whether the intervention remained useful when introduced to learners at different stages of healthcare education.

The Experimental Study-1 was particularly important in demonstrating that the value of game-based learning need not be restricted to test-based learning outcomes. By embedding data-collection activities within a location-based game, the intervention connected learning with the actual working environment of CHWs. The improvement in the Measurement Efficiency Index suggests that the game-based approach may have encouraged CHWs to engage more consistently in routine data collection. The large effect size also indicates that the difference was not merely statistically significant but potentially meaningful in practical terms. This was encouraging because it provided some evidence that the motivational mechanisms of games could extend into real-world healthcare activities.

At the same time, the delayed results complicated the findings. Although the game-based group retained substantially better performance than the conventional application group after three months, its score still declined from the immediate post-test. This suggests that introducing game elements may provide an initial motivational advantage without necessarily guaranteeing sustained engagement. In retrospect, this was an important finding for the research. It shifted my understanding of gamification from seeing it primarily as a mechanism for increasing motivation to viewing it as something that requires continued reinforcement, meaningful feedback, and contextual support. A game can make an activity more engaging, but engagement itself may decline as the novelty of the intervention wears off.

The location-based nature of the intervention also offered an important insight. In this setting, game mechanics were not simply added to an existing training application as decorative features. Location, exploration, task completion, and feedback were connected to the actual work that CHWs were expected to perform. This suggests that the effectiveness of gamification may depend not only on which game elements are used but also on how closely those elements are connected to the underlying activity. The findings therefore encouraged a more contextual understanding of game-based learning, where the game mechanics are designed around the realities of the users' work rather than applied as an additional layer.

The Experimental Study-2 provided a different kind of reflection. Initially, I expected the intervention to be reasonably transferable to pre-service healthcare trainees because nurs-

ing students and CHWs share several learning and practice requirements. However, the results showed that learner background mattered considerably. The first-year ANM students, who had little prior knowledge of immunization, were not an appropriate population for evaluating the intervention specifically as a refresher-training tool. Their experience highlighted an important distinction between using a game to introduce new knowledge and using it to reinforce knowledge already acquired.

The second-year ANM and third-year B.Sc. Nursing results were more closely aligned with the intervention's intended purpose. Both the physical and digital versions led to significant improvements among learners with prior exposure to immunization. Although the digital groups generally showed larger numerical gains, the between-group comparisons did not establish statistically significant superiority of the digital intervention. This was an important corrective to an assumption that digital delivery would necessarily be more effective than a physical game. The findings instead suggest that the underlying game-based learning experience may be more important than the delivery medium itself.

The relatively small sample sizes within the individual intervention groups also require caution when interpreting these findings. In particular, the digital groups were substantially smaller than the physical card groups in some cohorts. The absence of a statistically significant difference between the two formats should therefore not be interpreted as evidence that the formats are equivalent. Rather, the results provide preliminary evidence that both formats can support refresher learning, while leaving the question of their relative effectiveness open for further investigation.

Taken together, these studies changed the way I viewed the role of game-based interventions in healthcare training. The Experimental Study-1 suggested that game mechanics could potentially influence behavior and performance in real-world settings, while the Experimental Study-2 showed that the effectiveness of a game-based learning intervention is closely related to learners' prior knowledge and the purpose for which the intervention is introduced. These findings reinforced the idea that gamification should not be treated as a universal solution. Its value depends on the activity, the learner, the surrounding context, and the mechanisms through which game elements are connected to the desired behavior.

Perhaps the most important reflection from these studies was that improvement in immediate outcomes is only one part of evaluating a game-based intervention. Sustained engagement,

transfer to practice, learner background, and contextual fit are equally important. This realization shaped the subsequent development of the research and encouraged a shift from simply asking whether games work to asking when, for whom, and under what conditions game-based interventions are most useful.

Chapter 8

Development of the Game

8.1 From Guidelines to Game Concept

Initially, we began with content focusing on the health and nutrition of both the mother and child. Abstract nutrition advice in Indian guidelines, such as "balanced diet" or "gradual complementary feeding", inherently resists binary game outcomes (win/lose), as real counseling involves cultural variability and no fixed "correct" meal. Nutrition guidelines in India, such as those in ASHA Modules 6 and 7, often emphasize broad dietary mixes (e.g., cereals, pulses, greens) without precise quantities or unambiguous rules, which complicates the derivation of game rules. In contrast, ANC, PNC, and immunization follow prescriptive schedules (e.g., 4+ ANC visits, specific postnatal checks on days 3, 7, 14, 28, 42), which suit the incremental scaffolding of content and decision-making mechanics. The game-based learning literature notes similar issues, where ambiguous health content could lead to superficial mechanics unless simplified, thereby risking an ambiguous knowledge transfer to CHWs.

Health module as Scope of Gamification

During the course of this research, it became evident that gamification requires a well-

defined and unambiguous set of rules or protocols for participants to follow. In this regard, the procedures and protocols within the health module, such as immunization schedules and medication supplements for children and mothers, were found to be more clearly codified compared to those related to nutrition. Given this distinction, the gamification efforts were initially confined to the refresher training syllabus for ASHA workers, focusing on health, to ensure clarity and alignment with established health protocols.

The primary challenge was converting detailed healthcare guidelines on child immunization, ANC, and PNC into playable game mechanics without compromising clinical fidelity. The raw content was in the form of tables, checklists, and training slides, which do not map directly to meaningful card actions or win conditions, a difficulty also noted in other educational card-game projects and research. The design task was to identify "gameable" patterns, such as age windows, sequences, and common disruptions, and decide what could become card types, suits, or event triggers. Early drafts attempted to represent every nuance of the guidelines, resulting in a complex list of card types and conditions. The playable structure was missing. There was no clear sense of what a "round" was, what players were trying to optimize, or how a session would end.

We resolved this issue by revising the core metaphor of the game. Play rounds represent months (the time period during which the vaccine or care is administered), and the win condition is completing the correct care sequences for children and mothers with minimal missed opportunities. Guideline details that did not translate well to simple actions were shifted to facilitator notes and debrief discussions, rather than being forced into the game rules. All the information was categorized into four silos: children below 1 year, children above 1 year, ANC, and PNC, but with an important difference.

8.2 Development of the Augmented Reality Game

8.2.1 Idea of game

CHWs can play in single-player or multiplayer mode (based on group size), focusing on modules such as pregnancy (ANC) and child immunization. Before starting, players stick 10-20 printed child and mother cards from the kit onto room walls at a height easy to scan with smartphones, while carrying bags with physical vaccine and medicine cards.

In single-player mode, the goal is to immunize all children and care for mothers correctly and efficiently according to the schedule. Players scan a wall card (eg. 0-month child) to display a 3D model with age details, then scan matching cards (BCG, HepB, OPV, colostrum) for points and smiling sounds. Wrong cards result in crying sounds and point loss. Missed cards also have negative points from all players or teams. Play runs in 4 sequential rounds where children age (e.g., $1\frac{1}{2}$ to $2\frac{1}{2}$ months), with scores shown on a leaderboard.

In multiplayer mode, 4 players form teams of 2 with randomized card bags. Teams trade cards within or between groups to finish rounds; hoarding slows others, but cooperation leads to higher team scores and wins. Players choose roles freely, such as one scanning while the other trades cards.

8.2.2 Design of the card deck

Unity served as the primary game development platform, with Vuforia integrated for augmented reality (AR) functionality. Card images were uploaded to the Vuforia server as image markers, requiring unique visual designs to enable distinct detection by the vision model. Text modifications alone proved insufficient for reliable recognition, necessitating graphic variations for each vaccine or information card to ensure accurate AR tracking.

Cards were designed to integrate textual statements with illustrations for effective information delivery, featuring graphics, titles, and brief descriptions. Images proved more effective

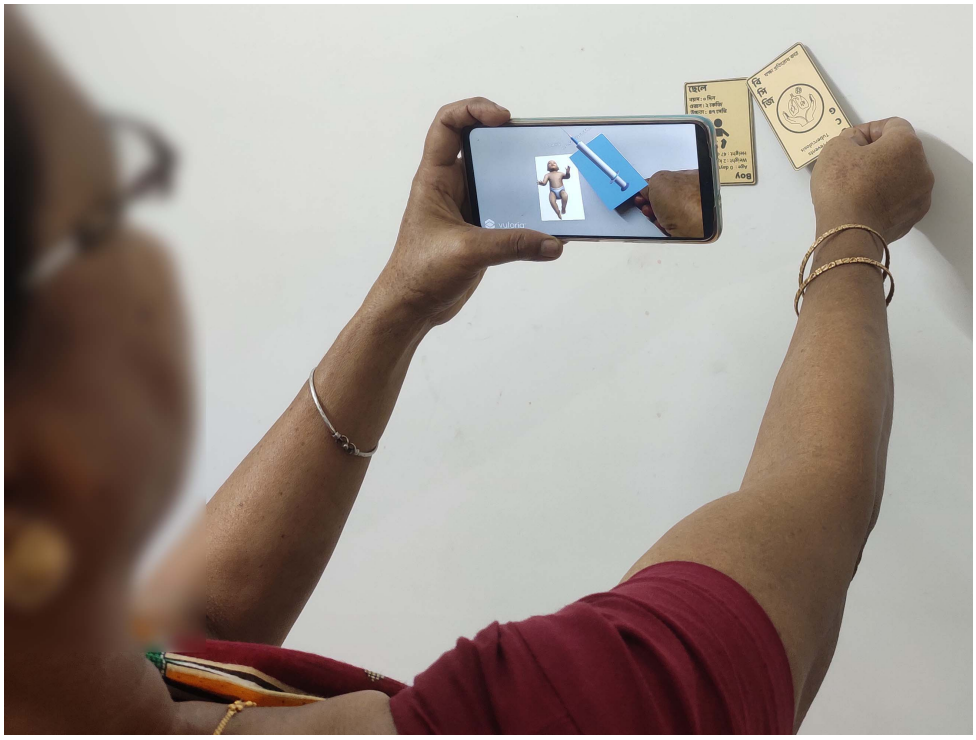


Figure 8.1: ASHAs play-testing the Tikakaran-AR play app

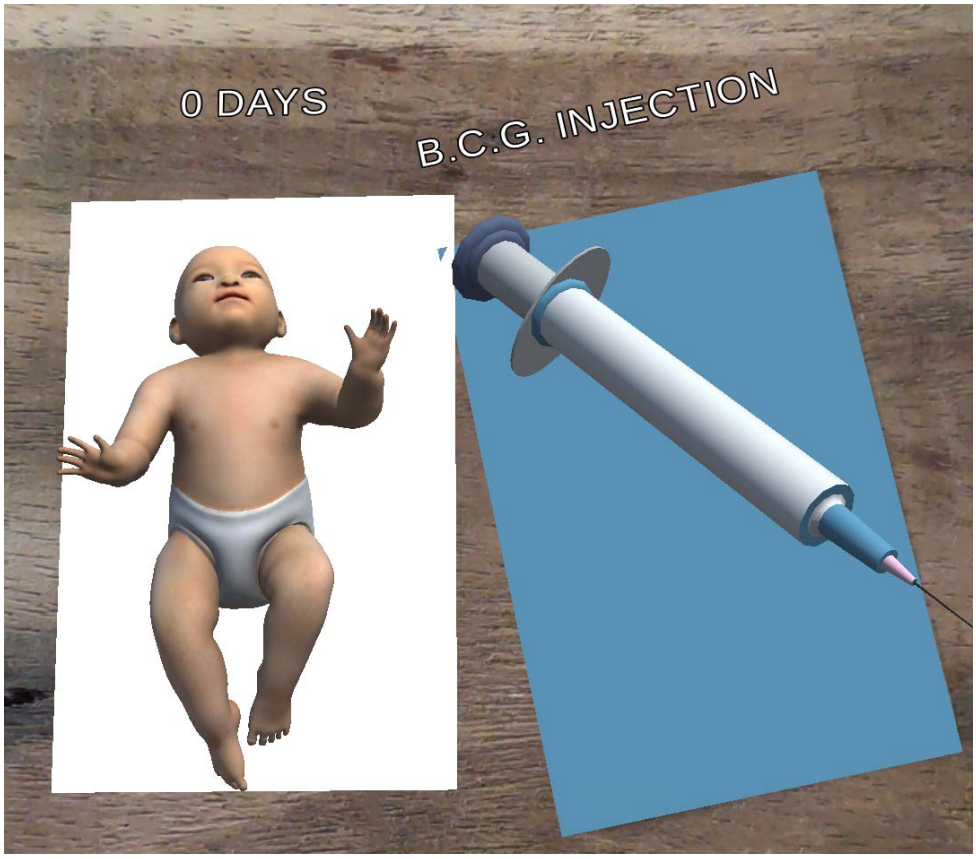


Figure 8.2: Child 3D model and Vaccine representation in Augmented Reality

Figure 8.3: CHWs playing in teams of two - one holds the smartphone while the other shows the card

Figure 8.4: CHWs playing in teams of two with physical playing cards

than text alone for straightforward communication, enhancing recall, reducing language barriers, and increasing self-explanatory value. In focus group discussions (FGDs), participatory design exercises involved Community Health Workers (CHWs) selecting royalty-free illustrations resembling specific vaccines, medicines, or conditions. The majority's preferences determined final choices, supplemented by CHW sketches for contextual relevance.

Bengali was selected as the primary language for card text, targeting field studies in rural and suburban West Bengal. The deck comprised 20 unique cards: 10 on immunization, 5 on child medicines/supplements, 3 on maternal medications/immunization, and 2 sets depicting boys/girls across ages, heights, and weights. Cards measured 5.5 × 8.5 cm (standard playing card or credit card size), printed with graphics and information.

Figure 8.5: Card design for AR play and physical card play

8.3 Development of the Sequence Game

Insights from the Augmented Reality game prototype indicated that matching age-appropriate vaccines should be retained across various game formats. However, the games must remain simple for use by novice smartphone users and elderly women.

The sequencing game presents bins representing different age groups with vaccines in randomized order; players drag vaccines into the correct bins. On dragging and dropping the items, the list automatically rearranges, and correctly positioned items change to green to provide immediate visual feedback to the learner. A five-minute countdown timer is displayed to create a sense of urgency and encourage the timely completion of the task.

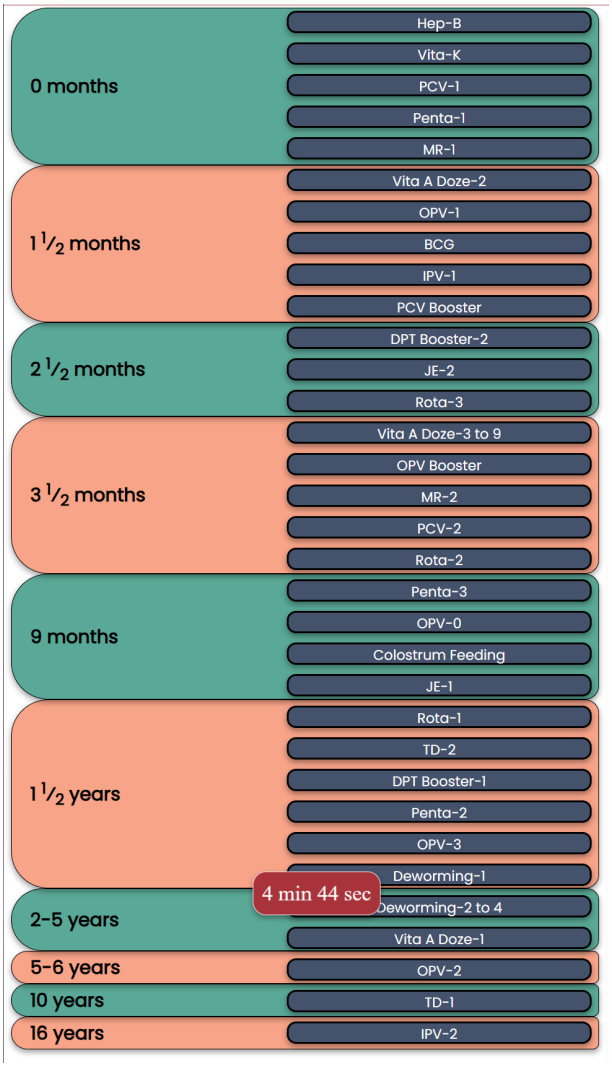


Figure 8.6: Gamestart - Screenshot of the Sequencing Game

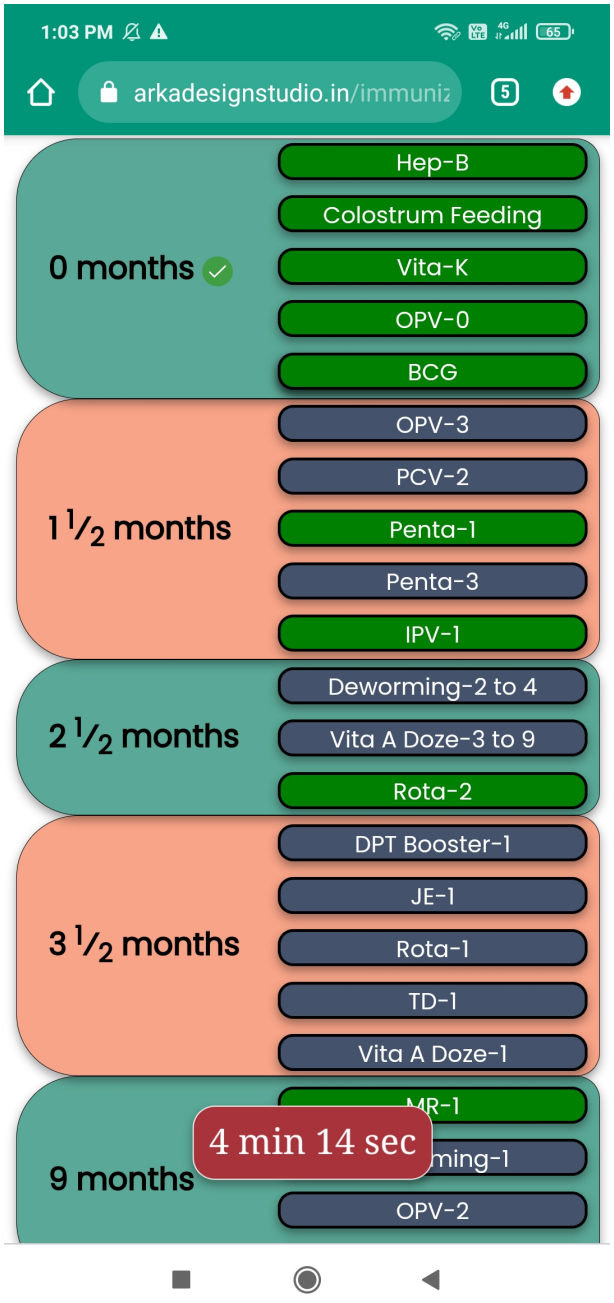


Figure 8.7: Upon sequencing the first bin of birth age (0 months) - Screenshot of the Sequencing Game

Out of the full code, we will discuss a few important sections that make the major decisions in the game design. The link to the GitHub repository and the Playable Live link to the game are in the footnote [1] [2].

A large interactive container was designed to hold smaller draggable items representing different vaccines. These items can be repositioned and arranged according to the immunization schedule. The schedule consists of ten stages of interventions, and each stage may include multiple vaccines, denoted as A, B, C, D, and E.

```
<div id="draggableContainer">

    <div id = "1A" draggable="true">OPV-0</div>
    <div id = "1B" draggable="true">Hep-B</div>
    <div id = "1C" draggable="true">BCG</div>
    <div id = "1D" draggable="true">Vita-K</div>
    <div id = "1E" draggable="true">Colostrum Feeding</div>

    <div id = "2A" draggable="true">OPV-1</div>
    <div id = "2B" draggable="true">Penta-1</div>
    <div id = "2C" draggable="true">Rota-1</div>
    <div id = "2D" draggable="true">PCV-1</div>
    <div id = "2E" draggable="true">IPV-1</div>

    <div id = "3A" draggable="true">OPV-2</div>
    <div id = "3B" draggable="true">Penta-2</div>
    <div id = "3C" draggable="true">Rota-2</div>

    <div id = "4A" draggable="true">OPV-3</div>
    <div id = "4B" draggable="true">Penta-3</div>
    <div id = "4C" draggable="true">Rota-3</div>
    <div id = "4D" draggable="true">PCV-2</div>
    <div id = "4E" draggable="true">IPV-2</div>

    <div id = "5A" draggable="true">MR-1</div>
    <div id = "5B" draggable="true">JE-1</div>
    <div id = "5C" draggable="true">PCV Booster</div>
    <div id = "5D" draggable="true">Vita A Doze-1</div>
```

[1] Link to the repository: https://github.com/arkamajhi/1000days/tree/main/Child_Imm_Lvl2

[2] Link to the Live Url: http://1000days.in/Child_Imm_Lvl2/index.html

```
29
30      <div id = "6A" draggable="true">MR-2</div>
31      <div id = "6B" draggable="true">JE-2</div>
32      <div id = "6C" draggable="true">OPV Booster</div>
33      <div id = "6D" draggable="true">DPT Booster-1</div>
34      <div id = "6E" draggable="true">Vita A Doze-2</div>
35      <div id = "6F" draggable="true">Deworming-1</div>
36
37      <div id = "7A" draggable="true">Vita A Doze-3 to 9</div>
38      <div id = "7B" draggable="true">Deworming-2 to 4</div>
39
40      <div id = "8A" draggable="true">DPT Booster-2</div>
41
42      <div id = "9A" draggable="true">TD-1</div>
43
44      <div id = "10A" draggable="true">TD-2</div>
45
46  </div>
```

Listing 8.1: Draggable Items Initialization

8.4 Development of the Card Game

8.4.1 Idea of game

Insights from the sequencing game prototype confirmed the need to retain age-appropriate vaccine matching. However, presenting information modules in a list format was less intuitive, leading to the exploration of card-based interfaces, which are more familiar to users. Cards were tested to determine if sequencing them proves more user-friendly while maintaining similar learning outcomes.

The design process followed an iterative, design-based research journey with recurring bottlenecks around content translation, rule complexity, materials, and institutional fit, each addressed through cycles of simplification, re-framing, and co-design with CHWs and trainers.

8.4.1.1 Unequal "Suits" from Endogenous Content

In classic card game, the standard deck consists of four suits with exactly 13 ranks each, which underpins core mechanics like balanced distribution, complete sequences, and symmetry between suits. In this game, the four "suits" (silos/foundations) emerged directly from health content: Child under 1 year, Child above 1 year, ANC, and PNC.

Because the immunization schedule and care protocols do not distribute evenly across these four categories, each suit ended up with a different number of cards. Some age bands and care periods are critical, characterized by a high density of interventions, while others have relatively few. Some "suits" (e.g., under-1 vaccines) had long, dense sequences, while others (e.g., PNC) had relatively few cards, upsetting expectations of symmetry. This pattern is also observed in other endogenous educational game designs that derive their suits from content structures rather than abstract symmetry. Respecting domain fidelity means giving up some of the mathematical neatness of traditional card decks. As a result, standard 4×13 assumptions behind Solitaire-like or Badam Satti-like rules could not simply be imported.

The first instinct was to treat these four content-based suits like the four suits in a regular 52-card deck and apply familiar patterns from Solitaire and Badam Satti. In Badam Satti,

for example, balanced suits allow sequences to grow outward from the 7 in both directions, with reasonably similar possibilities across suits. Using unmodified Badam Satti rules could make certain suits structurally advantaged or disadvantaged, because they offered more or fewer playable extension points. Concerns arose that the imbalance might break game flow or make the game feel "rigged" or incomplete to players familiar with standard decks. In Solitaire, equal suit lengths support the predictable completion of foundations.

Through playtesting, it became clear that strict numerical symmetry was not necessary for the game to support Solitaire-like or Badam Satti-like modes. What mattered more was preserving the relational structure. Other card-game design work similarly notes that non-standard decks can still support familiar patterns, as long as the ordering and adjacency relations within suits remain coherent (Palmer and Denelsbeck, 2024; Yan et al., 2005). In our case, the realization was that unequal suit lengths did not prevent:

- A Badam Satti mode, where any intervention can act as an anchor (analogous to 7s), and players extend care sequences up/down in time within each category, even if some sequences are shorter or longer.
- A Solitaire mode, where players build foundations of correct sequences for each category (under-1, above-1, ANC, PNC) and manage a "stock" of remaining interventions, with completion defined as finishing all required sequences rather than filling 13 slots per suit.

Thus, instead of retrofitting content into a 4×13 grid, we preserved endogenous suit sizes and reinterpreted the victory conditions and extension rules. Because suits are unequal, there is no expectation that every suit will reach the same length. The completion condition occurs when players exhaust all the cards in their hands or a team empties its hand. This approach aligns with broader serious-game design guidance, which suggests that mechanics can be adapted from existing games, but they must be flexibly re-engineered to fit the logic and scope of the educational content (Stefany et al., 2024; Arnab et al., 2015).

This psychological transition from the standard 4×13 card format to variable numbers of cards in each suit, though straightforward in hindsight, took considerable time. One probable reason could be that I was not an avid card player in my childhood, when one typically develops an intuitive understanding of card games and of the process of modifying the "standard" rules to suit a group's convenience.

8.4.1.2 Low-Fidelity Prototyping and Rule Explosion

Once a basic game concept was drafted, we created low-fidelity paper prototypes by designing handwritten cards, sketched icons, and a one-page rules sheet. This stage mirrored pen-and-paper prototyping practices in card-game design, where the emphasis is on testing ideas quickly rather than refining graphics. Early playtests with colleagues and a small number of CHWs exposed the problem of rule explosion.

Rules were too complex to hold in working memory. Initial rules included a range of possible actions (draw, play, swap, reserve, "counsel," "refer," etc.), multiple scoring categories, and various exceptions. Players spent cognitive effort on remembering what they were allowed to do, rather than thinking about the immunization schedule, ANC, and PNC sequences. Structural complexity might overwhelm the intended learning focus. Thus, simplifications of rules were made. The action set per turn was reduced to a small, fixed repetitive moves (e.g., draw 1, play or discard 1, optionally support a teammate), following serious games design guidance that early prototypes should test the minimal rules necessary for making meaningful decisions.

Rules were also reorganized into a progression according to difficulty levels for first-time players. Introductory rounds used only child–vaccine sequences. Only after players became fluent were ANC and PNC event cards introduced, echoing phased complexity and scaffolding recommendations in educational game design, a strategy that came in handy even in the digital version of the game.

8.4.1.3 Translating Tactile Play to Touch Interaction

In the physical game, CHWs spread cards on a table, point, rearrange, and jointly sequence, which naturally supports discussion and distributed cognition. Moving this to Android required redefining what "play a card," "show a card," or "build a sequence" looks like in a single-screen touch interface. Similar issues are reported in other research on translating physical card games to digital platforms (Van Rozen et al., 2023).

The major bottleneck is the loss of spatial flexibility. A phone screen cannot show the same dense layout of multiple cases and long sequences without becoming cramped. Designing responsive layouts that keep cards legible on small, low-resolution devices while still showing

enough context for reasoning became a core UI challenge. We resolved this issue by utilizing stacked views and "zoom" or "detail" screens, rather than attempting to display the entire table at once, an approach commonly employed by other digitalized card-game platforms.

The next issue was regarding gesture mapping. deciding whether players tap, drag-and-drop, or use long-press menus, and ensuring these gestures are easily discoverable for CHWs with varying smartphone familiarity. We resolved this by choosing a small, consistent set of gestures (tap to select, drag to place) and adding subtle animations and highlights to show valid targets. Drag-and-drop functionality for playing or reordering cards was implemented with robust hit detection, collision targets, and snap zones that are tolerant of finger imprecision, especially for CHWs who are unused to fine-grained touch gestures. Handling conflicting gestures (scroll vs. drag, tap vs. long press) required clear priority rules and visual feedback to prevent accidental moves or frustration.

Social cues, such as physical pointing and eye contact, are replaced by on-screen indicators, which can weaken the sense of collaborative play. We resolved this by adding explicit indicators of turn, hints, chat, and prompt mechanisms to support social signals.

8.4.1.4 Encoding Rules, Turn Logic, and State

Physical play relies on human judgment. Players know whose turn it is, whether a move is legal, and when a sequence is complete. In the physical game, players enforce rules. In digital play, the app must track the state and verify the legality of every move, automatically enforcing rules in real-time. Encoding all constraints (correct sequence, no skipping required care, suit compatibility) into a rule engine was both technically and conceptually demanding. Multiple suits (under-1, above-1, ANC, PNC) and their asymmetric sequences were transformed into data structures that can be checked efficiently. We implemented "adjacency check" (correct next vaccine or visit) as code, rather than simple numeric adjacency like 6–7–8, which many standard card engines assume. We created explicit rule engines or lookup tables that codify allowed transitions between interventions, similar to logic-model-driven health apps that embed behaviour and content rules.

Handling incomplete cases, such as skipped visits, catch-up schedules, or deliberate "wrong" moves for learning, required the system to distinguish between illegal moves (blocked) and ped-

agogically allowed but incorrect moves (allowed but flagged with feedback). This distinction complicates validation logic and UI prompts (warnings vs. hard blocks vs. auto-correction of moves). Rule exceptions can be negotiated informally at the table in a physical card game. However, for the digital card game, we built "soft rules" (warnings rather than hard blocks) so that CHWs can explore "wrong" moves and then receive feedback, which is common in educational game apps.

While designing, we faced a dilemma about the potential inclusion of a 'joker' card, familiar from rummy games, which substitutes for any card. Although feasible mechanically, this feature posed a significant risk of creating a misconception among CHWs that vital immunization steps, like the measles vaccine at 9 months or the tetanus toxoid booster for pregnant women, were optional or skippable. While real-world practices occasionally bypass checks (e.g., the four antenatal care visits or multiple pentavalent/OPV doses), formal health protocols prohibit this. After brief deliberation, the Joker card idea was abandoned to avoid undermining procedural rigor. This episode highlights the need for caution when adapting traditional games for health training, as symbolic flexibility may inadvertently compromise adherence to evidence-based knowledge or practices.

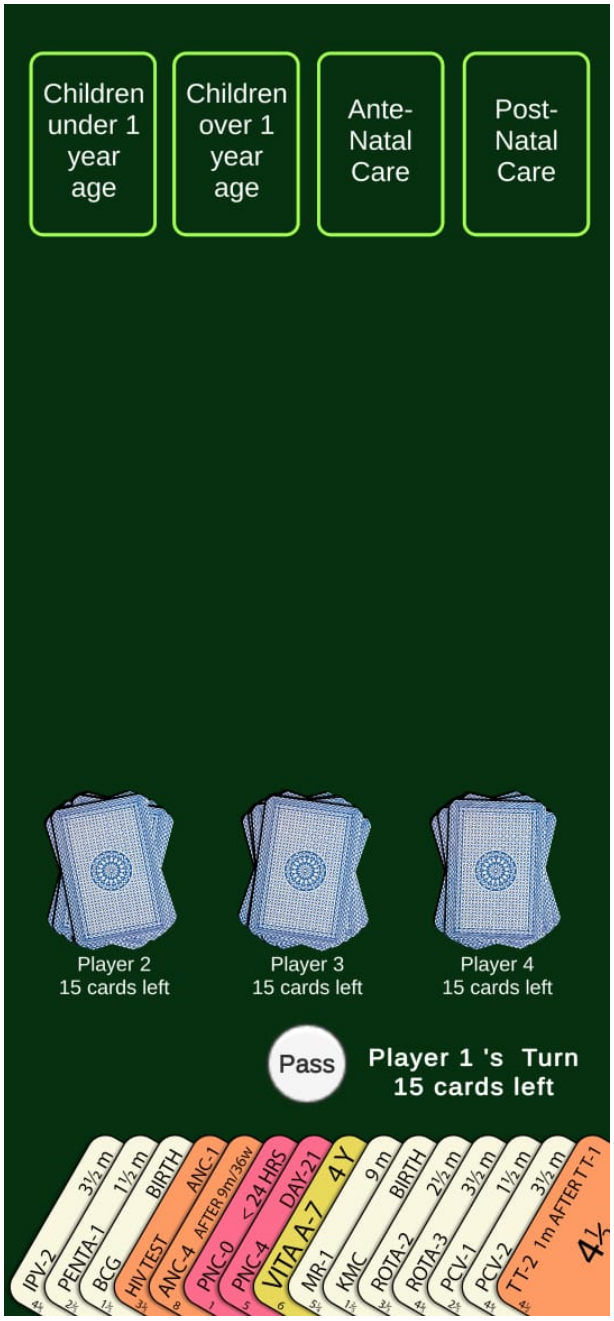


Figure 8.8: Game start Screenshot

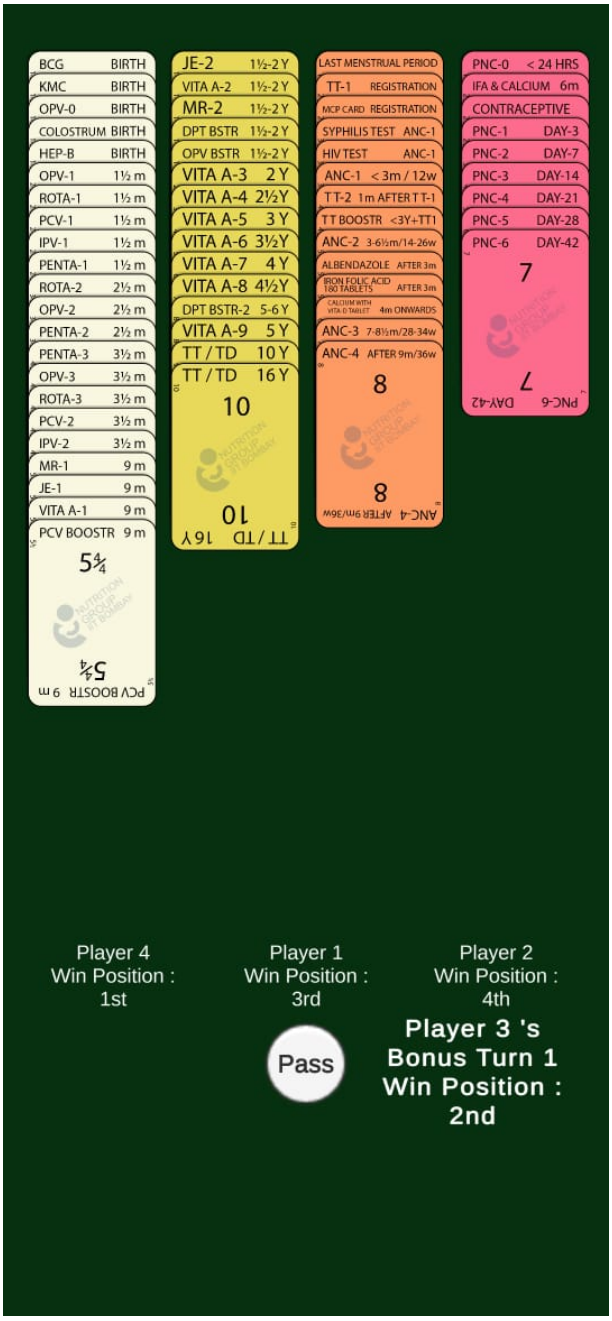


Figure 8.9: Game Over Screenshot

8.4.2 Design of the Card Deck

The design of the cards evolved through many iterations. We began with a minimal card design, featuring only the name of the vaccine and the time of administration. We realised that there was not enough information to communicate. Also, it is not convenient to turn cards upside down after shuffling. In the second iteration, we added relevant images of dosage, diseases that the vaccines prevent, and other relevant information. We attempted to cater to local people by using the local language.

Then we realised that in some time periods, multiple vaccines are administered simultaneously. We added a mixed fraction to denote the time of intervention. In some cases, there were two mixed fractions in the card, one for time and one for order. CHWs found it more complex and were unable to figure it out on their own. So we dropped the idea of mixed fractions for time period display. Instead, we placed a set of small white boxes at the card corner below time period to depict multiple doses of vaccines at a time. We made two versions of the card, focusing on digital app usage and a physical card deck. The design is iterated based on what information can be shown better while stacked physically or digitally.

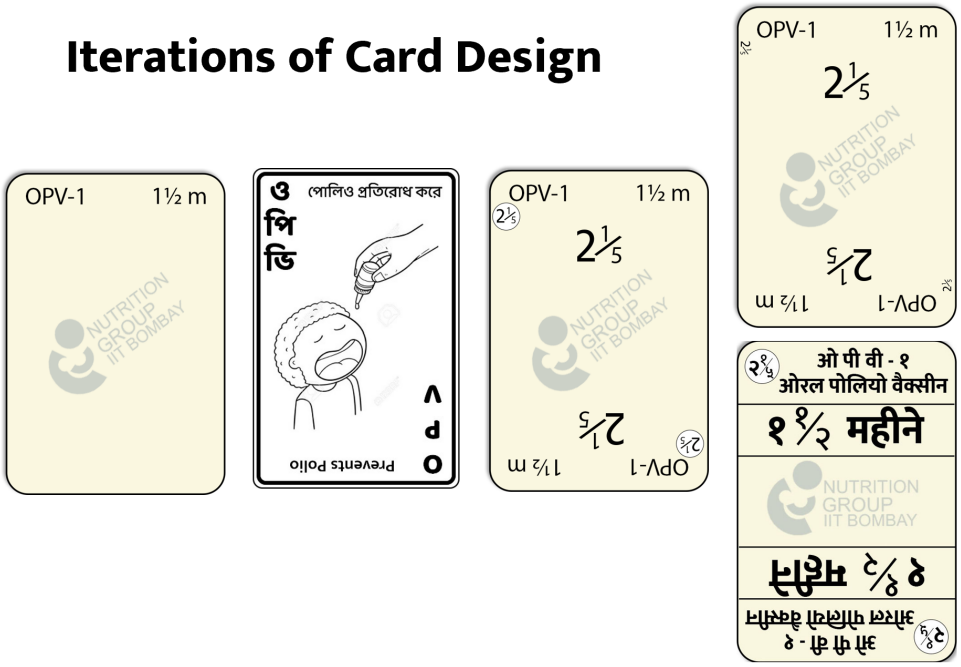


Figure 8.10: Iterations of Card Design

Figure 8.11: New Card Design

Figure 8.12: Card design finalized

Figure 8.13: Immunization cards for children below 1 year (continued on next page)

Figure 8.14: Immunization cards for children below 1 year (continued from previous page)

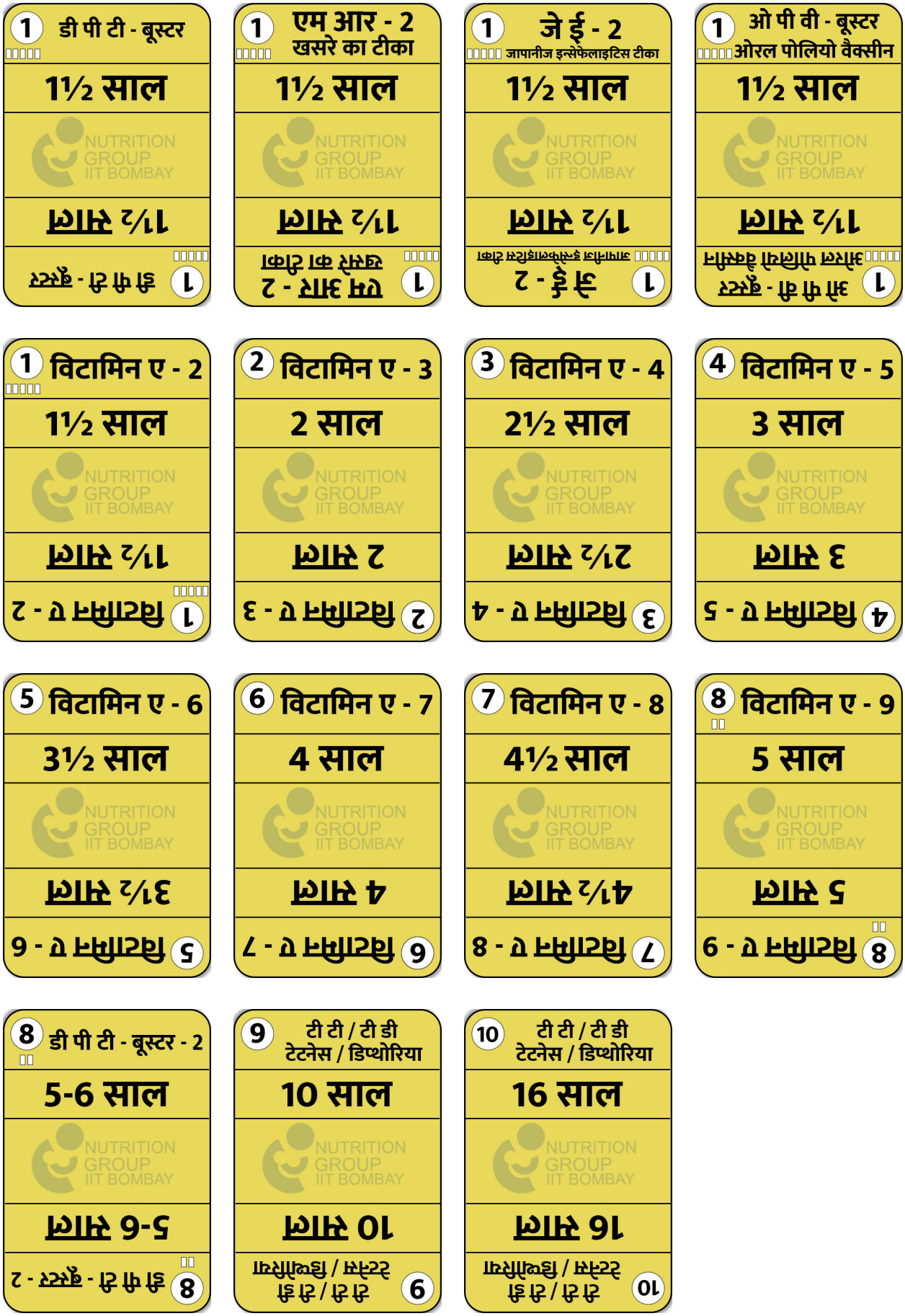


Figure 8.15: Immunization cards for children above 1 year

Figure 8.16: Ante-Natal-Care (ANC)

Figure 8.17: Post-Natal-Care (PNC)

8.4.3 Coding of the game

Out of all the scripts, we will discuss about a few important ones which makes the major decisions in the game design. The link to the repository is in the footnote [3].

8.4.3.1 Game Manager Script

The Game Manager script executes upon loading the game scene and initializes cards across four foundations. Four string arrays define the cards using prefixes: "c" for children under one year, "C" for children over one year, "A" for antenatal care (ANC), and "P" for postnatal care (PNC), with unequal card counts per category. The second character denotes intervention timing (e.g., "c1" for birth, "c2" for $1\frac{1}{2}$ months, "c3" for $2\frac{1}{2}$ months, "c4" for $3\frac{1}{2}$ months, "c5" for 9 months), while a third character sequentially identifies multiple vaccines or interventions within each period (e.g., "c10" for exclusive breastfeeding, "c11" for OPV-0, "c12" for hepatitis B, "c13" for BCG).

```
1 public static string[] child0_1 = new string[] { "c10", "c11", "c12", "
    c13", "c14", "c20", "c21", "c22", "c23", "c24", "c30", "c31", "c32",
    "c40", "c41", "c42", "c43", "c44", "c50", "c51", "c52", "c53" };
2 public static string[] child1_16 = new string[] { "C00", "C01", "C02", "
    C03", "C04", "C10", "C20", "C30", "C40", "C50", "C60", "C70", "C71",
    "C80", "C90" };
3 public static string[] anc = new string[] { "A10", "A20", "A21", "A30",
    "A31", "A32", "A40", "A41", "A50", "A51", "A52", "A60", "A70", "A80"
    };
4 public static string[] pnc = new string[] { "P10", "P20", "P21", "P22",
    "P30", "P40", "P50", "P60", "P70" };
```

Listing 8.2: Cards Initialization

The four foundations are then initialized to store cards in a Badam Satti-style structure, with each foundation requiring three distinct stacks: top ("T"), center ("C"), and bottom ("B"). The foundations correspond to "c", "C", "A", and "P", resulting in $4 \times 3 = 12$ total stacks for organizing the vaccine cards.

[3] Link to the repository: `https://github.com/arkamajhi/Card-Game-for-ASHAs`

```
Foundations = new List<string>[] {
    FoundationcT, FoundationcC, FoundationcB,
    FoundationCT, FoundationCC, FoundationCB,
    FoundationAT, FoundationAC, FoundationAB,
    FoundationPT, FoundationPC, FoundationPB
};
```

Listing 8.3: Foundation and Stacks Initialization

A new deck is then generated by sequentially adding cards from each of the 12 stacks across the four foundations.

```
public static List<string> GenerateDeck()
{
    List<string> newDeck = new List<string>();
    foreach (string a in child0_1)
    {
        newDeck.Add(a);
    }
    foreach (string a in child1_16)
    {
        newDeck.Add(a);
    }
    foreach (string a in anc)
    {
        newDeck.Add(a);
    }
    foreach (string a in pnc)
    {
        newDeck.Add(a);
    }
    return newDeck;
}
```

Listing 8.4: Generate Deck Function

The deck is then shuffled to randomize card order by iterating 60 times, swapping each of the 60 cards with a randomly selected position using a temporary variable. This Fisher-Yates-inspired algorithm ensures thorough randomization for the sequencing task. The process prepares the playable deck with vaccine cards in an unpredictable order to challenge players'

recall of the immunization schedule.

```
void Shuffle<T>(List<T> list)
{
    System.Random random = new System.Random();
    int n = list.Count;
    while (n > 1)
    {
        int k = random.Next(n);
        n--;
        T temp = list[k];
        list[k] = list[n];
        list[n] = temp;
    }
}
```

Listing 8.5: Shuffling of Deck Function

The deck is then dealt among the players, with HandTransform defining each player's initial hand position. These hand positions are synchronized across all players' devices using PhotonNetwork for consistent multiplayer gameplay. This distribution enables collaborative or competitive sequencing of vaccine cards.

```
IEnumerator Deal()
{
    foreach (string card in deck)
    {
        yield return new WaitForSeconds(0.01f);

        Vector3 position = handTransform.position + new Vector3(cardposx
    * gapx, 0f, -1 * (cardposx) * gapz);

        GameObject cardObj = PhotonNetwork.Instantiate(Path.Combine("
   PhotonPrefabs", "cardPrefab"), position, Quaternion.Euler(0, 0, 60));

        PhotonView cardPhotonView = cardObj.GetComponent<PhotonView>();

        cardObj.GetComponent<DynamicObjectName>().UpdateObjectName(card)
   ;
        cardObj.GetComponent<Selectable>().faceUp = true;
```

```
15          cardPhotonView.TransferOwnership(PhotonNetwork.PlayerList[
    playernumber]);
16
17          cardIndex++;
18
19          playernumber++;
20          if (playernumber == playerCount)
21          {
22              playernumber = 0;
23              cardposx++;
24          }
25      }
26
27      Vector3 PBposition = Vector3.zero + new Vector3(2, -2, 0);
28
29      GameObject PButton = PhotonNetwork.Instantiate(Path.Combine("
    PhotonPrefabs", "PassButton"), PBposition, Quaternion.Euler(0, 0, 0))
    ;
30      Debug.Log("Instantiated PassButton");
31
32  }
```

Listing 8.6: Dealing of Deck Function

8.4.3.2 User Input Script

After dealing cards to players' hands, the game commences with Player 1 taking the first turn. This script manages all user inputs by first verifying Photon Server connectivity and the player's host/client status before initializing turns. Correct moves grant two bonus turns; after these, or upon passing, the turn advances to the next player.

```
1  void UpdateText()
2  {
3      if (Photon.Pun.PhotonNetwork.IsConnected)
4      {
5          p_no.text = "Player Number: " + playerNumber.ToString();
6      }
7      else
```

```
    {
        p_no.text = "Player Number: N/A";
    }

    if (PhotonNetwork.IsMasterClient)
    {
        m_c.text = "This is Master";
    }
    else
    {
        m_c.text = "This is Client";
    }

    foreach (GameObject g in GameObject.FindGameObjectsWithTag("PassButton"))
    {
        int ownernumber = g.GetComponent<PhotonView>().Owner.ActorNumber;
        owner.text = "Owner : " + ownernumber;
        if (ownernumber == playerNumber)
        {
            if (countstack == 0)
            {
                displaytext.text = ("Your Turn now");
            }
            else if(countstack==1)
            {
                displaytext.text = ("Your First Bonus Turn");
            }
            else if (countstack == 2)
            {
                displaytext.text = ("Your Second Bonus Turn");
            }
        }
        else
        {
            displaytext.text = (PhotonNetwork.PlayerList[ownernumber - 1].NickName + "'s turn now \n Wait for your Turn");
        }
```

```
        break;
    }
}
```

Listing 8.7: User Input Script

This code segment validates whether a card can be stacked onto the foundations and identifies the precise position (top, center, or bottom stack) where it fits. It enforces the Badam Satti-style stacking rules aligned with the immunization sequence across the four foundations.

```
    void Stackable(GameObject selected)
{
    //Check if can be stacked
    string symbol = selected.name.Substring(0, 1);

    for(int i = 0;i<FoundationsCharArray.Length;i++)
    {
        if (symbol == FoundationsCharArray[i])
        {
            /*
            //if center has zero cards
                //Stack (by default it happens at top)
            //else
                //(if center has cards)
                //if second letter is one less than the first
                    //Stack (by default it happens at top)
                //else if second letter is one more than the last
                    //Stack (by default it happens at below)
            */

            if (Foundations[(i*3)+1].Count==0)
            {
                if (Foundations[(i * 3)].Count == 0)//if no top cards
                {
                    Stack(selected);
                    Debug.Log("Stacked at center (initialization) of
  foundation : " + symbol);
                }
                else
                {
```

```
30                      if (selected.name.Substring(1, 1) == Foundations[(i
 * 3)][0].Substring(1, 1)) //Ifcenter is blank but cards are at top
initially
31                      {
32                          Stack(selected);
33                          Debug.Log("Center empty and Stacked at top of
foundation : " + symbol);
34                      }
35                  }
36
37              }
38              else
39              {
40                  if(int.Parse(selected.name.Substring(1, 1))== int.Parse
(Foundations[(i * 3) + 1][0].Substring(1, 1))-1)
41                  {
42                      Stack(selected);
43                      Debug.Log("Stacked at top of foundation : " +
symbol);
44                  }
45                  else if (int.Parse(selected.name.Substring(1, 1)) ==
int.Parse(Foundations[(i * 3) + 1][Foundations[(i * 3) + 1].Count-1].
Substring(1, 1)) + 1)
46                  {
47                      Stack(selected);
48                      Debug.Log("Stacked at bottom of foundation : " +
symbol);
49                  }
50                  else
51                  {
52                      Debug.Log("Can't be Stacked on foundation : " +
symbol);
53                      source.PlayOneShot(cantstack);
54                  }
55              }
56          }
57      }
```

```
58 }
```

Listing 8.8: User Input Script - Check of Stackable

8.4.3.3 Update Sprite

This script manages card visibility states—displaying faces, turning backs, or hiding cards based on ownership and context. Cards in the player's hand or foundations appear face up, while those in other players' hands remain hidden for multiplayer fairness. During a player's turn, clicking a hand card raises it above the deck to indicate selection. Subsequent clicks lower the prior card and elevate the new one.

```
1  public class UpdateSprite : MonoBehaviourPun
2  {
3      void Start()
4      {
5          int i = 0;
6          foreach (string card in deck)
7          {
8              if (this.name == card)
9              {
10                 cardFace = gameManager.cardFaces[i];
11                 break;
12             }
13             i++;
14         }
15         state = "inactive";
16     }
17
18
19     // Update is called once per frame
20     void Update()
21     {
22         if(selectable.faceUp==true)
23         {
24             spriteRenderer.sprite = cardFace;
25         }
26         else
```

```
27          {
28              spriteRenderer.sprite = cardBack;
29          }
30
31          if (name == userInput.slot1.name && tag != "FoundationCard")
32          {
33              Vector3 pos = transform.position;
34              pos.y = -3.15f;
35
36              if (cardrotated == 0)
37              {
38                  cardx = pos.x;
39                  pos.x -= 0.6f;
40                  cardrotated = 1;
41
42                  transform.rotation = Quaternion.Euler(0, 0, 3);
43              }
44
45              transform.position = pos;
46
47              //Debug.Log(name+" is active");
48
49              state = "active";
50          }
51          else
52          {
53              //spriteRenderer.color = Color.white;
54              if (state == "active")
55              {
56                  state = "inactive";
57                  //Debug.Log(name + " is inactive");
58
59                  if (tag != "FoundationCard")
60                  {
61                      Vector3 pos = transform.position;
62                      pos.y = -4.7f;
63
64                      if (cardrotated == 1)
65                      {
```

```
                        pos.x = cardx;
                        cardrotated = 0;
                    }

                    transform.position = pos;
                    transform.rotation = Quaternion.Euler(0, 0, 60);
                }
            }

        }



        if (PV.IsMine && this.CompareTag("Card"))
        {
            spriteRenderer.enabled = true;
        }
        else
        {
            spriteRenderer.enabled = false;
        }

        if (this.CompareTag("Foundations")|| this.CompareTag("
    FoundationCard"))
        {
            spriteRenderer.enabled = true;
        }


        int i = 0;
        foreach (string card in deck)
        {
            if (this.name == card)
            {
                cardFace = gameManager.cardFaces[i];
                break;
            }
            i++;
        }
```

```
104         }
105 }
```

Listing 8.9: Update Sprite Script

Chapter 9

Discussion and Conclusion

9.1 Connecting the Dots: Synthesizing Insights from the Three Studies

The three empirical studies presented in this dissertation were designed as a progression rather than as isolated experiments. Study-1 (Chapter-4) explored whether refresher-training content for Community Health Workers (CHWs) could be transformed into a playful learning activity and examined the role of augmented reality (AR) in supporting this experience. Study-2 (Chapter-5) moved towards a more direct comparison of physical and digital card-based gameplay, conventional classroom training, and a control condition, while also introducing delayed assessment of knowledge retention. Study-3 (Chapter-6) extended the digital intervention into a smartphone-based refresher training format and examined learning progression and retention through repeated interactions. This progression reflects the methodology of the dissertation from feasibility and interaction, to comparative evaluation, and finally to questions of repeatability and sustained learning.

The studies provide evidence that game-based interaction can support CHW refresher

training, but they also show that the effect of a game cannot be understood in isolation from its modality, social context, learner characteristics, and opportunities for reinforcement. The findings, therefore, do not support a simple conclusion that "games are better than conventional training." Rather, they suggest that game-based approaches can provide useful learning affordances that are different from, and potentially complementary to, those offered by conventional training.

9.1.1 Immediate Knowledge Gain and Subsequent Retention

A recurring pattern across the experimental studies was the distinction between immediate knowledge acquisition and knowledge retained after the intervention. In Study-1 (Chapter-4), the playful intervention produced measurable learning gains, while the comparison between the physical and AR-supported intervention provided an initial indication that technological enhancement alone should not be treated as synonymous with learning effectiveness. In Study-2 (Chapter-5), the digital card-game condition produced the largest immediate improvement, followed by the physical card condition, while the classroom condition showed a smaller improvement. The delayed assessment, however, reduced the distinction between the digital and physical formats. Study-3 (Chapter-6) similarly found that the digital format could produce stronger short-term gains while longer-term retention was more comparable between digital and physical game-based approaches.

This distinction is important because a refresher-training intervention is successful not only when participants perform well immediately after training. The practical value of refresher learning depends on whether relevant knowledge remains available when CHWs subsequently encounter situations in their work. The decline between immediate and delayed assessments, therefore, is an important finding that suggests that a single game session, regardless of how engaging it is, may not be sufficient to produce retention in learning.

The results consequently point towards a reinforcement-learning approach of game-based refresher training. Games may be particularly effective at producing initial knowledge gains, but knowledge retention may require repeated exposure, retrieval, application, and feedback. Future versions of the intervention could therefore incorporate periodic replay, progressively more

difficult scenarios, cumulative challenges, and short, knowledge-retrieval-based assessments. The objective should not simply be to make a single training session more engaging, but to create opportunities for continued interaction with the content.

9.1.2 Complementary Affordances of Digital and Physical Gameplay

One of the most important findings emerging across the studies is that digital and physical formats need not be treated as competing alternatives. The digital format provided rapid feedback, automated progression, and interactive presentation. These features appeared particularly useful for immediate engagement and knowledge acquisition. The physical card format, in contrast, encouraged discussion, negotiation, verbal explanation, and shared problem-solving. In the delayed assessments, the difference between the two formats was less significant than in the immediate post-test results.

This pattern led me to conceptualize the two formats as providing complementary affordances. The digital format can support rapid interaction and individualized feedback, while the physical format can support social interaction and deliberative reasoning. This does not mean that every learner will benefit from a single format in a predictable way, nor does it imply that digital and physical games necessarily yield equivalent learning. Rather, it suggests that the choice of modality should depend on the learning objective and implementation context.

A hybrid learning ecosystem may therefore be more appropriate than an exclusive preference for either a digital or a physical mode of playful learning. Digital games may be particularly useful when scalability, automated feedback, and progress tracking are important. Physical games may remain valuable where internet access, device availability, or digital confidence is limited, or where peer discussion is an important part of the learning process. The findings from Study-3 (Chapter-6) similarly indicate that physical cards can be accessible and compatible with everyday CHW routines, while smartphone-based games offer opportunities for automated feedback and scalable dissemination of knowledge.

9.1.3 The Importance of Co-Design

The findings also reinforce the importance of involving CHWs in the design of learning interventions. The interventions developed in this dissertation were not designed solely from an external understanding of healthcare training. The iterative process of developing, testing, and refining the games enabled CHWs' experiences to influence the content, language, mechanics, and physical form of the interventions.

This was particularly important for the physical card game. Familiar card-game conventions made the intervention less dependent on technological infrastructure and allowed the learning activity to be incorporated into a socially recognizable form. Similarly, the use of local-language content helped address linguistic differences that might otherwise have become a barrier to participation.

Co-design should therefore not be viewed only as a usability technique. In this research, it served to establish contextual relevance. The intervention had to make sense not only as a game but also as a training activity that CHWs could understand, accept, and use within their existing social environment.

9.1.4 Differences in Comparison and Control Conditions

An important methodological consideration across the studies concerns the differences between their comparison conditions. Although the experimental groups were generally comparable at baseline, the comparison conditions were not identical across all studies. In some settings, the comparison involved conventional classroom teaching, whereas in other settings, participants served as a control condition without the same training exposure. Furthermore, classroom training was delivered by trainers whose availability, teaching style, and interaction with participants varied across locations.

This variation limits the extent to which the results of different studies can be directly pooled or interpreted as though they were produced under identical experimental conditions. A difference between a game-based group and a classroom group may reflect the game's af-

fordances, but it may also partly reflect differences in instructor interaction, teaching intensity, group dynamics, or the amount of attention participants receive.

For this reason, the cross-study findings should be interpreted primarily as a pattern of converging evidence rather than as a single controlled comparison. Study-2 provided the strongest direct comparison between physical and digital game formats and conventional classroom training, while Study-3 extended the investigation into smartphone-based repeated learning. The methodological progression strengthens the overall argument, but it does not eliminate the contextual differences between individual experiments.

Future studies should therefore standardize the comparison conditions as far as possible. Where classroom training is used as a comparator, the duration, content, trainer preparation, and instructional exposure should be documented and controlled. This would make it easier to distinguish the effect of the game-based intervention from the effect of differences in implementation.

9.2 From Immediate Learning to Sustained Learning

The findings challenge a narrow understanding of training effectiveness that relies solely on post-intervention performance. Across the studies, the greatest improvements were generally observed immediately after gameplay, while some of these gains declined during delayed assessment. This pattern indicates that engagement and immediate learning are related to, but not sufficient for, retention.

The distinction also provides an important qualification to the argument that game-based learning is inherently more effective. The studies demonstrate that game-based interventions can produce stronger immediate learning outcomes than some conventional comparison conditions. However, the persistence of these gains depends on what happens after the initial intervention. Repetition, retrieval, opportunities for application, and continued motivation may all be necessary for maintaining knowledge.

This observation is particularly relevant for CHW refresher training because CHWs do not encounter all health protocols with equal frequency. A game can introduce or reactivate

knowledge, but the intervention must be embedded within a broader training system if that knowledge is expected to remain available over longer periods. The role of the game may therefore be better understood as one component of a continuing learning ecosystem rather than as a standalone replacement for training.

9.3 Learning by Doing and the Role of Play

The development and field evaluation of the prototypes also changed my understanding of what "learning by doing" means in this context. During the fieldwork, I observed a tension between the value of active engagement and the practical desire for fast and convenient technological solutions. Digital tools can reduce the effort required to access information, but that reduction does not necessarily lead to deeper learning.

The game-based activities were useful partly because they required participants to make decisions, discuss answers, manipulate objects, remember sequences, and respond to consequences. These actions made the learning process visible. Participants were not simply receiving information; they were repeatedly attempting to use it.

This became particularly evident in the physical card-game sessions, where participants sometimes discussed an answer before deciding on a move. Such discussion may have slowed the activity down, but it also created opportunities for explanation and correction. The resulting learning experience was therefore not always the most efficient way to complete the game, but it could be productive for constructing understanding.

The finding has implications for the design of future digital interventions. Digitizing the game should not mean removing the productive difficulty from the learning process. Automation should support learning rather than simply speed up the activity.

9.4 Learner Characteristics and Age

Across the field experiments, age appeared to be associated with differences in knowledge gain. Older CHWs generally showed smaller changes between pre-test and post-test scores than younger participants. This pattern may reflect differences in familiarity with new content or digital interfaces, although age should not be interpreted as a simple indicator of learning ability.

The qualitative observations also showed that some older CHWs were highly motivated and enthusiastic about engaging with the intervention. Thus, the quantitative association between age and learning gain should be interpreted as a tendency within the observed samples rather than as evidence that older CHWs are inherently less capable learners.

Years of formal education also did not consistently explain baseline knowledge differences. In the context of CHW work, practical experience and repeated exposure to maternal and child health activities may contribute substantially to knowledge that is not captured by formal educational attainment.

These findings suggest that future game-based training should avoid designing for a single assumed "typical" learner. Interface complexity, instructions, language, pacing, and interaction modes may need to be adapted to accommodate different levels of digital familiarity and prior experience.

9.5 Social, Cultural, and Linguistic Context

The experiments further demonstrated that learning does not occur independently of the social environment. Language, cultural familiarity, socioeconomic conditions, access to technology, and peer relationships influenced how participants interacted with the interventions.

India's linguistic diversity makes this particularly important for CHW training. A standardized training intervention that does not account for local language and communication practices may create barriers even when the underlying content is appropriate. Providing content in locally understandable forms can reduce unnecessary cognitive load and allow participants to

focus on the health content itself.

The physical card game also demonstrated the value of social interaction. Participants could discuss answers, explain rules, and collectively negotiate decisions. These interactions were not simply incidental to the game. They became part of the learning process.

The findings, therefore, support a situated understanding of game-based learning. The same game may not produce the same experience in every context. The meaning assigned to gameplay can change according to where it is played, who plays it, and how the surrounding group understands the activity. This observation is consistent with the findings reported in Study-3, in which the perceived seriousness of the game varied across rural, tribal, and urban contexts.

9.6 Implications for Game Design

The findings suggest several principles for designing game-based refresher training for CHWs.

First, game mechanics should be endogenous to the learning content wherever possible. In the card game, sequencing vaccination cards by age turned an abstract immunization schedule into a concrete decision-making activity. Resource constraints and simulated health events created situations that required participants to apply rather than merely recall information.

Second, difficulty should increase progressively. Scaffolding can allow learners to first understand the basic interaction before encountering more complex decisions. As competence increases, support can gradually be reduced.

Third, feedback should be immediate enough to allow participants to identify and correct misconceptions. However, feedback should not completely remove the need for reflection. If the game provides the answer too quickly, learners may optimize for completion rather than understanding.

Fourth, social interaction should be treated as a design feature rather than an accidental by-product. Physical games naturally created opportunities for discussion, but similar mechanisms could be incorporated into digital games through collaborative challenges, shared decision-making, and explanation-based tasks.

Finally, the game should remain subordinate to the learning objective. A visually attractive or technologically sophisticated intervention is not necessarily a better educational intervention. The findings from the comparison between digital and physical formats reinforce the importance of designing the learning experience first and selecting technology according to what that experience requires.

9.7 Expanding the Content of the Game

The present research focused primarily on maternal and child health content, including immunization, ANC, and PNC. The underlying game structure could be extended to other areas of CHW work, including adolescent health, maternal nutrition, breastfeeding, complementary feeding, disease prevention, and other community health topics.

However, expansion should not simply involve adding more information to the existing game. New content should be accompanied by mechanics that reflect the type of reasoning required by that content. For example, a nutrition module could involve constructing balanced meals from different food groups, while an adolescent-health module could involve identifying symptoms, choosing appropriate counseling strategies, or responding to common misconceptions.

This approach maintains the principle of endogenous game mechanics. The game should emerge from the structure of the health task rather than treating the health content as decoration placed inside an unrelated game.

At the same time, game-based refresher training should not be interpreted as a replacement for initial hands-on training. Some CHW competencies involve physical procedures, interpersonal communication, or situated judgment that cannot be adequately developed through a card or smartphone game alone. Games can reinforce, rehearse, and contextualize such knowledge, but they cannot replace all forms of experiential training.

9.8 Collaborative Learning Through Games

Collaboration is particularly relevant to CHW work because many health activities require coordination with other workers, supervisors, and community members. Future game designs could therefore include cooperative tasks in which players have access to different pieces of information and must combine their knowledge to make a decision.

For example, players could collectively develop a care plan for a malnourished child, with each player receiving different information about dietary practices, symptoms, family circumstances, or previous interventions. The group would then need to exchange information before reaching a decision.

Such mechanisms could reinforce communication, information sharing, and collective decision-making. Competitive elements can also be carefully incorporated, provided that competition does not undermine the cooperative nature of healthcare work.

9.9 Implications for Policy and Practice

The findings have several implications for CHW training systems.

9.9.1 Integrating Game-Based Refreshers into Existing Training

The evidence from Chapters-**??**, **??**, and **??** suggests that game-based activities can function effectively as refresher-learning tools. Their most appropriate role is likely to be complementary rather than substitutive. Conventional instruction and hands-on training can establish foundational competencies, while game-based activities can provide repeated opportunities for retrieval, practice, and reinforcement.

9.9.2 Maintaining Multiple Delivery Formats

A single delivery format is unlikely to serve all CHWs equally well. Digital games offer scalability, automated feedback, and the ability to track learner progress. Physical games require fewer technological resources and can support peer interaction. Maintaining both options may therefore be more equitable than assuming that digital delivery is always preferable.

9.9.3 Addressing Digital and Resource Barriers

Digital interventions need to account for differences in smartphone access, internet connectivity, device familiarity, and digital confidence. Offline functionality should be considered where reliable connectivity cannot be assumed. Physical alternatives can provide a practical fallback where digital deployment is difficult.

9.9.4 Language and Cultural Adaptation

Training content should be available in locally understandable languages and should be developed with the involvement of CHWs, trainers, and relevant stakeholders. Co-design can help ensure that examples, terminology, interaction patterns, and game conventions are recognizable to the intended users.

9.9.5 Continuous Reinforcement

The decline observed between immediate and delayed assessments provides a strong argument for periodic reinforcement rather than one-time exposure. A game-based refresher could therefore be incorporated into recurring training schedules, with short sessions distributed over time rather than relying on a single intensive intervention.

9.9.6 Using Learning Data Responsibly

Digital versions of the intervention can potentially collect information about learner progression, commonly missed questions, and repeated errors. Such information could help trainers identify areas requiring further instruction. However, analytics should be used primarily to support learning and should not be automatically interpreted as measures of individual workers' competence or performance without appropriate validation.

9.10 Limitations and Interpretation of the Evidence

The limitations of this dissertation need to be considered alongside the findings rather than treated as a separate checklist of weaknesses. The studies provide useful evidence about the potential of game-based refresher learning, but the design and context of the experiments place boundaries on the conclusions that can be drawn.

9.10.1 Sample Size, Statistical Power, and Precision

The relatively small sample sizes in the individual experiments are an important limitation. Sample size directly affects statistical power, the precision of estimated effect sizes, and the probability of detecting genuine differences between groups. The presence of a statistically significant result in a small sample can indicate that an effect was sufficiently large or consistent to be detected in that particular experiment, but it does not by itself establish that the estimated effect is precise or that it will be reproduced in a larger population.

This issue is particularly important for interpreting non-significant findings. For example, the absence of a statistically significant difference between digital and physical game formats in delayed retention should not be interpreted as proof that the two formats are equivalent. A study with limited statistical power may fail to detect a meaningful difference. The appropriate interpretation is therefore that the present experiments did not provide sufficient evidence to establish a statistically significant difference under the conditions tested.

The same consideration applies to subgroup analyses. Observed relationships involving age, educational background, or other participant characteristics should be regarded as exploratory unless the studies were specifically powered to test those effects. These patterns may be useful for generating hypotheses, but stronger conclusions require larger and adequately powered samples.

The limitation, therefore, affects the strength and precision of the conclusions rather than invalidating the observed findings. Future studies should conduct prospective power calculations using effect-size estimates informed by the present work and comparable previous research. Larger samples would also allow more reliable examination of interaction effects and subgroup differences.

9.10.2 Differences in Comparison Conditions

The comparison conditions varied across the three studies. Study-2 (Chapter-5) included digital, physical, classroom, and control conditions, while the other studies addressed somewhat different comparisons. In addition, classroom delivery depended on trainer availability and teaching approach.

These differences make direct numerical comparison across studies difficult. The results should therefore be interpreted as evidence accumulated through a sequence of related experiments rather than as observations from one homogeneous experimental design.

Future research should use standardized comparator conditions where possible, including comparable training duration, instructional content, facilitator preparation, and exposure time.

9.10.3 Ecological Validity

The experiments were conducted in organized training contexts rather than fully embedded within the everyday workflow of CHWs. This distinction matters because behavior in an experimental training session may differ from behavior under routine field conditions, where workload, supervisory structures, competing responsibilities, and community demands influ-

ence how learning is applied.

The results therefore provide stronger evidence for learning outcomes than for changes in routine healthcare practice. Testing the interventions during actual CHW activities would help determine whether improved knowledge translates into improved performance.

9.10.4 Technological Literacy and Access

Differences in digital familiarity may have influenced participants' experiences with smartphone and AR-based interventions. A participant who is already comfortable with smartphone interaction may be able to focus on the health content, whereas another participant may devote considerable cognitive effort to understanding the interface.

This creates a potential interaction between intervention effectiveness and technological familiarity. Future studies should measure baseline digital proficiency more systematically and examine whether it moderates learning outcomes.

9.10.5 Duration of Follow-Up

The delayed assessments provided evidence about retention beyond the immediate training session, but the follow-up periods were still relatively short. Consequently, the dissertation cannot establish whether the observed gains persist for several months or whether they influence longer-term professional practice.

Longer longitudinal studies are needed to examine whether repeated game-based reinforcement produces durable changes in knowledge, behavior, or service delivery.

9.10.6 Contextual and Geographical Transferability

The studies were conducted within particular linguistic, cultural, and institutional contexts. Although using multiple locations broadens the research's contextual scope, it does not establish

universal applicability.

The findings should therefore be transferred cautiously to CHWs working in substantially different contexts. Replication across additional regions, languages, healthcare systems, and worker populations would help determine which aspects of the intervention are robust and which require adaptation.

9.11 Future Research Directions

The limitations and findings together suggest several directions for future research.

9.11.1 Adequately Powered Comparative Experiments

Future studies should use prospective power calculations and sufficiently large samples to provide more precise estimates of intervention effects. Multi-site designs would be particularly useful because they could separate intervention effects from location-specific factors.

9.11.2 Longitudinal Evaluation

Future research should extend follow-up beyond the few weeks examined in the present studies. Repeated measurements over several months could determine whether periodic gameplay produces more durable retention than one-time exposure.

9.11.3 From Knowledge to Practice

The present research primarily evaluated learning outcomes. Future studies should examine whether improved knowledge translates into observable changes in CHW behavior, task accuracy, service efficiency, counseling quality, or community-level outcomes. Such research would

move evaluation beyond learning outcomes towards the higher levels of the training-impact chain.

9.11.4 Personalization and Adaptation

Digital interventions provide opportunities to adapt difficulty, content, and feedback according to learner performance. Future systems could identify recurring errors and provide targeted practice rather than requiring every learner to complete identical content.

However, personalization should be evaluated empirically rather than assumed to improve learning. Adaptive systems may introduce additional complexity and should be tested against simpler interventions.

9.11.5 Further Comparison of Digital and Physical Modalities

The present findings suggest that digital and physical games have different strengths. Future research should investigate the conditions under which each modality performs best rather than only asking which modality is "better." This could include comparisons based on learning objectives, learner age, digital proficiency, group size, setting, and frequency of reinforcement.

9.11.6 Expansion to Other Health Topics

The same design principles can be investigated in areas beyond immunization and maternal-child health. Potential topics include nutrition, adolescent health, chronic disease management, mental health, and community disease prevention.

The extension should retain the principle that game mechanics emerge from the underlying task. This would allow future research to examine whether the design principles identified in this dissertation generalize beyond the specific health content studied here.

9.12 Overall Discussion

The central contribution of this dissertation is not that games can replace conventional CHW training. Rather, it is that game-based interaction can provide an additional mode of learning that addresses some limitations of one-time, predominantly passive refresher training.

The three studies demonstrate a progression in this argument. Study-1 (Chapter-4) established the feasibility of transforming CHW training content into playful interaction and examined the contribution of AR. Study-2 (Chapter-5) provided a stronger comparison between physical and digital game-based learning and the classroom and control conditions. Study-3 (Chapter-6) extended the intervention towards smartphone-based repeated learning and provided further evidence concerning knowledge acquisition and retention.

Throughout this progression, one finding became increasingly clear. The educational value of a game depends less on technological novelty alone and more on how it structures interaction with the learning content. Digital technology can provide useful affordances, but physical interaction, social discussion, repetition, and contextual relevance can also contribute substantially to learning.

The findings, therefore, support a multimodal view of game-based refresher training. Rather than asking whether CHWs should receive digital or physical game-based training, the more useful question may be which combination of modalities provides the appropriate opportunities for learning, practice, and reinforcement in a particular context.

At the same time, the evidence should be interpreted cautiously. The sample sizes, differences between comparison conditions, limited follow-up periods, and context-specific nature of the experiments mean that the dissertation does not establish a universal effect of game-based learning. The strongest claim supported by the evidence is more specific: under the conditions examined in these studies, game-based refresher activities produced meaningful improvements in immediate learning and, in several comparisons, maintained advantages over conventional training on delayed assessments. The consistency of this pattern across related experiments provides a basis for further research, but larger, more rigorously powered studies are required to estimate the magnitude and durability of these effects with greater confidence.

9.13 Conclusions

This dissertation investigated whether game-based approaches could support refresher learning among CHWs, with particular attention to knowledge acquisition and retention, as well as differences between physical and digital modalities. Across three empirical studies, the research moved from exploratory AR-supported playful learning to comparative physical and digital card-based interventions and finally to smartphone-based refresher learning.

The results indicate that game-based interventions can support immediate knowledge acquisition and can provide advantages over conventional training under the conditions examined. However, the decline observed during delayed assessment also demonstrates that engagement during a single intervention is not sufficient to guarantee durable learning. Continuous reinforcement and opportunities for repeated retrieval are therefore important components of an effective refresher-training system.

The comparison between physical and digital formats further suggests that technological sophistication should not be treated as the primary determinant of learning effectiveness. Digital games offer scalability, automated feedback, and progression tracking, whereas physical games offer accessibility, tangible interaction, and opportunities for peer discussion. In several delayed comparisons, these differences in immediate performance were reduced, suggesting that the quality and structure of the learning experience may matter as much as the medium through which it is delivered.

The dissertation, therefore, argues for a complementary and context-sensitive approach to game-based CHW training. Game-based interventions can complement, rather than replace, conventional instruction and hands-on training. Their greatest potential may lie in providing low-cost, repeatable opportunities for practice and reinforcement between formal training sessions.

Future research should move beyond immediate knowledge outcomes and examine whether game-based refresher learning produces durable changes in professional practice and healthcare delivery. Such research should use adequately powered experimental designs, longer follow-up periods, and field-based evaluation. It should also continue to involve CHWs and other stakeholders in the design process so that technological interventions remain responsive to the social

and practical contexts in which they are used.

9.14 Personal Reflections

Coming to maternal and child healthcare from a Master's degree in Design was initially unfamiliar. I had to learn not only the content of public health but also the structures within which CHWs work, the constraints of government health programs, and the practical realities of designing for communities with very different technological and educational experiences.

The three studies changed the way I think about the relationship between design, technology, and learning. At the beginning of the research, it was tempting to think of technology as the main source of innovation. The progression from the AR prototype in Study-1 (Chapter-4), through the physical and digital comparison in Study-2 (Chapter-5), to the smartphone intervention in Study-3 (Chapter-6) gradually challenged that assumption. I came to see that the important question is not how much technology can be introduced into a training intervention, but what the technology allows people to do that would otherwise be difficult to achieve.

Working with CHWs also made me more aware of the difference between designing for users and designing with users. Their feedback repeatedly challenged assumptions that seemed reasonable from a designer's perspective. A feature that appeared innovative could become distracting in practice, while a simple physical card could become useful precisely because it was familiar and easy to share.

The research also changed my understanding of efficiency. During the fieldwork, I sometimes observed a preference for solutions that made learning faster and more convenient. Yet some of the most meaningful learning moments occurred when participants slowed down, discussed an answer, disagreed with one another, and eventually reached a shared understanding. This made me reconsider the assumption that good educational technology should always reduce effort. Sometimes the effort is part of the learning.

Perhaps the most important lesson from this research is that CHWs should not be treated simply as recipients of training interventions. They are experienced practitioners working within complex systems, and their knowledge comes from both formal training and years of

engagement with communities. A game can contribute to that knowledge, refresh it, and create opportunities to practice it, but it cannot substitute for the broader experience through which professional competence develops.

This realization also shapes my view of the limitations of the present research. The experiments provide useful evidence, but they are not the final answer to whether game-based learning improves healthcare delivery. The relatively limited sample sizes, short follow-up periods, differences between experimental contexts, and focus on knowledge outcomes mean that there is still considerable work to be done. Rather than weakening the dissertation's contribution, these limitations define the next questions the research should address.

Ultimately, this dissertation has made me more interested in designing learning systems than in designing individual learning technologies. The game is only one component of such a system. Its value depends on when it is used, who uses it, how often it is revisited, what happens around the gameplay, and whether the knowledge gained can be carried into everyday practice.

I hope that the work presented here contributes, even in a modest way, to that broader conversation. The aim is not to suggest that games are a universal solution for CHW training, but to demonstrate that carefully designed playful experiences can provide another pathway for CHWs to learn, practice, discuss, and revisit important knowledge. The next step is to understand how these pathways can be integrated into real training and healthcare systems in ways that are sustainable, inclusive, and genuinely useful to the people who use them.

Appendix A

Designing the Refresher Training Activity

A.1 Content for designing Refresher Training Activity

Conducting a Focused Group Discussion (FGD) among a group of AWWs and ASHAs, it was found that most CHWs lacked complete knowledge of the immunization schedule, growth monitoring chart, and other aspects of the Mother and Child Protection card (MCP card). The findings of other researchers from different parts of India were also similar to our (Bag and Datta, 2017). A study found that the issuing of MCP cards to pregnant mothers was strongly associated with partial immunization in children (Kizhatil et al., 2019). The child immunization table from the MCP card was chosen as content for the game. This schedule also follows the Indian Academy of Paediatrics (IAP) guidelines (Kasi and et.al., 2021). A reference to the immunization schedule is in Table A.1 in the Appendix.

Vaccine Name	Birth	$1\frac{1}{2} months$	$2\frac{1}{2} months$	$3\frac{1}{2} months$	9 months	$1\frac{1}{2} - 2 years$
BCG	✓					
Hepatitis B	✓					
OPV	✓	✓	✓	✓		✓
IPV		✓		✓		
Penta		✓	✓	✓		
PCV		✓		✓	✓	
Rota		✓	✓	✓		
MR					✓	✓
JE					✓	✓
DPT						✓

Table A.1: Child Immunization schedule

A.2 Physical Card Play as a Learning Tool

Initial thoughts of game design were that the information about vaccines is displayed to players through blocks of information. Then the player has to make a decision based on the information and the current scenario. Out of many physical tokens and objects of play, card gameplay was chosen as the game artifact. The possibilities of playing through card games were explored. Physical card games are ubiquitous. The primary reasons for this are its cost-effectiveness and handling ease. A deck of cards can be played in numerous different ways just by altering the rules. Different groups of cards can be added to the existing deck to enable new ways of playing with it. Multiplayer card game researchers (Bochennek et al., 2007) have shown significant learning advantages through engagement and collaboration. However, published research papers on using card games for training community healthcare workers were sparse.

The core learning objective of the game is that the community health workers memorize the sequence and dosage of immunizations and supplements required by the child and mother. Initially, there were thoughts of modding the classic physical card games like Rummy, Solitaire, Bridge, Teen Patti, etc. It was easier to think along the lines of combining the rules and mechanics of a popular card game with educational content. However, in several instances of

design iterations, the associated traditional rules of play restricted the fulfillment of our learning objectives. As a result, cherry-picking rules and mechanics were performed from several popular card games, which would address and fulfill the learning objectives of the ASHAs.

A.3 Designing the cards

The content can be categorized into 4 silos depending on the time of delivering the care to the stakeholders. 4 silos of cards were made each symbolizing an immunization, medicine supplement or a care activity or event. They are categorized as follows: Children below 1yr (n=22), Children above 1yr (n=15), ANC (n=14) and PNC (n=9) combined to form a deck of 60 cards. The deck of card design is displayed through Figures : 8.13, 8.14, 8.15, 8.16 and 8.17 in the Appendix section.

These playing cards are made the size of a standard playing card or credit card (5.5 X 8.5 cm), made of card paper (200 GSM), and have printed information and graphics on them.

A.4 Combination of cards

These 60 playing cards can be equally divided between 2, 3, 4, 5, or 6 players, creating multiplayer team combinations. The most common of them is a 4-player match. Theoretically, all the 60 cards could be arranged in a total of 60! ways. But for each of those arrangements, there are many duplicate arrangements in which cards 1 to 15 are the same, but in a different order. Similarly for cards 16 to 30, and so on. For each group of 15, there are 15! ways to rearrange them and not change the cards in each player's hands. That's why four factors of 15! would be in the denominator, considering a 4-player round. The mathematical expression of the number of permutations possible for each player's hand, considering a 4-player round, is as follows: 60! / (15! X 15! X 15! X 15!) or simplified $\frac{60!}{(15!)^4}$. Google Sheet Expression =FACT(60)/POW(FACT(15),4) which calculates to an astronomical value of 2,845,616,726,065,970,000,000,000,000,000,000

A.5 Designing Challenges, Mechanics and Rules of Play

In the process of game design, after brainstorming the basic game idea of stacking cards as per the schedule, we focused on methods to translate content elements of immunization into game elements. We can call this as translation phase. This helped towards achieving endogenousness in design, where game mechanics and resources emerged directly from the immunization content itself.

We started with reimagining content elements as game elements. In the designed game, the immunization content was represented by the vaccines, which act as resources in the game. Each vaccine (such as polio, measles, or tetanus) becomes a game resource that players must acquire and administer at the correct time, simulating the real-world immunization schedule.

The designed game involved players stacking "vaccine cards" in chronological order based on when each vaccine needs to be administered according to a child's age. Players would have to strategically decide which vaccines to prioritize, mirroring real-world challenges faced by CHWs in ensuring timely immunizations. These vaccine cards were color-coded or labeled to correspond to specific diseases, time of intervention (in months), and players received points or rewards for correctly stacking the cards in line with a virtual child's immunization needs.

Our strategy of choosing Game Mechanics, firstly involved identifying player actions that can be performed with the game's resources, while the other strategy mapped established mechanics onto content elements. By linking well-known mechanics with content-derived resources on immunization, we tried to develop a system that feels more naturally connected to immunization or the game's subject matter.

We also focused on opposition mechanics, obstructions, or opposing forces within the context of immunization. In the moderately advanced level of the game, the mechanics were included in the form of unforeseen challenges like a vaccine shortage, requiring players to trade or manage limited resources, or a simulated disease outbreak that necessitates quick decision-making to administer booster shots. This created dynamic decision-making and action elements that reflect the content of immunization, transforming real-world health information into an engaging and educational game format. Challenges often arise when opposition mechanics are not apparent in the content, leading to exogenous mechanics. However, these challenges can

also inspire creative solutions and novel opposition mechanics.

Endogenous design arises when resources or mechanics are organically derived from the content. However, when we were unable to synthesize mechanics or resources directly from content, we resorted to exogenous mechanics, using designs like races or other standard systems. These choices might lead to a more disjointed relationship between content and gameplay.

Once a core or opposition mechanic (blocking cards to restrict completing the sequence) has been identified, we followed the Mechanics-Dynamics-Aesthetics (MDA) framework by (Fabricatore Carlo, 2007). The translation of content to game elements, particularly through mechanics and resources, steered whether the design remains endogenous or shifts towards exogenous approaches.

In game-based learning, effective translation of content into game elements enables the delivery of learning through a variety of channels. These include visual elements (4 foundation layouts demonstrating 4 silos of mother and child care, mainly focusing on immunization), action elements (game mechanics like stacking of cards and ordering), information elements (Card details like name of immunization and time of intervention), and decision elements (player choices during the gameplay). Each of these elements contributes to how players interact with the game, allowing learning to emerge organically from engagement with the game system.

However, knowledge acquisition through game-based learning presents significant challenges, especially when introducing novel concepts. Research, including our previous studies (Majhi et al., 2021, 2022, 2024c,a,b), shows that activating prior knowledge is meaningful and sometimes easier than facilitating the acquisition of new knowledge. Designing games for novel knowledge is particularly difficult in complex subjects like immunization concepts.

Thus, while games can be highly effective for knowledge reinforcement and application, they are less frequently successful in facilitating the acquisition of entirely new knowledge, particularly in cases where the content is abstract or requires sensory engagement for full comprehension. However, in our designed game, we restricted our content to only factual information, and learning through simple game mechanics was an appropriate choice.

A.6 Rules of Play (Based on the rules of Badam Satti)

- The game starts by shuffling the deck of 60 cards, and distributing it equally to 4 players, 15 cards each (2 players 30 each or 3 players 20 each).

- Each color belongs to one silo or foundation. There are 4 foundations. Children below 1 year, above 1 year, antenatal care, and postnatal care. Make 4 spaces for 4 foundations (similar to solitaire)

- Then, at each player's turn, they need to put a card of one type onto the foundation of the same silo on the board.

- Let's say they put the cards of 2 years (eg. $2^1/_5$). This means there are 5 cards or immunizations in year 2. This might also be shown in 5 white rectangles in the new version to avoid confusion. They have to complete the 5 cards of 2 years in their turns. After completing the 2 years, they can either move up or down $1^1/_2$

- The cards are numbered in order of the months and year. If there are multiple immunizations/cards in an age group, they are numbered $2^1/_5$, $2^2/_5$, $2^3/_5$ and so on. This acts as a cue for the players to know how many more cards are left to complete the current sequence so that they don't miss a card and jump to the next month or sequence.

- If a player successfully places a card in the placeholder silo, the player gets 2 more chances or play rounds as a bonus.

- If there is no possibility of cards being placed or 2 bonus rounds are complete, then the turn goes to the next/following player.

- After one age group is completed, the following players, in their turn, can place cards in one age group above the upper set or below the lower set on the board.

- The rounds of play continue until all cards are exhausted from a player's hand. The first player to exhaust the holding cards wins the round and is awarded the first rank, followed by the next players until all cards are exhausted by all players.

A.7 Limitations of playing with physical cards

- The rules of the play need to be followed in order to achieve the intended learning. However, players need to constantly monitor the activities in order to catch deviations from the rules. This problem can be solved digitally through rules written in the form of algorithms. Every wrong move would be announced through voice cues.
- In India, there are 22 official languages. Different groups have different number literacy levels. Some are comfortable with the script of Devnagri numbers or regional ones, while some are with English numbers. Making customized decks cater to each group won't be scalable.
- The content or the knowledge to be delivered changes with new findings in medical science, which needs to be implemented over the state or country. This translates to adding new cards or scraping off the distributed cards, printing and resupplying new decks to all ASHAs, which is not a scalable solution.
- Mundane chores of play, like shuffling cards, equally distributing cards between the number of players, and packing them up, take up a lot of time for players to do manually. These can be done digitally in almost no time. It can also be done in an animated sequence to mimic the physical action sequence of shuffling, distributing, and packing up, which are very satisfying to watch in an interface.

A.8 Designing an app-based interface of the same activity

In the digital app, there are 4 foundations on the top which represent 4 silos. The players click on the desired card to pick it and click on the empty foundation to drop it. If there are multiple cards to that sequence (like $2^1/_5$, $2^2/_5$, $2^3/_5 \ldots$), The cards keep stacking over. As soon as the sequence is complete, they collapse into the foundation. It opens up the possibility to stack cards over or under, the bigger and smaller cards in sequence, respectively. A reference to the app's User Interface is shown in Figure **??** in the Appendix section.

A.9 Multiplayer Playing Variation

During previous field studies, it was observed that more than half of the ASHAs had access to smartphones but did not own them. Usually, their husbands own it and share it with their wives, the ASHA. The resource constraints of owning a phone individually could be potentially solved if one smartphone could be used by 4 players, turn by turn, while 4 players can physically arrange themselves sitting in a circle. A pass button in the User Interface to skip the turn to the next player was introduced. After each player plays her move in her turn and ends successfully stacking a card, the next player gets the turn. But if there are no moves possible, then the player can click on the Pass button to pass the turn to the next player. The player, after completing her turn, orients the phone by turning it towards the next player for her turn to play. Also, an assistive voice would be played announcing the current player's name, to reduce the confusion of which player's turn it is currently.

During previous field studies, it was also observed that the Community Healthcare Workers are comparatively busy during their daytime serving the community, going door to door. It becomes hard for them to manage time to physically get together, sit in a circle, and play the game either digitally or physically. A multiplayer remotely connected play would potentially solve this problem. In this case, one of the players creates a server generally called a "Room" having a meaningful name representing the team. The rest of the players connect to that room and the creator of the room starts the game. Similar mechanics are followed. The players get an opportunity to put cards in their turn. When the turn is passed, the moves get frozen until the turn comes around again, completing all the other players.

During trials, with remotely connected players, we observed that there were a lot of misunderstandings due to the lack of communication between players. They felt frustrated if they couldn't perform a task due to the restrictions enforced by the rules of play, for example, when the turn was missed or they couldn't place a card in their turn, etc. So we introduced audio communication similar to a radio which broadcasts audio of one player to all other players connected in the room or with the other player forming a group with the player. We observed peer learning happening through the discussions between the players while explaining the rules to one another and in general team conversations.

Following the same game mechanics, rules, and deck of cards, two variations of the Android game app were developed. The gameplay analytics of each player were saved offline and asynchronously stored in an online database whenever the internet was available, for further analysis. ASHAs having an active data connection would update the scores instantly. It was observed that a number of ASHAs didn't have an active data pack in their smartphone. For them, there would be asynchronous syncing multiple times within a week, whenever she would come to the state institution for reporting and would connect her smartphone to the Wi-Fi network of the institution (which is usually available).

Appendix B

Survey Questionnaire

मैं अपनी हस्ताक्षर के माध्यम से स्वीकार करती हूं कि इस शोध अध्ययन में मेरी भागीदारी स्वैच्छिक है। मैं इस तथ्य से अवगत हूं कि एकत्र की गई व्यक्तिगत जानकारी का उपयोग केवल शैक्षिक उपयोग के लिए किया जाएगा।

_____ ____ / ____ / २०२२

हस्ताक्षर **तिथि**

नीचे दी हुई जगह पर अपना पूरा नाम लिखिए :

_____

अपनी उम्र लिखिए : _____ **साल**

आपकी स्कूली शिक्षा कौनसी श्रेणी तक हुई है :

☐ ८ वि ☐ ९ वि ☐ १० वि ☐ ११वी
☐ १२ वि ☐ ग्रेजुएट ☐ पोस्ट-ग्रेजुएट

आप कितने वर्षो से आशा कार्यकर्ता हैं :

_____ **साल**

कृपया निम्नलिखित प्रश्नों के उत्तर खाली बक्सों में ☑ निशान लगाकर दीजिये । प्रत्येक प्रश्न का केवल एक ही सही उत्तर है ।

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१. ओ पी वी - बूस्टर शिशु को किस आयु मै दिया जता है?

☐ ९ महीने ☐ ११/२ साल
☐ २ साल ☐ ३ १/२ साल

२. पेंटावैलेंट - १ टीका शिशु को किस आयु मै दिया जता है?

☐ ११/२ महीने ☐ २ १/२ महीने
☐ ३ १/२ महीने ☐ ९ महीने

३. एम आर - १ (खसरे का पहला टीका) शिशु को किस आयु मै दिया जता है?

☐ ९ महीने ☐ ११/२ साल
☐ २ साल ☐ ३ १/२ साल

४. पहला डीपीटी बूस्टर शिशु को किस आयु मै दिया जता है?

☐ ९ महीने ☐ ११/२ साल
☐ २ साल ☐ ३ १/२ साल

५. दूसरा डीपीटी बूस्टर शिशु को किस आयु मै दिया जता है?

☐ ९ महीने ☐ ११/२ साल
☐ ५-६ साल ☐ ३ १/२ साल

६. टी टी / टी डी (टेटनेस / डिप्थोरिया) का पहला टीका शिशु को कौनसी आयु मे दिया जाता है?

☐ १० साल ☐ १६ साल
☐ ६ साल ☐ ५ साल

७. टी टी / टी डी (टेटनेस / डिप्थोरिया) का दूसरा टीका शिशु को कौनसी आयु मे दिया जाता है?

☐ १० साल ☐ १६ साल
☐ ६ साल ☐ ५ साल

८. पेंटावैलेंट - २ टीका शिशु को किस आयु मै दिया जता है?

☐ ११/२ महीने ☐ २ १/२ महीने
☐ ३ १/२ महीने ☐ ९ महीने

Figure B.1: Questionnaire Page-1

९. विटामिन ए सबसे पहले बच्चों को कब दिया जाता है?

⬜ ९ महीने ⬜ १ १/२ साल

⬜ २ साल ⬜ ३ १/२ साल

१०. विटामिन ए दूसरी बार बच्चों को कब दिया जाता है?

⬜ ९ महीने ⬜ १ १/२ साल

⬜ २ साल ⬜ ३ १/२ साल

११. दूसरी बार के बाद विटामिन ए बच्चों को किस अंतराल में दिया जाता है?

⬜ ३ महीने ⬜ ४ महीने

⬜ ५ महीने ⬜ ६ महीने

१२. एम आर - २ (खसरे का दूसरा टीका) शिशु को कब दिया जता है?

⬜ ९ महीने ⬜ १ १/२ साल

⬜ २ साल ⬜ ३ १/२ साल

१३. जे ई - १ (जापानीज इन्सेफेलाइटिस) का पहला टीका शिशु को कब दिया जाता है?

⬜ ९ महीने ⬜ १ १/२ साल

⬜ २ साल ⬜ ३ १/२ साल

१४. जे ई - २ (जापानीज इन्सेफेलाइटिस) का दूसरा टीका शिशु को कब दिया जाता है?

⬜ ९ महीने ⬜ १ १/२ साल

⬜ २ साल ⬜ ३ १/२ साल

१५. पी सी वी - १ (न्यूमोकोकल कंजुगेट) का पहला टीका शिशु को कब दिया जाता है?

⬜ १ १/२ महीने ⬜ २ १/२ महीने

⬜ ३ १/२ महीने ⬜ ९ महीने

१६. पी सी वी - २ (न्यूमोकोकल कंजुगेट) का दूसरा टीका शिशु को कब दिया जाता है?

⬜ १ १/२ महीने ⬜ २ १/२ महीने

⬜ ३ १/२ महीने ⬜ ९ महीने

१७. पी सी वी - बूस्टर शिशु को कब दिया जता है?

⬜ १ १/२ महीने ⬜ २ १/२ महीने

⬜ ३ १/२ महीने ⬜ ९ महीने

१८. पेंटावैलेंट - ३ टीका शिशु को किस आयु मै दिया जता है?

⬜ १ १/२ महीने ⬜ २ १/२ महीने

⬜ ३ १/२ महीने ⬜ ९ महीने

१९. हेपेटाईटिस बी टीका शिशु को कब दिया जता है?

⬜ जन्म ⬜ १ १/२ महीने

⬜ २ महीने ⬜ ३ १/२ महीने

२०. रोटा - १ (रोटावायरस) का पहला टीका शिशु को कब दिया जता है?

⬜ जन्म ⬜ १ १/२ महीने

⬜ २ महीने ⬜ ३ १/२ महीने

२१. बी सी जी का टीका शिशु को कब दिया जता है?

⬜ जन्म ⬜ १ १/२ महीने

⬜ २ महीने ⬜ ३ १/२ महीने

२२. रोटा - २ (रोटावायरस) का दूसरा टीका शिशु को कब दिया जता है?

⬜ जन्म ⬜ १ १/२ महीने

⬜ २ महीने ⬜ ३ १/२ महीने

२३. ओ पी वी - ० (ओरल पोलियो वैक्सीन) का पहला टीका शिशु को कब दिया जता है?

⬜ जन्म ⬜ १ १/२ महीने

⬜ २ महीने ⬜ ३ १/२ महीने

२४. रोटा - ३ (रोटावायरस) का तीसरा टीका शिशु को कब दिया जता है?

⬜ जन्म ⬜ १ १/२ महीने

⬜ २ महीने ⬜ ३ १/२ महीने

२५. जन्म के तुरंत बाद शिशु को क्या दिया जाना चाहिए?

⬜ कोलोस्ट्रम ⬜ ओ पी वी - ०

⬜ हेपेटाईटिस बी ⬜ बी सी जी का टीका

Figure B.2: Questionnaire Page-2

२६. ओ पी वी - १ (ओरल पोलियो वैक्सीन) का दूसरा टीका शिशु को कब दिया जता है?

⬜ जन्म ⬜ १ १/२ महीने

⬜ २ महीने ⬜ ३ १/२ महीने

२७. आई पी वी - १ (इनएक्टिवेटेड पोलियो वैक्सीन) का पहला टीका शिशु को कब दिया जता है?

⬜ जन्म ⬜ १ १/२ महीने

⬜ २ महीने ⬜ ३ १/२ महीने

२८. ओ पी वी - २ (ओरल पोलियो वैक्सीन) का दूसरा टीका शिशु को कब दिया जता है?

⬜ जन्म ⬜ १ १/२ महीने

⬜ २ महीने ⬜ ३ १/२ महीने

२९. पेंटावैलेंट - २ टीका शिशु को किस आयु मै दिया जता है?

⬜ ९ महीने ⬜ १ १/२ साल

⬜ २ साल ⬜ ३ १/२ साल

३०. ओ पी वी - ३ (ओरल पोलियो वैक्सीन) का दूसरा टीका शिशु को कब दिया जता है?

⬜ जन्म ⬜ १ १/२ महीने

⬜ २ महीने ⬜ ३ १/२ महीने

३१. आई पी वी - २ (इनएक्टिवेटेड पोलियो वैक्सीन) का पहला टीका शिशु को कब दिया जता है?

⬜ जन्म ⬜ १ १/२ महीने

⬜ २ महीने ⬜ ३ १/२ महीने

३२. एक गर्भवती माँ को अपनी पूरी गर्भावस्था में कितने न्यूनतम प्रसवपूर्व देखभाल सत्रों की आवश्यकता होती है ?

⬜ १ ⬜ २

⬜ ३ ⬜ ४

३३. किस प्रसवपूर्व देखभाल पर गर्भवती माँ को टी टी - १ (टिटनेस टॉक्साइड) का पहला इंजेक्शन दिया जाता है ?

⬜ पहला ⬜ दूसरा

⬜ तीसरा ⬜ चौथा

३४. किस प्रसवपूर्व देखभाल पर गर्भवती माँ को टी टी - २ (टिटनेस टॉक्साइड) का दूसरा इंजेक्शन दिया जाता है ?

⬜ पहला ⬜ दूसरा

⬜ तीसरा ⬜ चौथा

३५. टीटी बूस्टर उन गर्भवती माताओ को दी जाती है जिन्होने पिछले कितनी सालो में टीटी के २ टीके लगवा चुकी है ?

⬜ १ ⬜ २

⬜ ३ ⬜ ४

३६. किस प्रसवपूर्व देखभाल पर गर्भवती माँ को आयरन फोलिक एसिड (आई एफ ए) की गोलियां सेवन चालू करनी चाहिए ?

⬜ पहला ⬜ दूसरा

⬜ तीसरा ⬜ चौथा

३७. गर्भवती महिलाओं को कौन सी कृमिनाशक गोली दी जाती है ?

⬜ कैल्शियम ⬜ एल्बेंडाजोल

⬜ आई एफ ए ⬜ विटामिन डी

३८. आई एफ ए गोलि के साथ किस गोलि का सेवन नहीं करना चाहिए ?

⬜ कैल्शियम ⬜ एल्बेंडाजोल

⬜ आई एफ ए ⬜ विटामिन डी

३९. सिफलिस टेस्ट और एचआईवी टेस्ट किस प्रसवपूर्व देखभाल पर गर्भवती माँ को करवाना चाहिए ?

⬜ पहला ⬜ दूसरा

⬜ तीसरा ⬜ चौथा

४०. पहला प्रसव पश्चात देखभाल कितने घंटों के भीतर करना चाहिए?

⬜ ४ ⬜ ६

⬜ १२ ⬜ २४

Figure B.3: Questionnaire Page-3

	A	B	D	E	F	G
1	Sl	English	IndiaTyping	Google	Corrected Human	Reverse Translated
2	1	I felt content	मला समाधान वाटले	मला सामग्री वाटली	मला समाधान वाटलं	I was satisfied
3	2	I felt skillful	मला कुशल वाटले	मला कुशल वाटले	मला खेळामुळे खूप कुशल वाटलं	I felt very skilled because of the game
4	3	I was interested in the game's story	मला खेळाच्या कथेत रस होता	मला खेळाच्या कथेत रस होता	या खेळामधील गोष्टी साठी मी उत्सुक होते	I was looking forward to the things in this game
5	4	I thought it was fun	मला वाटलं गंमत आहे	मला वाटले की ते मजेदार आहे	मला हा खेळ गम्मत वाटली	I had this game of fun
6	5	I was fully occupied with the game	मी खेळात पूर्णपणे गुंतले होते	मी खेळाने पूर्णपणे ताब्यात घेत होतो	मी खेळामध्ये पूर्ण दंग झालेले	I was full of timber in the game
7	6	I felt happy	मला आनंद वाटला	मला आनंद झाला	मला आनंद वाटला	I was happy
8	7	It gave me a bad mood	मला वाईट मूड दिला	यामुळे मला एक वाईट मूड दिला	याच्यामुळे मला वाईट अनुभव आला	I had a bad experience because of this
9	8	I thought about other things	मी इतर गोष्टींचा विचार केला	मी इतर गोष्टींबद्दल विचार केला	मी खेळताना दुसऱ्या गोष्टींचा विचार करता होते	I was thinking of other things while playing
10	9	I found it tiresome	मला ते कंटाळवाणे वाटले	मला ते कंटाळवाणे वाटले	मला हा खेळ थकवणारा वाटला	I found this game tiring
11	10	I felt competent	मला सक्षम वाटले	मला सक्षम वाटले	मला सक्षम असल्यासारख वाटलं	I felt like I was able to
12	11	I thought it was hard	मला वाटले ते कठीण आहे	मला वाटले ते कठीण आहे	मला वाटले ते कठीण आहे	I thought it was hard
13	12	It was aesthetically pleasing	ते सौंदर्यानि सुखावणारे होते	हे सौंदर्यात्मकदृष्ट्या आनंददायक होते	खेळ दिसायला सुंदर आणि आकर्षक होता	The game was pretty and attractive to see
14	13	I forgot everything around me	मी माझ्या आजूबाजूचे सर्व काही विसरलो	मी माझ्या सभोवतालचे सर्व काही विसरलो	मी सगळं विसरून खेळात आकंठ बुडाले	I forgot all about the game
15	14	I felt good	मला बरं वाटलं	मला बरे वाटले	मला छान वाटल	I would like to feel nice
16	15	I was good at it	मी त्यात चांगला होतो	मी त्यात चांगला होतो	मला खेळात गती होती	I had a speed in the game
17	16	I felt bored	मला कंटाळा आला	मला कंटाळा आला	मला कंटाळा आला	I got bored
18	17	I felt successful	मला यशस्वी वाटले	मला यशस्वी वाटले	मला यशस्वी वाटलं	I felt successful
19	18	I felt imaginative	मला कल्पनारम्य वाटले	मला काल्पनिक वाटले	मझ्या कल्पनाशक्ती ला चालना मिळाली	My imagination got a walk
20	19	I felt that I could explore things	मला वाटले की मी गोष्टी शोधू शकतो	मला वाटले की मी गोष्टी शोधू शकतो	मला असा वाटलं कि मी पण गोष्टी शिकू शकते	I think I can learn things but
21	20	I enjoyed it	मला मजा आली	मी त्याचा आनंद घेतला	मी इंजॉय केला	I did the engoy
22	21	I was fast at reaching the game's targets	मी खेळाचे लक्ष्य गाठण्यात जलद होते	मी खेळाच्या लक्ष्यात पोहोचण्यात वेगवान होतो	मी खूप वेगाने एक की टप्पा पार करत होते	I was crossing a key phase very fast
23	22	I felt annoyed	मला चीड वाटली	मला राग आला	मला खूप त्रासदायक वाटलं	I feel very annoying
24	23	I felt pressured	मला दडपण जाणवले	मला दबाव वाटला	मला दबाव जाणवला	I felt the pressure
25	24	I felt irritable	मला चिडचिड वाटली	मला चिडचिडे वाटले	खेळताना माझी चिडचिड झाली	I got irritated while playing
26	25	I lost track of time	मी वेळेचा मागोवा गमावला	मी वेळेचा मागोवा गमावला	मी खेळताना वेळेचा भान विसरले	I forgot about the idea of time while playing
27	26	I felt challenged	मला आव्हान वाटले	मला आव्हान वाटले	मला खेळ आव्हानात्मक वाटला	I found the game challenging
28	27	I found it impressive	मला ते प्रभावी वाटले	मला ते प्रभावी वाटले	मला खेळ आकर्षक वाटला	I found the game attractive
29	28	I was deeply concentrated in the game	मी खेळात मनापासून एकाग्र होतो	मी गेममध्ये खोलवर लक्ष केंद्रित केले होते	मी खेळ खेळताना खूप एकाग्र झालेले	I
30	29	I felt frustrated	मला निराश वाटले	मी निराश झालो	मला खूप निराश वाटले	I felt very frustrated
31	30	It felt like a rich experience	एक समृद्ध अनुभव वाटला	हा एक समृद्ध अनुभव वाटला	हा खूप आल्हाददायक अनुभव होता	It was a very pleasant experience
32	31	I lost connection with the outside world	माझा बाहेरच्या जगाशी संपर्क तुटला	मी बाह्य जगाशी संबंध गमावला	मी बाकीच्या जगाचा भान विसरले	I forgot about the rest of the world
33	32	I felt time pressure	मला वेळेचा दबाव जाणवला	मला वेळ दबाव वाटला	मला वेळेत पूर्ण करण्याचा दबाव जाणवला	I felt the pressure to complete in time
34	33	I had to put a lot of effort into it	त्यासाठी मला खूप मेहनत घ्यावी लागली	मला त्यात बरेच प्रयत्न करावे लागले	मला खूप मेहनत घ्यावी लागली	I had to work a lot of hard work
35						
36	34	not at all	अजिबात नाही	अजिबात नाही	अजिबात नाही	Not at all
37	35	slightly	किंचित	किंचित	किंचित प्रमाणात	Quantitatively
38	36	moderately	माफक प्रमाणात	माफक प्रमाणात	मध्यम प्रमाणात	To a moderate amount
39	37	fairly	प्रामाणिकपणे	प्रामाणिकपणे	चांगल्या प्रमाणात	Well
40	38	extremely	अत्यंत	अत्यंत	खूप जास्त	Too much

Figure B.4: GEQ Translation Processs

Appendix C

Institute Review Board Approval

IRCC, Office of the Dean R&D, IIT Bombay

INDIAN INSTITUTE OF TECHNOLOGY BOMBAY

Institutional Review Board (IRB)

January 17, 2023

To
Prof. Satish Agnihotri
CTARA
IIT Bombay

Ref: Proposal No. IITB-IRB/2022/051

Sub: Review of the above-mentioned project proposal

Dear Professor,

Thank you for submitting your proposal to the IITB Institutional Review Board (IRB) for review. The IRB has reviewed the proposal submitted by you and the following proposal is approved:

Proposal number	: **IITB-IRB/2022/051**
Title	: **Refresher Training through Games for Frontline Healthcare Workers on Maternal and Newborn care**

The IRB approval is for the ethical conduct of the study. The study is approved for the entire duration and a closure report should be submitted within 2 months of the completion of the study.

Further, it is also confirmed that neither you, nor any of the study team members have participated in the decision-making process of the committee.

In case there are any changes in the proposed work (which is not limited to scope, dates, participants and methodology etc.), please communicate to IRB within 15 days of such a change.

Thank you.
With Best wishes from IRB for your study,

D Parthasarathy
Chairperson IITB-IRB

Figure C.1: IRB Approval

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List of Publications

1. **Majhi, A.**, Mondal, A., Joshi, A., Agnihotri, S. B., 2021. Refresher Training through Quiz App for capacity building of Community Healthcare Workers or Anganwadi Workers in India. *In Proceedings of the Asian CHI Symposium 2021 (Asian CHI '21). Association for Computing Machinery, New York, NY, USA, 80–88.*
   (Best Paper award in Long Paper category)
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2. **Majhi, A.**, Agnihotri, S. B., Mondal, A., 2022. Physical and Augmented Reality based Playful Activities for Refresher Training of ASHA Workers in India. *In Proceedings of the Asian HCI Symposium 2022 (Asian HCI '22). Association for Computing Machinery, New York, NY, USA, 16–22.*
   (Best Paper and Best Presentation award in Long Paper category)
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3. **Majhi, A.**, Agnihotri, S. B., Mondal, A., 2024. Replay, Revise, and Refresh: Smartphone-Based Refresher Training for Community Healthcare Workers in India. *HCI International 2024, Washington DC, USA*
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4. **Majhi, A.**, Agnihotri, S. B., Mondal, A., 2024. Mapping Child Malnutrition and Measuring Efficiency of Community Healthcare Workers through Location Based Games in India. International Conference on Information Technology for Social Good (GoodIT '24, Bremen, Germany)
   https://dl.acm.org/doi/10.1145/3677525.3678685

5. **Majhi, A.**, Agnihotri, S. B., Mondal, A., 2024. Refresher Training through Digital and Physical, Card-Based Game for Accredited Social Health Activists (ASHAs) and Anganwadi Workers (AWWs) in India. CHI PLAY Companion '24, October 14–17, 2024, Tampere, Finland
   https://dl.acm.org/doi/10.1145/3665463.3678819

6. **Majhi, A.**, Agnihotri, S. B., Mondal, A., 2026. Design Guidelines for Game-Based Refresher Training of Community Health Workers in Low-Resource Contexts. Interactive Health Conference (IH '26), July 05–08, 2026, Porto, Portugal. https://doi.org/10.1145/3786579.3804932

7. **Majhi, A.**, Agnihotri, S. B., Mondal, A., 2026. Healthcare App Design in Low-Resource Contexts: Challenges, Practices, and Opportunities. Interactive Health Conference (IH '26), July 05–08, 2026, Porto, Portugal. https://doi.org/10.1145/3786579.3803814

Author's Contributions

The following list presents the research contributions of the included publications' authors using the Contributor Roles Taxonomy (CRediT).

1-7: The Publication Numbers

AM1: Arka Majhi

AM2: Aparajita Mondal (Her contributions to research are not part of her academic pursuits)

SBA: Satish B. Agnihotri

AJ: Anirudha Joshi

	AM1	AM2	SBA	AJ
Conceptualization	1-7		1-7	1
Data curation	1-7	1-7		
Formal analysis	1-7			
Funding acquisition	1-7		1-7	1
Investigation	1-7	1-7	1-7	1
Field Work	1-7	1-7		
Methodology	1-7		1-7	1
Project administration	1-7	1-7	1-7	1
Resources	1-7			
Software	1-7			
Supervision			1-7	1
Validation	1-7			
Visualization	1-7			
Writing – original draft	1-7			
Writing – review & editing			2-7	1

Table 9.1: Author's Contributions

Reference of definition and details of contributor roles (https://credit.niso.org/)

1. **Conceptualization** – Ideas; formulation or evolution of overarching research goals and aims.

2. **Data curation** – Management activities to annotate (produce metadata), scrub data and maintain research data (including software code, where it is necessary for interpreting the data itself) for initial use and later re-use.

3. **Formal analysis** – Application of statistical, mathematical, computational, or other formal techniques to analyze or synthesize study data.

4. **Funding acquisition** - Acquisition of the financial support for the project leading to this publication.

5. **Investigation** – Conducting a research and investigation process, specifically performing the experiments, or data/evidence collection.

6. **Methodology** – Development or design of methodology; creation of models.

7. **Project administration** – Management and coordination responsibility for the research activity planning and execution.

8. **Resources** – Provision of study materials, reagents, materials, patients, laboratory samples, animals, instrumentation, computing resources, or other analysis tools.

9. **Software** – Programming, software development; designing computer programs; implementation of the computer code and supporting algorithms; testing of existing code components.

10. **Supervision** – Oversight and leadership responsibility for the research activity planning and execution, including mentorship external to the core team.

11. **Validation** – Verification, whether as a part of the activity or separate, of the overall replication/reproducibility of results/experiments and other research outputs.

12. **Visualization** – Preparation, creation and/or presentation of the published work, specifically visualization/data presentation.

13. **Writing – original draft** – Preparation, creation and/or presentation of the published work, specifically writing the initial draft (including substantive translation).

14. **Writing – review & editing** – Preparation, creation and/or presentation of the published work by those from the original research group, specifically critical review, commentary or revision – including pre- or post-publication stages.

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